Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title:	**CALCULATION** **49PR POWER PLANT PIPE RACK** **DESIGN OF FOUNDATION**

Document Number:	**MD-216-49PR-EG-CV-CAL-2001**
Revision:	**D01**
Revision Date:	**26.10.2023**
Status:	**AS BUILT**

Outotec Document Number:	900707	GAU01	WG01	CED01	45015	OU100682341	2
L&T Document Number:	O19045	008	C	K	24291	005	2

REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
A01	16.07.2020	FIRST ISSUE	SVK	UR	AG	
C01	14.08.2020	ISSUED FOR CONSTRUCTION	SVK	UR	AG	
D01	26.10.2023	AS BUILT	SVK	UR	AG	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
STAAD	:	Commonly used analysis and design software package named STAAD.Pro
Sec.	:	Section
max.	:	Maximum
ref.	:	Reference
No.	:	Number
cl.	:	Clause
nos.	:	Numbers
Col.	:	Column
approx.	:	Approximate
eqn.	:	Equation
mkd.	:	Marked
Dia.	:	Diameter
Spac.	:	Spacing
Rebar	:	Reinforcement Bar

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1. INTENT

The purpose of this document is to analysis and design of foundations for 49PR power plant pipe rack cum cable rack supporting trestles.The foundation sizes are calculated to ensure structural integrity and safety for the design load combinations. The analysis and design is carried out in accordance with project specifications, codes and standards.
Road cross over galleries RCO-1 & RCO-2 covered in this design document.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and STAAD. Load are taken from super structure STAAD file of RCO-1 & RCO-2. Sliding & overturning check are done in excel sheet. STAAD modelling has been done with plate elements.
Modulus of Subgrade reaction of 54000 kN/m^2/m and net allowable bearing capacity of 450 kN/m^2 have been considered as per BH-06 of power plant area as per soil report.

4. DESIGN PARAMETER:

- Concrete Strength of 28 days : For foundation & pedestals = 35MPa
  Binding concrete below foundation base = 15 MPa
- Reinforcement Grade : Deformed Bar : ASTM A 615/ A 615M Grade 60 (fy= 420MPa)
  Welded Wire Fabric : ASTM A 185 (Plain)/ A 497 (Deformed)
- Grouting : Grout shall be 30 mm cementitious of Strength 40 Mpa

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5. PRIMARY LOAD CASES CONSIDERED :

Primary Load Case	Description	Direction	Remarks	STAAD Nomenclature	Load Case Number
Dead Load	-	Downward	-	D	1
Live Load	-	Downward	-	L	2
	-	Downward	-	L1	3
Wind Load	-	Along +Z	-	WL (+Z)	11
	-	Along -Z	-	WL (-Z)	12
	-	Along +X	-	WL (+X)	13
	-	Along -X	-	WL (-X)	14
Earthquake Load	-	Along +Z	-	EL (+Z)	30
	-	Along -Z	-	EL (-Z)	31
	-	Along +X	-	EL (+X)	32
	-	Along -X	-	EL (-X)	33
Temperature Load	-	-	FOR +25°C	TL1	20
	-	-	FOR -25°C	TL2	21

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6. LOAD COMBINATIONS CONSIDERED:

Load combinations are determined accordance with SBC 301, SBC 304 and Civil, Structural Design Criteria document number MD-216-1000-EG-ST-CRT-1001.

Factored Load Combinations for strength design:

[ref. cl. 2.3.2 : SBC 301-2007]

Serial No.	STAAD Load Combination Number	Description
1	100	1.4xD
2	101	1.2xD + 1.2xTL1 + 1.6xL
3	102	1.2xD + 1.2xTL1 + 1.6xL1
4	103	1.2xD + 1.2xTL2 + 1.6xL
5	104	1.2xD + 1.2xTL2 + 1.6xL1
6	105	1.2xD + 0.8xWL(+Z)
7	106	1.2xD + 0.8xWL(+X)
8	107	1.2xD + 0.8xWL(-Z)
9	108	1.2xD + 0.8xWL(-X)
10	109	0.9xD + 1.6xWL(+Z)
11	110	0.9xD + 1.6xWL(+X)
12	111	0.9xD + 1.6xWL(-Z)
13	112	0.9xD + 1.6xWL(-X)
14	113	1.2xD + 0.5xL + 1.6xWL (+Z)
15	114	1.2xD + 0.5xL + 1.6xWL (+X)
16	115	1.2xD + 0.5xL + 1.6xWL (-Z)
17	116	1.2xD + 0.5xL + 1.6xWL (-X)
18	117	1.2xD + 0.5xL1 + 1.6xWL (+Z)
19	127	1.2xD + 0.5xL1 + 1.6xWL (+X)
20	118	1.2xD + 0.5xL1 + 1.6xWL (-Z)
21	119	1.2xD + 0.5xL1 + 1.6xWL (-X)
22	128	1.2xD + 0.5xL + 1xEL (+Z)
23	120	1.2xD + 0.5xL + 1xEL (+X)
24	121	1.2xD + 0.5xL + 1xEL (-Z)
25	122	1.2xD + 0.5xL + 1xEL (-X)
26	129	1.2xD + 0.5xL1 + 1xEL (+Z)
27	123	1.2xD + 0.5xL1 + 1xEL (+X)
28	124	1.2xD + 0.5xL1 + 1xEL (-Z)
29	125	1.2xD + 0.5xL1 + 1xEL (-X)
30	126	0.9xD + 1xEL(+Z)
31	130	0.9xD + 1xEL(+X)
32	131	0.9xD + 1xEL(-Z)
33	132	0.9xD + 1xEL(-X)

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Factored Load Combinations for serviceability design:

[ref. cl. 2.4.1 : SBC 301-2007]

Serial No.	STAAD Load Combination Number	Description
1	201	D + L + TL1
2	202	D + L1 + TL1
3	203	D + L + TL2
4	204	D + L1 + TL2
5	205	D + 0.75xL + 0.75xTL1
6	206	D + 0.75xL1 + 0.75xTL1
7	207	D + 0.75xL + 0.75xTL2
8	208	D + 0.75xL1 + 0.75xTL2
9	209	D + WL(+Z)
10	210	D + WL(+X)
11	211	D + WL(-Z)
12	212	D + WL(-X)
13	213	D + 0.7xEL(+Z)
14	214	D + 0.7xEL(+X)
15	215	D + 0.7xEL(-Z)
16	216	D + 0.7xEL(-X)
17	217	D + 0.75xL + 0.75xWL(+Z)
18	218	D + 0.75xL1 + 0.75xWL(+Z)
19	219	D + 0.75xL + 0.75xWL(-Z)
20	220	D + 0.75xL1 + 0.75xWL(-Z)
21	221	D + 0.75xL + 0.75xWL(+X)
22	222	D + 0.75xL1 + 0.75xWL(+X)
23	223	D + 0.75xL + 0.75xWL(-X)
24	224	D + 0.75xL1 + 0.75xWL(-X)
25	225	D + 0.75xL + 0.525xEL(+Z)
26	226	D + 0.75xL1 + 0.525xEL(+Z)
27	227	D + 0.75xL + 0.525xEL(+X)
28	228	D + 0.75xL1 + 0.525xEL(+X)
29	229	D + 0.75xL + 0.525xEL(-Z)
30	230	D + 0.75xL1 + 0.525xEL(-Z)
31	231	D + 0.75xL + 0.525xEL(-X)
32	232	D + 0.75xL1 + 0.525xEL(-X)
33	233	0.6xD + WL(+Z)
34	234	0.6xD + WL(+X)
35	235	0.6xD + WL(-Z)
36	236	0.6xD + WL(-X)
37	237	0.6xD + 0.7xEL(+Z)
38	238	0.6xD + 0.7xEL(+X)
39	239	0.6xD + 0.7xEL(-Z)
40	240	0.6xD + 0.7xEL(-X)

7. ALLOWABLE SOIL BEARING CAPACITY:

Net safe bearing capacity of soil considered as 450kN/m2 and vertical modulus of subgrade reaction considered as 54000 kN/m2/mat at 1.5m below EGL as per BH-6 [refer table-5 of MD-216-3000-EG-CV-RPT-1008].

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8. DESIGN CALCULATION OF FOOTING:

- **DESIGN CALCULATION OF FOOTING RCO-1/ F1:**

Size of Columns at node	Along-X	Along-Z
1	0.60m	0.60m
2	0.60m	0.60m
3	0.60m	0.60m
4	0.60m	0.60m
Size of fdn. raft:	5.00m	5.00m

Thickness of raft:	0.50	m	RL (+) 913.60
Original Gr. Level:	0.00	m	RL (+) 913.95
Finished Gr. Level:	0.00	m	
U/S of B/Pl Above GL:	0.35	m	
U/S level of fdn:	-1.50	m	
Unit weight of conc:	24	kN/m^3	
Unit weight of soil:	20	kN/m^3	

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

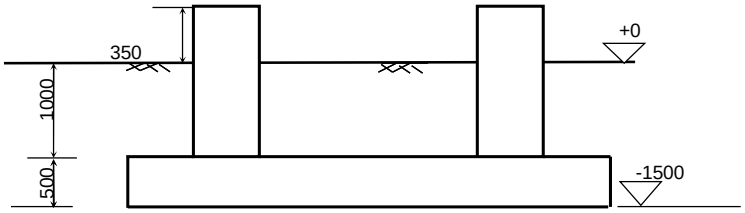
Calculation of Weight of Foundation :

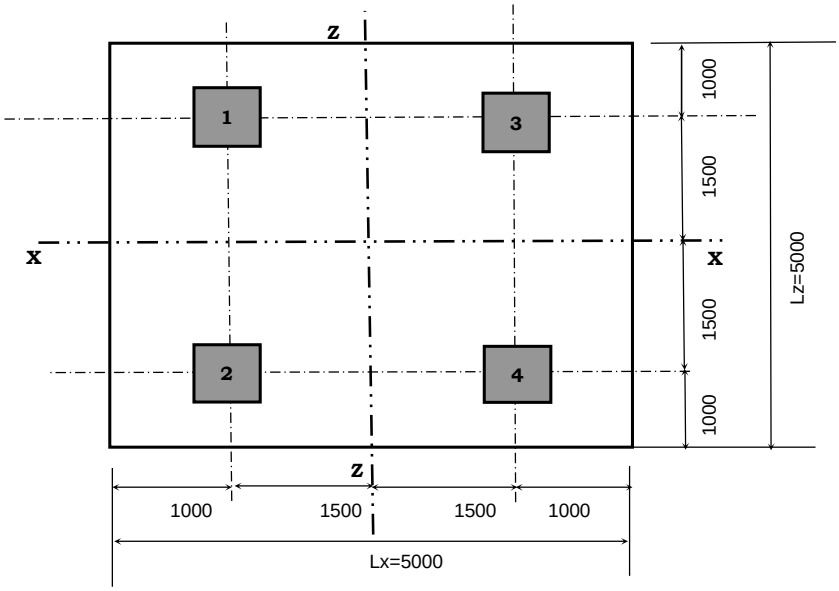
Weight of raft = (5*5*0.5*24) = 300.00KN

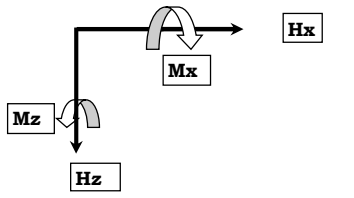
Weight of soil = ((5*5)-4*(0.6*0.6))*1*20 = 471.20KN

Total = 771.20KN

Submerged wt. = 771.2-5*5*1.5*10 = 396.20KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING RCO-1/ F1:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	72.646	0
Min Px	23	201	0	72.646	0
Max Py	1312	202	0	98.693	0
Min Py	1334	220	0	-8.715	0
Max Pz	23	201	0	72.646	0
Min Pz	23	201	0	72.646	0

Gross Maximum Base pressure, p_{max} = 98.7 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 480 KN/m^2
> 98.7 KN/m^2
Hence OK

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DESIGN CALCULATION OF FOOTING RCO-1/ F1:

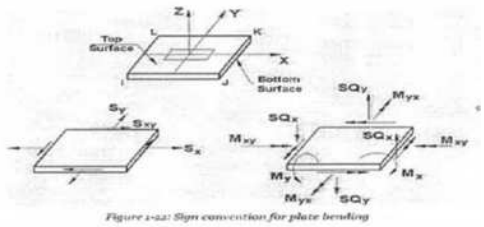
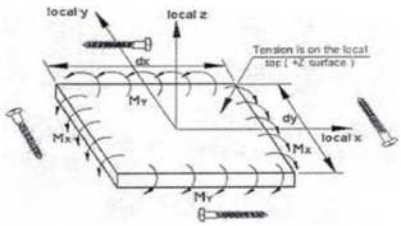


Figure 1-32: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; FDN_12_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
102 1.2X	879	0.219	-0.014	0	0	0	-19.4	-29.299	5.601	-25.003	-34.9	102 1.2XD + 1.2XTL1 + 1.6XL1
102 1.2X	876	-0.186	-0.035	0	0	0	-5.115	-22.624	0	-5.115	-22.624	102 1.2XD + 1.2XTL1 + 1.6XL1
102 1.2X	884	0.027	0.227	0	0	0	-35.65	-21.581	6.143	-41.79	-27.724	102 1.2XD + 1.2XTL1 + 1.6XL1
103 1.2X	852	-0.013	-0.195	0	0	0	-15.91	-12.621	2.773	-18.678	-15.394	103 1.2XD + 1.2XTL2 + 1.6XL
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
102 1.2X	1190	-0.021	0.001	0	0	0	17.77	-0.165	1.585	19.355	-1.75	102 1.2XD + 1.2XTL1 + 1.6XL1
102 1.2X	884	0.027	0.227	0	0	0	-35.65	-21.581	6.143	-41.79	-27.724	102 1.2XD + 1.2XTL1 + 1.6XL1
101 1.2X	1047	0.007	0	0	0	0	-0.866	25.023	1.037	-1.903	26.06	101 1.2XD + 1.2XTL1 + 1.6XL
104 1.2X	879	0.219	-0.014	0	0	0	-19.38	-29.375	5.609	-24.987	-34.984	104 1.2XD + 1.2XTL2 + 1.6XL1
102 1.2X	864	-0.053	-0.076	0	0	0	-2.421	3.561	18.748	-21.169	22.309	102 1.2XD + 1.2XTL1 + 1.6XL1
101 1.2X	110	-0.041	0.058	0	0	0	0.22	5.277	-14.894	15.114	20.171	101 1.2XD + 1.2XTL1 + 1.6XL

PLATE CENTER STRESS ; FDN_12_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
104 1.2X	1208	0.04	0.001	0	0	0	3.138	-0.072	1.495	4.633	-1.567	104 1.2XD + 1.2XTL2 + 1.6XL1
102 1.2X	1183	-0.084	0.018	0	0	0	-11.3	-0.817	0	-11.298	-0.817	102 1.2XD + 1.2XTL1 + 1.6XL1
102 1.2X	1145	0.009	0.075	0	0	0	-0.513	6.186	0.092	-0.605	6.278	102 1.2XD + 1.2XTL1 + 1.6XL1
101 1.2X	1152	0.01	-0.083	0	0	0	-0.601	5.065	0.068	-0.669	5.133	101 1.2XD + 1.2XTL1 + 1.6XL
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.116	-0.124	0.082	-0.198	-0.206	100 1.4XD
102 1.2X	1190	-0.021	0.001	0	0	0	17.77	-0.165	1.585	19.355	-1.75	102 1.2XD + 1.2XTL1 + 1.6XL1
102 1.2X	1185	-0.012	0.031	0	0	0	-25.11	-0.745	2.124	-27.238	-2.869	102 1.2XD + 1.2XTL1 + 1.6XL1
101 1.2X	1047	0.007	0	0	0	0	-0.866	25.023	1.037	-1.903	26.06	101 1.2XD + 1.2XTL1 + 1.6XL
104 1.2X	1154	0.029	-0.025	0	0	0	-0.635	-19.129	1.59	-2.225	-20.719	104 1.2XD + 1.2XTL2 + 1.6XL1
102 1.2X	894	-0.024	-0.018	0	0	0	11.423	1.848	10.728	22.151	12.576	102 1.2XD + 1.2XTL1 + 1.6XL1
101 1.2X	170	-0.015	0.016	0	0	0	10.635	3.264	-7.358	17.993	10.622	101 1.2XD + 1.2XTL1 + 1.6XL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

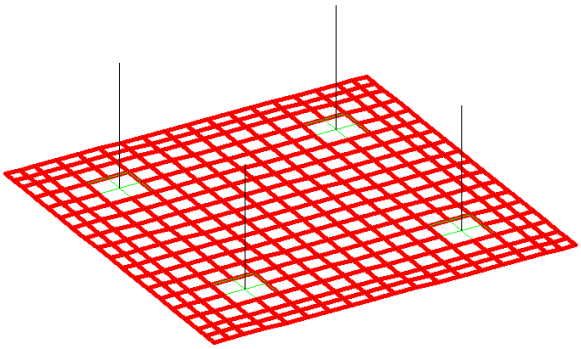
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102 1.2X	19.36	top face	50	442	19.355	0.13	0.0003	139.3	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	-41.79	bottom face	50	442	41.79	0.29	0.0007	301.6		16	252.82	200	1006	14.18	17.73	0.072	tension control
102 1.2X	-21.17	top face	50	442	21.169	0.14	0.0003	152.4		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
101 1.2X	15.11	top face	50	442	15.114	0.10	0.0002	108.7		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	104 1.2X	0.04	1	20	4.633	0.002	0.442	fc'451	fc'435.5	fc'758.3	435.48	1621	> 20	ok	326.6	φVc>Vu	not required
	102 1.2X	0.084	1	42	11.298			fc'446.5			435.48	1621	> 42	ok	326.6	φVc>Vu	not required

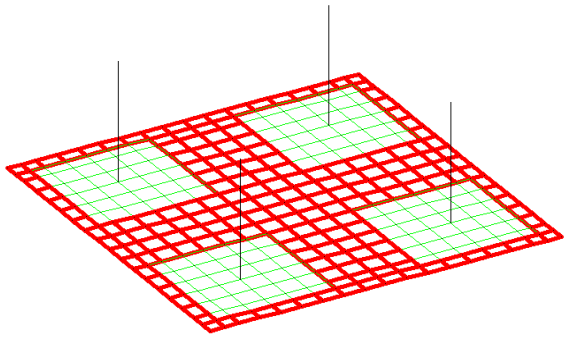
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
101 1.2X	26.06	top face	50	442	26.06	0.18	0.0004	187.7	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
104 1.2X	-34.98	bottom face	50	442	34.984	0.24	0.0006	252.3		16	252.82	200	1006	14.18	17.73	0.072	tension control
102 1.2X	22.31	top face	50	442	22.309	0.15	0.0004	160.6		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
101 1.2X	20.17	top face	50	442	20.171	0.14	0.0003	145.2		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	kN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	102 1.2X	0.075	1	37.5	6.278	0.002	0.442	463.528	435.484	fc'758.3	435.48	456	> 37.5	ok	326.6	φVc>Vu	not required
	101 1.2X	0.083	1	41.5	5.133			479.488			435.48	456	> 41.5	ok	326.6	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING RCO-1/ F1:

Load at bottom of fooring	Node 1					Node 3				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	1.4	-18.7	0.0	-3.8	0.0	5.4	-19.2	0.0	-12.3	0.0
202 D + L1 + TL1	3.2	-8.2	0.0	3.8	0.0	7.2	-8.7	0.0	16.3	0.0
203 D + L + TL2	1.4	-18.7	0.0	-4.1	0.0	5.4	-19.2	0.0	-12.6	0.0
204 D + L1 + TL2	3.2	-8.2	0.0	3.6	0.0	7.2	-8.7	0.0	16.0	0.0
205 D + 0.75XL + 0.75XTL1	1.9	-13.9	0.0	-2.8	0.0	4.9	-14.4	0.0	-9.1	0.0
206 D + 0.75XL1 + 0.75XTL1	3.2	-6.1	0.0	2.9	0.0	6.3	-6.6	0.0	12.3	0.0
207 D + 0.75XL + 0.75XTL2	1.9	-14.0	0.0	-3.0	0.0	4.9	-14.4	0.0	-9.3	0.0
208 D + 0.75XL1 + 0.75XTL2	3.2	-6.1	0.0	2.7	0.0	6.3	-6.6	0.0	12.1	0.0
209 D + WL(+Z)	2.6	-4.0	0.0	-4.0	0.0	2.7	-4.5	0.0	-12.2	0.0
210 D + WL(+X)	3.3	-1.8	0.0	0.2	0.0	3.8	-2.3	0.0	0.3	0.0
211 D + WL(-Z)	4.3	4.6	0.0	4.3	0.0	4.6	4.1	0.0	12.9	0.0
212 D + WL(-X)	3.6	2.4	0.0	0.2	0.0	3.5	1.9	0.0	0.3	0.0
213 D + 0.7XEL(+Z)	3.4	0.2	0.0	0.0	0.0	3.6	-0.3	0.0	0.0	0.0
214 D + 0.7XEL(+X)	3.4	0.1	0.0	0.2	0.0	3.7	-0.4	0.0	0.3	0.0
215 D + 0.7XEL(-Z)	3.4	0.4	0.0	0.3	0.0	3.7	-0.1	0.0	0.6	0.0
216 D + 0.7XEL(-X)	3.4	0.5	0.0	0.2	0.0	3.6	0.1	0.0	0.3	0.0
217 D + 0.75XL + 0.75XWL(+Z)	1.3	-17.2	0.0	-6.0	0.0	4.2	-17.7	0.0	-18.7	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	2.6	-9.3	0.0	-0.3	0.0	5.6	-9.8	0.0	2.8	0.0
219 D + 0.75XL + 0.75XWL(-Z)	2.6	-10.7	0.0	0.2	0.0	5.6	-11.2	0.0	0.2	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	3.9	-2.9	0.0	5.9	0.0	7.0	-3.4	0.0	21.6	0.0
221 D + 0.75XL + 0.75XWL(+X)	1.8	-15.5	0.0	-2.9	0.0	5.1	-16.0	0.0	-9.2	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	3.1	-7.6	0.0	2.8	0.0	6.4	-8.1	0.0	12.2	0.0
223 D + 0.75XL + 0.75XWL(-X)	2.0	-12.4	0.0	-2.9	0.0	4.8	-12.9	0.0	-9.2	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	3.4	-4.5	0.0	2.8	0.0	6.2	-5.0	0.0	12.2	0.0
225 D + 0.75XL + 0.525XEL(+Z)	1.9	-14.0	0.0	-3.0	0.0	4.9	-14.5	0.0	-9.5	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	3.2	-6.1	0.0	2.7	0.0	6.3	-6.6	0.0	12.0	0.0
227 D + 0.75XL + 0.525XEL(+X)	1.9	-14.1	0.0	-2.9	0.0	4.9	-14.6	0.0	-9.3	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	3.2	-6.3	0.0	2.8	0.0	6.3	-6.8	0.0	12.2	0.0
229 D + 0.75XL + 0.525XEL(-Z)	1.9	-13.9	0.0	-2.8	0.0	4.9	-14.4	0.0	-9.0	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	3.3	-6.0	0.0	2.9	0.0	6.3	-6.5	0.0	12.4	0.0
231 D + 0.75XL + 0.525XEL(-X)	1.9	-13.8	0.0	-2.9	0.0	4.9	-14.2	0.0	-9.2	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	3.3	-5.9	0.0	2.8	0.0	6.3	-6.4	0.0	12.2	0.0
233 0.6XD + WL(+Z)	1.2	-4.1	0.0	-4.0	0.0	1.3	-4.4	0.0	-12.4	0.0
234 0.6XD + WL(+X)	1.9	-1.9	0.0	0.1	0.0	2.4	-2.2	0.0	0.2	0.0
235 0.6XD + WL(-Z)	2.9	4.5	0.0	4.3	0.0	3.1	4.2	0.0	12.7	0.0
236 0.6XD + WL(-X)	2.2	2.3	0.0	0.1	0.0	2.0	2.0	0.0	0.2	0.0
237 0.6XD + 0.7XEL(+Z)	2.0	0.1	0.0	0.0	0.0	2.2	-0.2	0.0	-0.1	0.0
238 0.6XD + 0.7XEL(+X)	2.0	-0.1	0.0	0.1	0.0	2.2	-0.4	0.0	0.2	0.0
239 0.6XD + 0.7XEL(-Z)	2.1	0.3	0.0	0.3	0.0	2.2	0.0	0.0	0.5	0.0
240 0.6XD + 0.7XEL(-X)	2.1	0.4	0.0	0.1	0.0	2.2	0.1	0.0	0.2	0.0

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Load at bottom of fooring	Node 2					Node 4				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	2.9	-10.6	0.0	-4.5	0.0	6.7	-11.1	0.0	-13.1	0.0
202 D + L1 + TL1	1.1	-21.1	0.0	3.1	0.0	4.9	-21.6	0.0	15.5	0.0
203 D + L + TL2	2.9	-10.6	0.0	-4.3	0.0	6.7	-11.1	0.0	-12.8	0.0
204 D + L1 + TL2	1.1	-21.1	0.0	3.3	0.0	4.9	-21.5	0.0	15.8	0.0
205 D + 0.75XL + 0.75XTL1	3.0	-7.9	0.0	-3.5	0.0	6.0	-8.4	0.0	-9.8	0.0
206 D + 0.75XL1 + 0.75XTL1	1.7	-15.8	0.0	2.2	0.0	4.6	-16.3	0.0	11.6	0.0
207 D + 0.75XL + 0.75XTL2	3.0	-7.9	0.0	-3.3	0.0	6.0	-8.4	0.0	-9.6	0.0
208 D + 0.75XL1 + 0.75XTL2	1.7	-15.8	0.0	2.4	0.0	4.6	-16.2	0.0	11.8	0.0
209 D + WL(+Z)	4.3	4.5	0.0	-4.4	0.0	4.6	4.0	0.0	-12.7	0.0
210 D + WL(+X)	3.3	-1.9	0.0	-0.3	0.0	3.8	-2.4	0.0	-0.2	0.0
211 D + WL(-Z)	2.6	-4.1	0.0	3.9	0.0	2.7	-4.6	0.0	12.4	0.0
212 D + WL(-X)	3.6	2.3	0.0	-0.3	0.0	3.5	1.8	0.0	-0.2	0.0
213 D + 0.7XEL(+Z)	3.4	0.3	0.0	-0.4	0.0	3.7	-0.2	0.0	-0.5	0.0
214 D + 0.7XEL(+X)	3.4	0.0	0.0	-0.3	0.0	3.7	-0.5	0.0	-0.2	0.0
215 D + 0.7XEL(-Z)	3.4	0.1	0.0	-0.1	0.0	3.6	-0.4	0.0	0.1	0.0
216 D + 0.7XEL(-X)	3.4	0.4	0.0	-0.3	0.0	3.6	-0.1	0.0	-0.2	0.0
217 D + 0.75XL + 0.75XWL(+Z)	3.6	-4.7	0.0	-6.5	0.0	6.7	-5.2	0.0	-19.2	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	2.3	-12.5	0.0	-0.8	0.0	5.3	-13.0	0.0	2.3	0.0
219 D + 0.75XL + 0.75XWL(-Z)	2.4	-11.1	0.0	-0.3	0.0	5.2	-11.6	0.0	-0.3	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	1.0	-19.0	0.0	5.5	0.0	3.9	-19.5	0.0	21.1	0.0
221 D + 0.75XL + 0.75XWL(+X)	2.9	-9.5	0.0	-3.4	0.0	6.1	-10.0	0.0	-9.7	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	1.5	-17.3	0.0	2.3	0.0	4.7	-17.8	0.0	11.7	0.0
223 D + 0.75XL + 0.75XWL(-X)	3.1	-6.4	0.0	-3.4	0.0	5.8	-6.8	0.0	-9.7	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	1.8	-14.2	0.0	2.3	0.0	4.4	-14.7	0.0	11.7	0.0
225 D + 0.75XL + 0.525XEL(+Z)	3.0	-7.9	0.0	-3.5	0.0	6.0	-8.3	0.0	-10.0	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	1.7	-15.7	0.0	2.2	0.0	4.6	-16.2	0.0	11.5	0.0
227 D + 0.75XL + 0.525XEL(+X)	3.0	-8.1	0.0	-3.4	0.0	6.0	-8.6	0.0	-9.8	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	1.7	-15.9	0.0	2.3	0.0	4.6	-16.4	0.0	11.7	0.0
229 D + 0.75XL + 0.525XEL(-Z)	3.0	-8.0	0.0	-3.3	0.0	5.9	-8.5	0.0	-9.5	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	1.7	-15.8	0.0	2.5	0.0	4.5	-16.3	0.0	11.9	0.0
231 D + 0.75XL + 0.525XEL(-X)	3.0	-7.7	0.0	-3.4	0.0	5.9	-8.2	0.0	-9.7	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	1.7	-15.6	0.0	2.3	0.0	4.5	-16.1	0.0	11.7	0.0
233 0.6XD + WL(+Z)	2.9	4.4	0.0	-4.3	0.0	3.1	4.1	0.0	-12.7	0.0
234 0.6XD + WL(+X)	1.9	-2.0	0.0	-0.2	0.0	2.4	-2.3	0.0	-0.1	0.0
235 0.6XD + WL(-Z)	1.2	-4.2	0.0	4.0	0.0	1.3	-4.5	0.0	12.4	0.0
236 0.6XD + WL(-X)	2.2	2.2	0.0	-0.2	0.0	2.0	1.9	0.0	-0.1	0.0
237 0.6XD + 0.7XEL(+Z)	2.1	0.2	0.0	-0.3	0.0	2.2	-0.1	0.0	-0.4	0.0
238 0.6XD + 0.7XEL(+X)	2.0	-0.1	0.0	-0.2	0.0	2.2	-0.4	0.0	-0.1	0.0
239 0.6XD + 0.7XEL(-Z)	2.0	0.0	0.0	0.0	0.0	2.2	-0.3	0.0	0.2	0.0
240 0.6XD + 0.7XEL(-X)	2.1	0.3	0.0	-0.2	0.0	2.2	0.0	0.0	-0.1	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + L + TL1	740.4	-59.58	11.7	-33.7	-4.2	158.3	439.3	5	8.8
202 D + L1 + TL1	740.4	-59.58	11.7	38.7	6.7	158.3	278.3	5	7.7
203 D + L + TL2	740.4	-59.58	11.7	-33.7	-4.2	158.2	439.3	5.0	8.8
204 D + L1 + TL2	740.4	-59.58	11.7	38.7	6.7	158.3	278.3	5.0	8
205 D + 0.75XL + 0.75XTL1	739.9	-44.68	8.9	-25.3	-3.2	206.8	586.1	6.6	11.7
206 D + 0.75XL1 + 0.75XTL1	739.9	-44.68	8.9	29.0	5.0	206.7	370.4	7	10.2
207 D + 0.75XL + 0.75XTL2	739.9	-44.68	8.9	-25.3	-3.2	206.7	586.1	6.6	11.7
208 D + 0.75XL1 + 0.75XTL2	739.9	-44.68	8.9	29.0	5.0	206.7	370.5	6.6	10
209 D + WL(+Z)	738.2	0.00	0.7	-33.4	-5.3	2629.1	346.6	NA	8.9
218 D + 0.75XL1 + 0.75XWL(+Z)	739.4	-23.40	5.3	2.0	1.7	345.9	1063.2	12.6	146.3
219 D + 0.75XL + 0.75XWL(-Z)	737.6	-30.65	4.8	16.6	4.3	381.3	425.1	9.6	17.7
220 D + 0.75XL1 + 0.75XWL(-Z)	742.1	-2.19	4.6	27.1	5.7	405.6	323.5	135.5	11.0
221 D + 0.75XL + 0.75XWL(+X)	738.0	-31.46	5.2	-13.1	-0.4	352.1	4522.3	9.4	22.6
222 D + 0.75XL1 + 0.75XWL(+X)	740.7	-16.34	5.4	14.5	3.7	343.3	496.2	18.1	20.4
223 D + 0.75XL + 0.75XWL(-X)	738.0	-25.54	4.5	-12.1	-0.3	409.8	6871.2	11.6	24.3
224 D + 0.75XL1 + 0.75XWL(-X)	740.7	-9.25	4.5	14.5	3.7	409.3	495.0	32.0	20.4
225 D + 0.75XL + 0.525XEL(+Z)	741.2	-38.35	9.1	-38.1	-5.2	204.7	355.7	7.7	7.8
226 D + 0.75XL1 + 0.525XEL(+Z)	741.2	-38.35	9.1	16.2	2.9	204.7	630.6	7.7	18.3
227 D + 0.75XL + 0.525XEL(+X)	738.5	-51.50	8.9	-12.7	-1.2	207.7	1602.8	5.7	23.2
228 D + 0.75XL1 + 0.525XEL(+X)	738.5	-51.50	8.9	41.6	7.0	207.7	263.9	5.7	7.1
229 D + 0.75XL + 0.525XEL(-Z)	739.9	-47.69	9.3	-24.9	-3.1	198.6	596.3	6.2	11.9
230 D + 0.75XL1 + 0.525XEL(-Z)	739.9	-47.69	9.3	29.4	5.0	198.5	366.5	6.2	10.1
231 D + 0.75XL + 0.525XEL(-X)	739.9	-41.20	8.5	-25.3	-3.2	216.9	585.3	7.2	11.7
224 D + 0.75XL1 + 0.75XWL(-X)	740.7	-9.25	4.5	14.5	3.7	409.3	495.0	32.0	20.4
225 D + 0.75XL + 0.525XEL(+Z)	741.2	-38.35	9.1	-38.1	-5.2	204.7	355.7	7.7	7.8
226 D + 0.75XL1 + 0.525XEL(+Z)	739.9	-44.68	8.9	28.3	4.9	206.7	377.9	6.6	10.4
227 D + 0.75XL + 0.525XEL(+X)	739.9	-45.38	9.0	-25.3	-3.2	204.5	586.1	7	11.7
228 D + 0.75XL1 + 0.525XEL(+X)	739.9	-45.38	9.0	29.0	5.0	204.5	370.5	6.5	10.2
229 D + 0.75XL + 0.525XEL(-Z)	739.9	-44.68	8.9	-24.6	-3.1	206.8	605.1	6.6	12.0
230 D + 0.75XL1 + 0.525XEL(-Z)	739.9	-44.68	8.9	29.7	5.1	206.7	363.2	6.6	10
231 D + 0.75XL + 0.525XEL(-X)	739.9	-43.99	8.8	-25.3	-3.2	209.0	585.8	6.7	11.7
232 D + 0.75XL1 + 0.525XEL(-X)	739.9	-43.99	8.8	29.0	5.0	209.0	370.4	6.7	10.2
233 0.6XD + WL(+Z)	732.6	0.00	0.4	-33.4	-5.3	4376.2	343.4	NA	9
234 0.6XD + WL(+X)	732.6	-8.30	1.4	0.0	0.0	1305.8	185640	35.3	10852.9
235 0.6XD + WL(-Z)	732.6	0.00	0.4	33.4	5.4	4360.6	342.0	NA	8.8
236 0.6XD + WL(-X)	732.6	8.30	-0.6	0.0	0.0	3247.3	162438	35.3	12740.6
237 0.6XD + 0.7XEL(+Z)	732.6	0.00	0.4	-0.9	-0.1	4360.6	15262.0	NA	325.2
238 0.6XD + 0.7XEL(+X)	732.6	-0.93	0.6	0.0	0.0	3326.9	185642	315.8	11721.2
239 0.6XD + 0.7XEL(-Z)	732.6	0.00	0.4	1.0	0.1	4360.6	12988.9	NA	307.5
240 0.6XD + 0.7XEL(-X)	732.6	0.93	0.3	0.0	0.0	6359.2	162437	316.1	11721.2
					Min.=	158.2	263.9	5.0	7.1

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DESIGN CALCULATION OF PEDESTAL FOR RCO-1/ F1:

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 600 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.8

Y_b = (b-d')/b = 0.8

Unsupported length of column parallel to X-X axis, l_{ux} = 2 m kl_{ux}/r_x = 13.33 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2 m kl_{uz}/r_z = 13.33

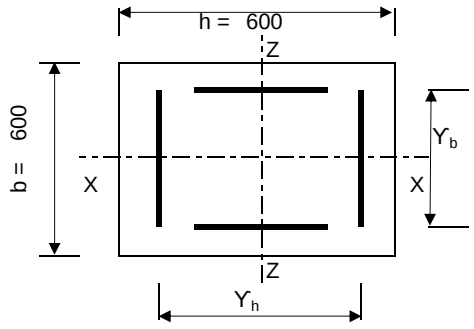
Radius of gyration r_x = 180 mm

Radius of gyration r_z = 180 mm

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 9746 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha = \log 0.5/\log\beta$	$(M_{nx}/M_{nx1})^{\alpha} + (M_{nz}/M_{nz1})^{\alpha}$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	350.1	47.9	25.7	539	73.7	39.5	0.043	0.090	680.4	0.11	0.090	680.4	0.06	1.609	0.04	safe
102	124.8	0.0	0.0	192	0.0	0.0	0.015	0.090	680.4	0.00	0.090	680.4	0.00		0.00	safe
102	151.0	46.2	63.6	232	71.1	97.9	0.018	0.090	680.4	0.10	0.090	680.4	0.14		0.07	safe
111	84.5	12.6	13.2	130	19.4	20.4	0.010	0.070	529.2	0.04	0.070	529.2	0.04		0.01	safe
102	350.1	47.9	29.2	539	73.7	45.0	0.043	0.065	491.4	0.15	0.065	491.4	0.09		0.07	safe
101	298.9	38.4	32.7	460	59.1	50.2	0.037	0.070	529.2	0.11	0.070	529.2	0.09		0.05	safe
100	53.9	0.4	0.8	83	0.7	1.2	0.007	0.060	453.6	0.00	0.060	453.6	0.00		0.00	safe
100	53.9	0.4	0.8	83	0.7	1.2	0.007	0.060	453.6	0.00	0.060	453.6	0.00		0.00	safe
101	298.9	38.4	32.7	460	59.1	50.2	0.037	0.070	529.2	0.11	0.070	529.2	0.09		0.05	safe
102	350.1	47.9	56.3	539	73.7	86.7	0.043	0.065	491.4	0.15	0.065	491.4	0.18		0.11	safe
102	151.0	46.2	63.6	232	71.1	97.9	0.018	0.090	680.4	0.10	0.090	680.4	0.14		0.07	safe
111	84.5	12.6	13.2	130	19.4	20.4	0.010	0.070	529.2	0.04	0.070	529.2	0.04		0.01	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3600	4	20	+	8	20	3772	1.05	10	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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- **DESIGN CALCULATION OF FOOTING MKD. RCO-1/ F2:**

Size of Columns at node	Along-X	Along-Z
Col.1	0.60m	0.85m
Col.2	0.60m	0.85m
Size of fdn. raft:	5.00m	2.50m
Thickness of raft:	0.50 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.35 m	
U/S level of fdn:	-1.50 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	20 kN/m^3	

Calculation of Weight of Foundation:

Weight of raft = (5*2.5*0.5*25) = 156.25KN

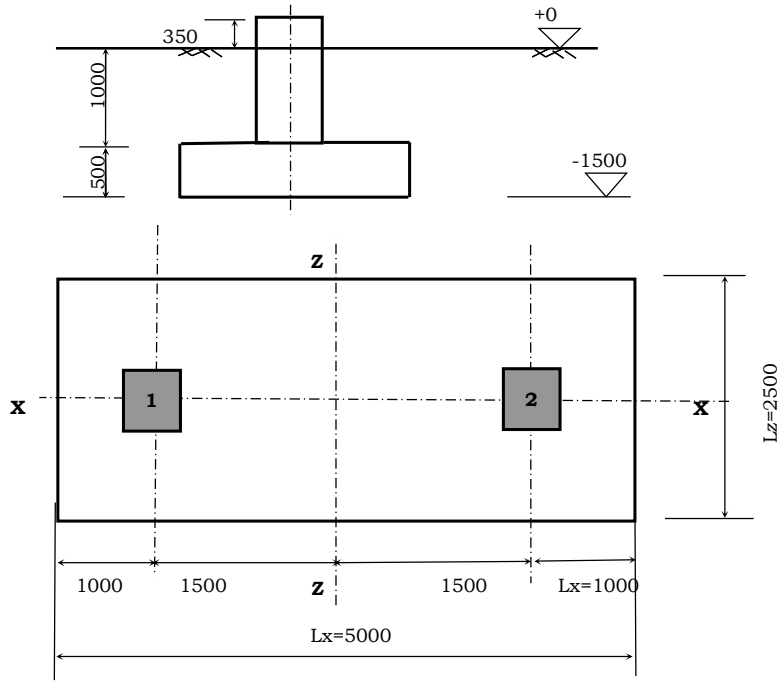
Weight of soil = ((5*2.5)-2*(0.6*0.85))*1*20 = 229.60KN

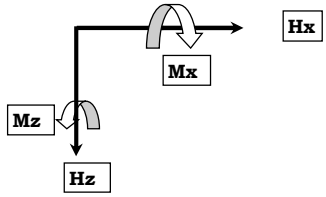
Total = 385.85KN

Submerged wt. = 385.85-5*2.5*1.5*10 = 198.35KN

Design Input:

Characteristic strength of concrete, f'_c =	35 MPa
Yield stress of steel reinforcement, f_y =	420 MPa
Clear cover for main reinforcement bars =	50mm





Hx=Force along X-X direction

Hz=Force along Z-Z direction

Mx=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING MKD. RCO-1/ F2:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	75.48	0
Min Px	23	201	0	75.48	0
Max Py	512	204	0	116.45	0
Min Py	498	220	0	-5.34	0
Max Pz	23	201	0	75.48	0
Min Pz	23	201	0	75.48	0

Gross Maximum Base pressure, p_{max} = 116.4 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

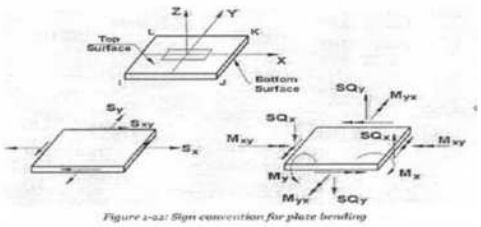
Gross Safe Bearing Capacity = 480 KN/m^2
> 116.4 KN/m^2
Hence OK

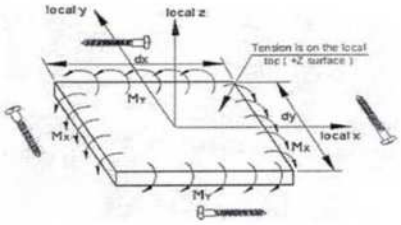
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DESIGN CALCULATION OF FOOTING MKD. RCO-1/ F2:





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F5_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	60	0.486	0.083	0	0	0	-96.63	-56.553	-3.964	-100.591	-60.517	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qx	112	-0.447	0.093	0	0	0	-91.18	-48.72	0	-91.176	-48.72	101 1.2XD + 1.2XTL1 + 1.6XL
Max Qy	58	0.247	0.329	0	0	0	-82.9	-35.354	12.063	-94.959	-47.417	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qy	61	0.42	-0.376	0	0	0	-89.86	-53.844	-1.622	-91.48	-55.466	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Sx	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Min Sx	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Max Sy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Min Sy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Max Sxy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Min Sxy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Max Mx	111	0.138	0.109	0	0	0	59.552	-0.61	-3.879	63.431	-4.489	104 1.2XD + 1.2XTL2 + 1.6XL1
Min Mx	60	0.486	0.083	0	0	0	-96.63	-56.553	-3.964	-100.591	-60.517	102 1.2XD + 1.2XTL1 + 1.6XL1
Max My	112	0.163	0.005	0	0	0	39.462	19.231	-2.492	41.954	21.723	111 0.9XD + 1.6XWL(-Z)
Min My	59	0.447	0.095	0	0	0	-94.96	-69.215	7.689	-102.649	-76.904	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Mxy	106	-0.059	-0.114	0	0	0	6.847	0.023	24.054	30.901	24.077	101 1.2XD + 1.2XTL1 + 1.6XL
Min Mxy	62	0.024	-0.092	0	0	0	3.33	-6.076	-26.163	29.493	-32.239	102 1.2XD + 1.2XTL1 + 1.6XL1

PLATE CENTER STRESS : FDN_F5_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	148	0.209	-0.025	0	0	0	-36.35	-34.355	-1.455	-37.802	-35.81	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qx	172	-0.191	-0.022	0	0	0	-34.52	-31.561	0	-34.523	-31.561	101 1.2XD + 1.2XTL1 + 1.6XL
Max Qy	150	0.105	0.061	0	0	0	-42.11	-13.154	12.294	-54.408	-25.448	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qy	152	0.14	-0.08	0	0	0	-34.97	-16.972	-11.609	-46.582	-28.581	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Sx	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Min Sx	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Max Sy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Min Sy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Max Sxy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Min Sxy	400	-0.001	-0.001	0	0	0	-0.013	-0.072	0.006	-0.019	-0.078	100 1.4XD
Max Mx	173	0.036	0.012	0	0	0	42.911	-2.13	-8.466	51.377	-10.596	104 1.2XD + 1.2XTL2 + 1.6XL1
Min Mx	293	-0.03	-0.061	0	0	0	-58.4	-0.878	0.631	-59.027	-1.509	102 1.2XD + 1.2XTL1 + 1.6XL1
Max My	171	0.087	-0.007	0	0	0	18.126	7.91	1.251	19.377	9.161	111 0.9XD + 1.6XWL(-Z)
Min My	148	0.209	-0.025	0	0	0	-36.35	-34.355	-1.455	-37.802	-35.81	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Mxy	150	0.105	0.061	0	0	0	-42.11	-13.154	12.294	-54.408	-25.448	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Mxy	152	0.14	-0.08	0	0	0	-34.97	-16.972	-11.609	-46.582	-28.581	102 1.2XD + 1.2XTL1 + 1.6XL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

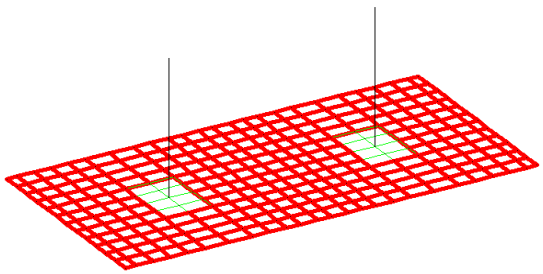
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104 1.2X	63.43	top face	50	442	63.431	0.43	0.0010	458.9	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	-100.59	bottom face	50	442	100.59	0.69	0.0017	731.0		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
101 1.2X	30.90	top face	50	442	30.901	0.21	0.0005	222.7		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	29.49	top face	50	442	29.493	0.20	0.0005	212.6		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.209	1	104.5	37.802	0.002	0.442	439.277	435.484	758.323	435.48	1621	> 104.5	ok	326.6	φVc>Vu	not required
	Min Qx	0.191	1	95.5	34.523			439.291			435.48	1621	> 95.5	ok	326.6	φVc>Vu	not required

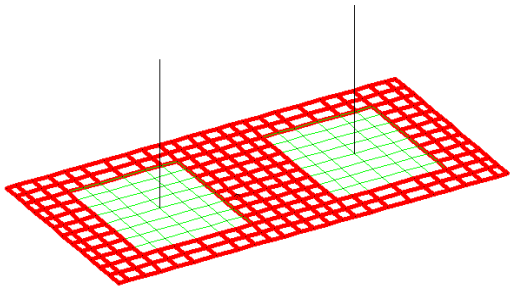
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
111 0.9X	21.72	top face	50	442	21.723	0.15	0.0004	156.4	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	-76.90	bottom face	50	442	76.904	0.52	0.0013	557.3		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
101 1.2X	24.08	top face	50	442	24.077	0.16	0.0004	173.4		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	-32.24	top face	50	442	32.239	0.22	0.0005	232.4		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	102 1.2X	0.061	1	30.5	25.448	0.002	0.442	427.443	435.484	fc'758.3	427.44	450	> 30.5	ok	320.6	φVc>Vu	not required
	102 1.2X	0.08	1	40	28.581			428.962			428.96	451	> 40	ok	321.7	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING MKD. RCO-1/ F2:

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	4.2	-26.8	0.0	10.2	0.0	6.6	-26.4	0.0	-63.1	0.0
202 D + L1 + TL1	7.4	-24.0	0.0	63.7	0.0	3.4	-29.3	0.0	-9.8	0.0
203 D + L + TL2	4.3	-26.8	0.0	-36.9	0.0	6.7	-26.4	0.0	-16.0	0.0
204 D + L1 + TL2	7.5	-24.0	0.0	16.6	0.0	3.5	-29.3	0.0	37.3	0.0
205 D + 0.75XL + 0.75XTL1	4.1	-20.1	0.0	8.6	0.0	6.0	-19.9	0.0	-48.3	0.0
206 D + 0.75XL1 + 0.75XTL1	6.5	-17.9	0.0	48.7	0.0	3.6	-22.0	0.0	-8.2	0.0
207 D + 0.75XL + 0.75XTL2	4.2	-20.1	0.0	-26.7	0.0	6.0	-19.9	0.0	-12.9	0.0
208 D + 0.75XL1 + 0.75XTL2	6.6	-17.9	0.0	13.4	0.0	3.7	-22.0	0.0	27.1	0.0
209 D + WL(+Z)	2.3	-0.2	0.0	-20.0	0.0	5.7	0.2	0.0	-27.3	0.0
210 D + WL(+X)	4.0	-1.3	0.0	3.7	0.0	4.0	-1.9	0.0	-3.6	0.0
211 D + WL(-Z)	5.7	0.8	0.0	27.3	0.0	2.3	-0.8	0.0	20.0	0.0
212 D + WL(-X)	4.0	1.9	0.0	3.7	0.0	4.0	1.3	0.0	-3.6	0.0
213 D + 0.7XEL(+Z)	4.0	0.3	0.0	3.4	0.0	4.0	-0.3	0.0	-3.9	0.0
214 D + 0.7XEL(+X)	4.0	0.0	0.0	3.7	0.0	4.0	-0.5	0.0	-3.6	0.0
215 D + 0.7XEL(-Z)	4.0	0.3	0.0	4.0	0.0	4.0	-0.3	0.0	-3.4	0.0
216 D + 0.7XEL(-X)	4.0	0.5	0.0	3.7	0.0	4.0	0.0	0.0	-3.6	0.0
217 D + 0.75XL + 0.75XWL(+Z)	2.9	-20.4	0.0	-26.8	0.0	7.3	-19.5	0.0	-48.3	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	5.3	-18.3	0.0	13.3	0.0	4.9	-21.6	0.0	-8.3	0.0
219 D + 0.75XL + 0.75XWL(-Z)	5.5	-19.7	0.0	8.7	0.0	4.7	-20.3	0.0	-12.9	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	7.9	-17.5	0.0	48.8	0.0	2.3	-22.4	0.0	27.1	0.0
221 D + 0.75XL + 0.75XWL(+X)	4.2	-21.2	0.0	-9.1	0.0	6.0	-21.1	0.0	-30.6	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	6.6	-19.1	0.0	31.0	0.0	3.6	-23.2	0.0	9.4	0.0
223 D + 0.75XL + 0.75XWL(-X)	4.2	-18.9	0.0	-9.1	0.0	6.0	-18.7	0.0	-30.6	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	6.6	-16.7	0.0	31.0	0.0	3.6	-20.8	0.0	9.4	0.0
225 D + 0.75XL + 0.525XEL(+Z)	4.2	-20.1	0.0	-9.3	0.0	6.0	-19.9	0.0	-30.8	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	6.6	-17.9	0.0	30.8	0.0	3.6	-22.0	0.0	9.2	0.0
227 D + 0.75XL + 0.525XEL(+X)	4.2	-20.3	0.0	-9.1	0.0	6.0	-20.1	0.0	-30.6	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	6.6	-18.1	0.0	31.0	0.0	3.6	-22.2	0.0	9.4	0.0
229 D + 0.75XL + 0.525XEL(-Z)	4.2	-20.1	0.0	-8.9	0.0	6.0	-19.9	0.0	-30.4	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	6.6	-17.9	0.0	31.2	0.0	3.6	-22.0	0.0	9.6	0.0
231 D + 0.75XL + 0.525XEL(-X)	4.2	-19.9	0.0	-9.1	0.0	6.0	-19.7	0.0	-30.6	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	6.6	-17.7	0.0	31.0	0.0	3.6	-21.8	0.0	9.4	0.0
233 0.6XD + WL(+Z)	0.7	-0.4	0.0	-21.4	0.0	4.1	0.4	0.0	-25.8	0.0
234 0.6XD + WL(+X)	2.4	-1.4	0.0	2.2	0.0	2.4	-1.8	0.0	-2.2	0.0
235 0.6XD + WL(-Z)	4.1	0.7	0.0	25.9	0.0	0.7	-0.7	0.0	21.4	0.0
236 0.6XD + WL(-X)	2.4	1.8	0.0	2.2	0.0	2.4	1.4	0.0	-2.2	0.0
237 0.6XD + 0.7XEL(+Z)	2.4	0.2	0.0	1.9	0.0	2.4	-0.2	0.0	-2.5	0.0
238 0.6XD + 0.7XEL(+X)	2.4	-0.1	0.0	2.2	0.0	2.4	-0.4	0.0	-2.2	0.0
239 0.6XD + 0.7XEL(-Z)	2.4	0.2	0.0	2.5	0.0	2.4	-0.2	0.0	-1.9	0.0
240 0.6XD + 0.7XEL(-X)	2.4	0.4	0.0	2.2	0.0	2.4	0.1	0.0	-2.2	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + L + TL1	396.7	-53.25	3.6	-52.9	0.0	274.9	NA	3.0	3.0
202 D + L1 + TL1	396.7	-53.25	-5.9	53.9	0.0	166.9	NA	3.0	2.9
203 D + L + TL2	396.9	-53.25	3.6	-52.9	0.0	275.0	NA	3.0	3.0
204 D + L1 + TL2	396.9	-53.25	-5.9	53.9	0.0	167.0	NA	3.0	2.9
205 D + 0.75XL + 0.75XTL1	396.0	-39.94	2.7	-39.7	0.0	365.6	NA	4.0	4.0
206 D + 0.75XL1 + 0.75XTL1	396.0	-39.94	-4.5	40.4	0.0	222.3	NA	4.0	3.9
207 D + 0.75XL + 0.75XTL2	396.1	-39.94	2.7	-39.7	0.0	366.0	NA	4.0	4.0
208 D + 0.75XL1 + 0.75XTL2	396.1	-39.94	-4.5	40.4	0.0	222.3	NA	4.0	3.9
209 D + WL(+Z)	393.8	0.00	5.1	-47.2	0.0	193.1	NA	NA	3.3
210 D + WL(+X)	393.8	-3.20	0.0	0.0	0.0	164100.0	NA	49.2	3750.9
211 D + WL(-Z)	393.8	0.00	-5.1	47.3	0.0	193.6	NA	NA	3.3
212 D + WL(-X)	393.8	3.20	0.0	0.0	0.0	131279.7	NA	49.2	4145.7
213 D + 0.7XEL(+Z)	393.8	0.00	0.0	-0.5	0.0	20512.5	NA	NA	321.5
214 D + 0.7XEL(+X)	393.8	-0.53	0.0	0.0	0.0	131279.7	NA	296.7	3938.4
215 D + 0.7XEL(-Z)	393.8	0.00	0.0	0.6	0.0	28539.1	NA	NA	276.4
216 D + 0.7XEL(-X)	393.8	0.53	0.0	0.0	0.0	131279.7	NA	296.7	3938.4
217 D + 0.75XL + 0.75XWL(+Z)	396.0	-39.94	6.5	-75.1	0.0	151.7	NA	4.0	2.1
218 D + 0.75XL1 + 0.75XWL(+Z)	396.0	-39.94	-0.6	5.0	0.0	1556.8	NA	4.0	31.6
219 D + 0.75XL + 0.75XWL(-Z)	396.0	-39.94	-1.1	-4.2	0.0	890.8	NA	4.0	37.3
220 D + 0.75XL1 + 0.75XWL(-Z)	396.0	-39.94	-8.3	75.9	0.0	119.7	NA	4.0	2.1
221 D + 0.75XL + 0.75XWL(+X)	396.0	-42.34	2.7	-39.7	0.0	365.7	NA	3.7	4.0
222 D + 0.75XL1 + 0.75XWL(+X)	396.0	-42.34	-4.5	40.4	0.0	222.2	NA	3.7	3.9
223 D + 0.75XL + 0.75XWL(-X)	396.0	-37.54	2.7	-39.7	0.0	365.7	NA	4.2	4.0
224 D + 0.75XL1 + 0.75XWL(-X)	396.0	-37.54	-4.5	40.4	0.0	222.2	NA	4.2	3.9
225 D + 0.75XL + 0.525XEL(+Z)	396.0	-39.94	2.7	-40.1	0.0	361.7	NA	4.0	4.0
226 D + 0.75XL1 + 0.525XEL(+Z)	396.0	-39.94	-4.4	40.1	0.0	223.8	NA	4.0	4.0
227 D + 0.75XL + 0.525XEL(+X)	396.0	-40.34	2.7	-39.7	0.0	365.7	NA	3.9	4.0
228 D + 0.75XL1 + 0.525XEL(+X)	396.0	-40.34	-4.5	40.4	0.0	222.2	NA	3.9	3.9
229 D + 0.75XL + 0.525XEL(-Z)	396.0	-39.94	2.7	-39.3	0.0	370.0	NA	4.0	4.0
230 D + 0.75XL1 + 0.525XEL(-Z)	396.0	-39.94	-4.5	40.8	0.0	220.8	NA	4.0	3.9
231 D + 0.75XL + 0.525XEL(-X)	396.0	-39.54	2.7	-39.7	0.0	365.7	NA	4.0	4.0
232 D + 0.75XL1 + 0.525XEL(-X)	396.0	-39.54	-4.5	40.4	0.0	222.2	NA	4.0	3.9
233 0.6XD + WL(+Z)	390.6	0.00	5.1	-47.2	0.0	191.7	NA	NA	3.3
234 0.6XD + WL(+X)	390.6	-3.20	0.0	0.0	0.0	325536.7	NA	48.8	6009.9
235 0.6XD + WL(-Z)	390.6	0.00	-5.1	47.3	0.0	191.9	NA	NA	3.3
236 0.6XD + WL(-X)	390.6	3.20	0.0	0.0	0.0	217023.9	NA	48.8	7102.6
237 0.6XD + 0.7XEL(+Z)	390.6	0.00	0.0	-0.5	0.0	21702.4	NA	NA	308.2
238 0.6XD + 0.7XEL(+X)	390.6	-0.53	0.0	0.0	0.0	217023.9	NA	294.3	6510.7
239 0.6XD + 0.7XEL(-Z)	390.6	0.00	0.0	0.6	0.0	26042.9	NA	NA	281.5
240 0.6XD + 0.7XEL(-X)	390.6	0.53	0.0	0.0	0.0	217023.9	NA	293.7	6510.7
					Min.=	119.7	0.0	3.0	2.1

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DESIGN CALCULATION OF PEDESTAL OF RCO-1/ F2:

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 850 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.77

Y_b = (b-d')/b = 0.84

Unsupported length of column parallel to X-X axis, l_{ux} = 1.85 m kl_{ux}/r_x = 7.255 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.85 m kl_{uz}/r_z = 10.28

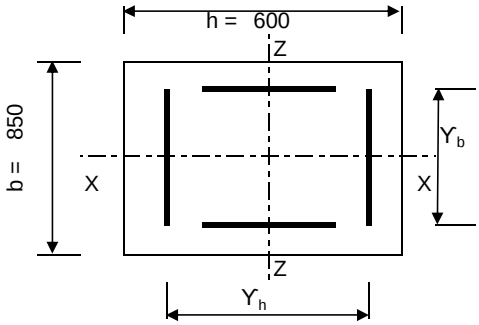
Radius of gyration r_x = 255 mm

Radius of gyration r_z = 180 mm

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 13708.05 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	488.6	213.9	90.4	752	329.0	139.0	0.042	0.075	1137.94	0.29	0.075	803.3	0.17	1.609	0.196	safe
215	185.2	360.3	32.0	285	554.3	49.3	0.016	0.060	910.35	0.61	0.060	642.6	0.08		0.467	safe
102	-15.1	11.2	109.9	-23	17.2	169.0	-0.001	0.075	1137.94	0.02	0.075	803.3	0.21		0.083	safe
112	61.3	7.8	6.6	94	12.0	10.1	0.005	0.070	1062.08	0.01	0.070	749.7	0.01		0.002	safe
102	488.6	213.9	90.4	752	329.0	139.0	0.042	0.070	1062.08	0.31	0.070	749.7	0.19		0.219	safe
101	411.4	211.9	99.1	633	326.0	152.4	0.035	0.060	910.35	0.36	0.060	642.6	0.24		0.291	safe
100	95.6	12.0	0.9	147	18.5	1.4	0.008	0.060	910.35	0.02	0.060	642.6	0.00		0.002	safe
100	95.6	12.0	0.9	147	18.5	1.4	0.008	0.060	910.35	0.02	0.060	642.6	0.00		0.002	safe
101	411.4	211.9	99.1	633	326.0	152.4	0.035	0.060	910.35	0.36	0.060	642.6	0.24		0.291	safe
102	488.6	213.9	90.4	752	329.0	139.0	0.042	0.060	910.35	0.36	0.060	642.6	0.22		0.28	safe
102	-15.1	11.2	109.9	-23	17.2	169.0	-0.001	0.075	1137.94	0.02	0.075	803.3	0.21		0.083	safe
112	61.3	7.8	6.6	94	12.0	10.1	0.005	0.070	1062.08	0.01	0.070	749.7	0.01		0.002	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5100	4	20	+	12	20	5029	0.99	10	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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- **DESIGN CALCULATION OF FOOTING RCO-1/ F3:**

Size of Columns at node	Along-X	Along-Z		
1	0.60m	0.60m		
2	0.60m	0.60m		
3	0.60m	0.60m		
4	0.60m	0.60m		
Size of fdn. raft:	4.50m	5.00m		
Thickness of raft:	0.50	m	RL (+)	913.60
Original Gr. Level:	0.00	m	RL (+)	913.95
Finished Gr. Level:	0.00	m		
U/S of B/Pl Above GL:	0.35	m		
U/S level of fdn:	-1.50	m		
Unit weight of conc:	24	kN/m^3		
Unit weight of soil:	20	kN/m^3		

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

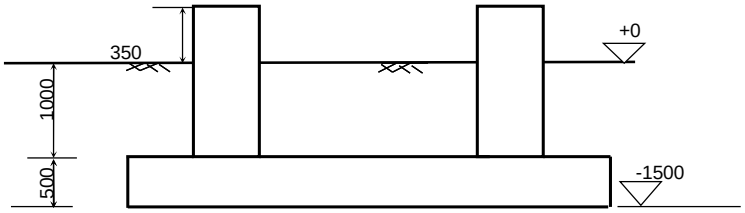
Calculation of Weight of Foundation :

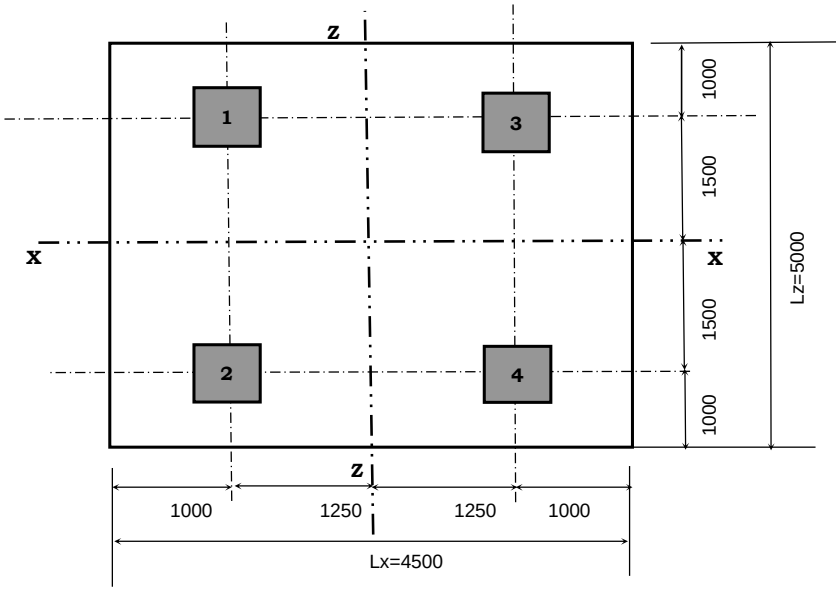
Weight of raft = (4.5*5*0.5*24) = 270.00KN

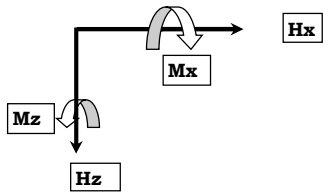
Weight of soil = ((4.5*5)-4*(0.6*0.6))*1*20 = 421.20KN

Total = 691.20KN

Submerged wt. = 691.2-4.5*5*1.5*10 = 353.70KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING RCO-1/ F3:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	42.949	0
Min Px	23	201	0	42.949	0
Max Py	1334	217	0	70.22	0
Min Py	1313	233	0	9.513	0
Max Pz	23	201	0	42.949	0
Min Pz	23	201	0	42.949	0

Gross Maximum Base pressure, p_{max} = 70.2 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

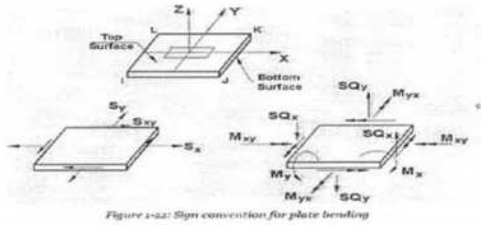
Gross Safe Bearing Capacity = 480 KN/m^2
> 70.2 KN/m^2
Hence OK

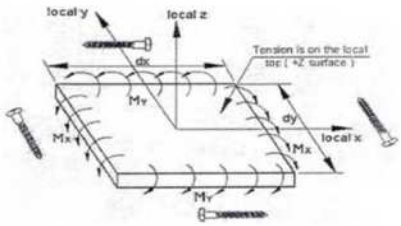
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DESIGN CALCULATION OF FOOTING RCO-1/ F3:





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; FDN_12_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
101 1.2X	60	0.116	0.02	0	0	0	-8.506	-12.532	4.734	-13.24	-17.266	101 1.2XD + 1.2XTL1 + 1.6XL
101 1.2X	64	-0.1	0.021	0	0	0	-8.774	-14.496	0	-8.774	-14.496	101 1.2XD + 1.2XTL1 + 1.6XL
102 1.2X	847	-0.026	0.116	0	0	0	-21.13	-11.972	-5.514	-26.641	-17.486	102 1.2XD + 1.2XTL1 + 1.6XL1
101 1.2X	54	-0.03	-0.122	0	0	0	-21.55	-13.067	5.446	-26.993	-18.513	101 1.2XD + 1.2XTL1 + 1.6XL
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
109 0.9X	848	-0.019	-0.056	0	0	0	8.851	7.381	4.043	12.894	11.424	109 0.9XD + 1.6XWL(+Z)
101 1.2X	54	-0.03	-0.122	0	0	0	-21.55	-13.067	5.446	-26.993	-18.513	101 1.2XD + 1.2XTL1 + 1.6XL
101 1.2X	244	-0.002	0.003	0	0	0	-2.377	10.061	4.838	-7.215	14.899	101 1.2XD + 1.2XTL1 + 1.6XL
104 1.2X	840	-0.099	-0.012	0	0	0	-8.756	-15.906	-5.116	-13.872	-21.022	104 1.2XD + 1.2XTL2 + 1.6XL1
101 1.2X	58	0.027	0.047	0	0	0	-4.665	2.037	12.663	-17.328	14.7	101 1.2XD + 1.2XTL1 + 1.6XL
102 1.2X	824	0.036	-0.052	0	0	0	-5.264	-0.103	-12.421	-17.685	-12.524	102 1.2XD + 1.2XTL1 + 1.6XL1

PLATE CENTER STRESS ; FDN_12_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
101 1.2X	1201	0.073	-0.006	0	0	0	-2.349	-0.537	1.557	-3.906	-2.094	101 1.2XD + 1.2XTL1 + 1.6XL
109 0.9X	1181	-0.053	-0.003	0	0	0	2.188	0.265	0	2.188	0.265	109 0.9XD + 1.6XWL(+Z)
101 1.2X	1159	-0.006	0.05	0	0	0	-0.368	-0.968	0.371	-0.739	-1.339	101 1.2XD + 1.2XTL1 + 1.6XL
102 1.2X	1169	-0.007	-0.05	0	0	0	-0.371	-2.844	-0.399	-0.77	-3.243	102 1.2XD + 1.2XTL1 + 1.6XL1
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
100 1.4X	1216	-0.001	-0.001	0	0	0	-0.19	-0.18	0.113	-0.303	-0.293	100 1.4XD
101 1.2X	173	0.009	-0.002	0	0	0	8.13	-0.868	5.657	13.787	-6.525	101 1.2XD + 1.2XTL1 + 1.6XL
101 1.2X	1197	0.016	-0.018	0	0	0	-15.93	-0.523	2.233	-18.167	-2.756	101 1.2XD + 1.2XTL1 + 1.6XL
101 1.2X	244	-0.002	0.003	0	0	0	-2.377	10.061	4.838	-7.215	14.899	101 1.2XD + 1.2XTL1 + 1.6XL
104 1.2X	1172	-0.014	-0.024	0	0	0	-0.312	-10.994	-1.626	-1.938	-12.62	104 1.2XD + 1.2XTL2 + 1.6XL1
101 1.2X	146	0.021	0.015	0	0	0	3.483	1.229	9.535	13.018	10.764	101 1.2XD + 1.2XTL1 + 1.6XL
102 1.2X	888	0.023	-0.015	0	0	0	3.511	0.188	-9.417	12.928	9.605	102 1.2XD + 1.2XTL1 + 1.6XL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

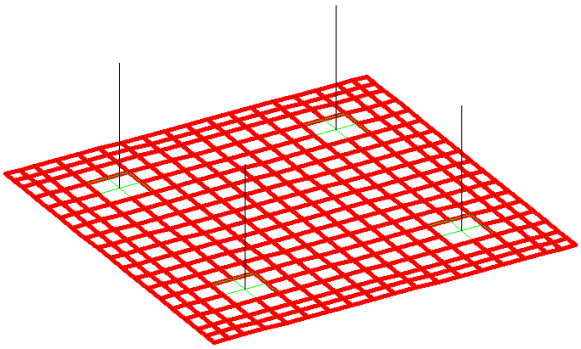
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	12.89	top face	50	442	12.894	0.09	0.0002	92.7	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
101 1.2X	-26.99	bottom face	50	442	26.993	0.18	0.0004	194.5		16	252.82	200	1006	14.18	17.73	0.072	tension control
101 1.2X	-17.33	top face	50	442	17.328	0.12	0.0003	124.7		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	-17.69	top face	50	442	17.685	0.12	0.0003	127.3		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	101 1.2X	0.073	1	36.5	3.906	0.002	0.442	fc'489	fc'435.5	fc'758.3	435.48	1621	> 36.5	ok	326.6	φVc>Vu	not required
	109 0.9X	0.053	1	26.5	2.188			fc'509.9			435.48	1621	> 26.5	ok	326.6	φVc>Vu	not required

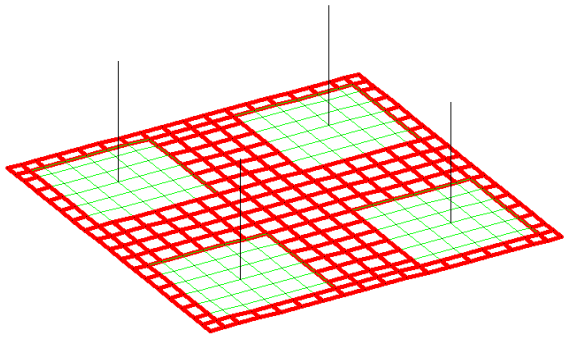
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
101 1.2X	14.90	top face	50	442	14.899	0.10	0.0002	107.2	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
104 1.2X	-21.02	bottom face	50	442	21.022	0.14	0.0003	151.4		16	252.82	200	1006	14.18	17.73	0.072	tension control
101 1.2X	14.70	top face	50	442	14.7	0.10	0.0002	105.8		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102 1.2X	-12.52	top face	50	442	12.524	0.09	0.0002	90.1		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	kN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	101 1.2X	0.05	1	25	1.339	0.002	0.442	559.49	435.484	fc'758.3	435.48	456	> 25	ok	326.6	φVc>Vu	not required
	102 1.2X	0.05	1	25	3.243			476.646			435.48	456	> 25	ok	326.6	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING RCO-1/ F3:

Load at bottom of fooring	Node 1					Node 3				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	3.5	0.6	0.0	-10.9	0.0	3.1	0.2	0.0	-0.1	0.0
202 D + L1 + TL1	5.0	-0.1	0.0	8.7	0.0	3.8	-0.6	0.0	0.8	0.0
203 D + L + TL2	3.5	0.6	0.0	-11.2	0.0	3.1	0.2	0.0	-0.4	0.0
204 D + L1 + TL2	5.0	-0.2	0.0	8.4	0.0	3.8	-0.6	0.0	0.5	0.0
205 D + 0.75XL + 0.75XTL1	3.6	0.5	0.0	-8.2	0.0	3.2	0.1	0.0	-0.1	0.0
206 D + 0.75XL1 + 0.75XTL1	4.7	-0.1	0.0	6.5	0.0	3.8	-0.5	0.0	0.7	0.0
207 D + 0.75XL + 0.75XTL2	3.6	0.5	0.0	-8.4	0.0	3.2	0.1	0.0	-0.2	0.0
208 D + 0.75XL1 + 0.75XTL2	4.7	-0.1	0.0	6.4	0.0	3.8	-0.5	0.0	0.5	0.0
209 D + WL(+Z)	2.8	0.6	0.0	-9.2	0.0	3.1	0.2	0.0	-1.1	0.0
210 D + WL(+X)	3.6	-0.7	0.0	0.2	0.0	3.5	-1.1	0.0	0.2	0.0
211 D + WL(-Z)	4.4	-0.2	0.0	9.5	0.0	3.9	-0.6	0.0	1.5	0.0
212 D + WL(-X)	3.7	1.1	0.0	0.2	0.0	3.4	0.7	0.0	0.2	0.0
213 D + 0.7XEL(+Z)	3.6	0.2	0.0	-0.2	0.0	3.5	-0.2	0.0	0.2	0.0
214 D + 0.7XEL(+X)	3.6	0.0	0.0	0.2	0.0	3.5	-0.4	0.0	0.2	0.0
215 D + 0.7XEL(-Z)	3.7	0.2	0.0	0.5	0.0	3.5	-0.2	0.0	0.3	0.0
216 D + 0.7XEL(-X)	3.7	0.4	0.0	0.2	0.0	3.4	0.0	0.0	0.2	0.0
217 D + 0.75XL + 0.75XWL(+Z)	3.0	0.8	0.0	-15.3	0.0	2.9	0.4	0.0	-1.1	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	4.1	0.2	0.0	-0.6	0.0	3.4	-0.2	0.0	-0.4	0.0
219 D + 0.75XL + 0.75XWL(-Z)	4.2	0.2	0.0	-1.3	0.0	3.5	-0.2	0.0	0.8	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	5.3	-0.3	0.0	13.5	0.0	4.1	-0.7	0.0	1.5	0.0
221 D + 0.75XL + 0.75XWL(+X)	3.5	-0.1	0.0	-8.3	0.0	3.3	-0.5	0.0	-0.1	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	4.7	-0.7	0.0	6.4	0.0	3.8	-1.1	0.0	0.6	0.0
223 D + 0.75XL + 0.75XWL(-X)	3.6	1.2	0.0	-8.3	0.0	3.2	0.8	0.0	-0.1	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	4.7	0.6	0.0	6.5	0.0	3.7	0.2	0.0	0.6	0.0
225 D + 0.75XL + 0.525XEL(+Z)	3.5	0.5	0.0	-8.5	0.0	3.2	0.1	0.0	-0.2	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	4.7	-0.1	0.0	6.2	0.0	3.7	-0.5	0.0	0.5	0.0
227 D + 0.75XL + 0.525XEL(+X)	3.5	0.4	0.0	-8.3	0.0	3.2	0.0	0.0	-0.2	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	4.7	-0.2	0.0	6.5	0.0	3.8	-0.6	0.0	0.6	0.0
229 D + 0.75XL + 0.525XEL(-Z)	3.6	0.5	0.0	-8.0	0.0	3.2	0.1	0.0	-0.1	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	4.7	-0.1	0.0	6.7	0.0	3.8	-0.5	0.0	0.6	0.0
231 D + 0.75XL + 0.525XEL(-X)	3.6	0.7	0.0	-8.3	0.0	3.2	0.3	0.0	-0.1	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	4.7	0.1	0.0	6.4	0.0	3.7	-0.3	0.0	0.6	0.0
233 0.6XD + WL(+Z)	1.4	0.5	0.0	-9.3	0.0	1.7	0.3	0.0	-1.2	0.0
234 0.6XD + WL(+X)	2.1	-0.8	0.0	0.1	0.0	2.1	-1.0	0.0	0.1	0.0
235 0.6XD + WL(-Z)	3.0	-0.2	0.0	9.5	0.0	2.5	-0.5	0.0	1.5	0.0
236 0.6XD + WL(-X)	2.2	1.0	0.0	0.1	0.0	2.0	0.8	0.0	0.1	0.0
237 0.6XD + 0.7XEL(+Z)	2.2	0.1	0.0	-0.2	0.0	2.1	-0.1	0.0	0.1	0.0
238 0.6XD + 0.7XEL(+X)	2.2	-0.1	0.0	0.1	0.0	2.1	-0.3	0.0	0.1	0.0
239 0.6XD + 0.7XEL(-Z)	2.2	0.1	0.0	0.4	0.0	2.1	-0.1	0.0	0.2	0.0
240 0.6XD + 0.7XEL(-X)	2.2	0.3	0.0	0.1	0.0	2.1	0.1	0.0	0.1	0.0

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Load at bottom of fooring	Node 2					Node 4				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	5.1	-0.2	0.0	-11.6	0.0	3.9	-0.6	0.0	-0.9	0.0
202 D + L1 + TL1	3.5	0.6	0.0	8.0	0.0	3.1	0.2	0.0	0.1	0.0
203 D + L + TL2	5.1	-0.2	0.0	-11.3	0.0	3.9	-0.6	0.0	-0.6	0.0
204 D + L1 + TL2	3.5	0.5	0.0	8.3	0.0	3.1	0.2	0.0	0.3	0.0
205 D + 0.75XL + 0.75XTL1	4.7	-0.1	0.0	-8.8	0.0	3.8	-0.5	0.0	-0.7	0.0
206 D + 0.75XL1 + 0.75XTL1	3.6	0.5	0.0	6.0	0.0	3.2	0.1	0.0	0.0	0.0
207 D + 0.75XL + 0.75XTL2	4.7	-0.1	0.0	-8.6	0.0	3.8	-0.5	0.0	-0.5	0.0
208 D + 0.75XL1 + 0.75XTL2	3.6	0.5	0.0	6.2	0.0	3.2	0.1	0.0	0.2	0.0
209 D + WL(+Z)	4.4	-0.2	0.0	-9.6	0.0	3.9	-0.6	0.0	-1.5	0.0
210 D + WL(+X)	3.6	-0.7	0.0	-0.2	0.0	3.5	-1.1	0.0	-0.2	0.0
211 D + WL(-Z)	2.8	0.6	0.0	9.1	0.0	3.1	0.2	0.0	1.1	0.0
212 D + WL(-X)	3.7	1.1	0.0	-0.2	0.0	3.4	0.7	0.0	-0.2	0.0
213 D + 0.7XEL(+Z)	3.7	0.2	0.0	-0.6	0.0	3.5	-0.2	0.0	-0.3	0.0
214 D + 0.7XEL(+X)	3.6	0.0	0.0	-0.2	0.0	3.5	-0.4	0.0	-0.2	0.0
215 D + 0.7XEL(-Z)	3.6	0.2	0.0	0.1	0.0	3.5	-0.2	0.0	-0.2	0.0
216 D + 0.7XEL(-X)	3.7	0.4	0.0	-0.2	0.0	3.5	0.0	0.0	-0.2	0.0
217 D + 0.75XL + 0.75XWL(+Z)	5.3	-0.4	0.0	-15.7	0.0	4.1	-0.8	0.0	-1.6	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	4.2	0.2	0.0	-1.0	0.0	3.5	-0.2	0.0	-0.9	0.0
219 D + 0.75XL + 0.75XWL(-Z)	4.1	0.2	0.0	-1.6	0.0	3.5	-0.2	0.0	0.4	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	3.0	0.7	0.0	13.1	0.0	2.9	0.4	0.0	1.1	0.0
221 D + 0.75XL + 0.75XWL(+X)	4.7	-0.8	0.0	-8.7	0.0	3.8	-1.2	0.0	-0.6	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	3.5	-0.2	0.0	6.1	0.0	3.3	-0.6	0.0	0.1	0.0
223 D + 0.75XL + 0.75XWL(-X)	4.7	0.5	0.0	-8.7	0.0	3.7	0.1	0.0	-0.6	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	3.6	1.1	0.0	6.1	0.0	3.2	0.7	0.0	0.1	0.0
225 D + 0.75XL + 0.525XEL(+Z)	4.7	-0.1	0.0	-8.9	0.0	3.8	-0.5	0.0	-0.7	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	3.6	0.5	0.0	5.8	0.0	3.2	0.1	0.0	0.1	0.0
227 D + 0.75XL + 0.525XEL(+X)	4.7	-0.2	0.0	-8.7	0.0	3.8	-0.6	0.0	-0.6	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	3.5	0.3	0.0	6.1	0.0	3.2	-0.1	0.0	0.1	0.0
229 D + 0.75XL + 0.525XEL(-Z)	4.7	-0.1	0.0	-8.4	0.0	3.8	-0.5	0.0	-0.6	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	3.5	0.5	0.0	6.3	0.0	3.2	0.1	0.0	0.1	0.0
231 D + 0.75XL + 0.525XEL(-X)	4.7	0.0	0.0	-8.7	0.0	3.8	-0.4	0.0	-0.6	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	3.6	0.6	0.0	6.1	0.0	3.2	0.2	0.0	0.1	0.0
233 0.6XD + WL(+Z)	3.0	-0.2	0.0	-9.5	0.0	2.5	-0.5	0.0	-1.5	0.0
234 0.6XD + WL(+X)	2.1	-0.8	0.0	-0.1	0.0	2.1	-1.0	0.0	-0.1	0.0
235 0.6XD + WL(-Z)	1.4	0.5	0.0	9.2	0.0	1.7	0.2	0.0	1.2	0.0
236 0.6XD + WL(-X)	2.2	1.0	0.0	-0.1	0.0	2.0	0.8	0.0	-0.1	0.0
237 0.6XD + 0.7XEL(+Z)	2.2	0.1	0.0	-0.5	0.0	2.1	-0.1	0.0	-0.2	0.0
238 0.6XD + 0.7XEL(+X)	2.2	-0.1	0.0	-0.1	0.0	2.1	-0.3	0.0	-0.1	0.0
239 0.6XD + 0.7XEL(-Z)	2.2	0.1	0.0	0.2	0.0	2.1	-0.1	0.0	-0.1	0.0
240 0.6XD + 0.7XEL(-X)	2.2	0.3	0.0	-0.1	0.0	2.1	0.1	0.0	-0.1	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + L + TL1	706.8	0.00	-2.0	-23.6	-3.4	796.6	517.8	NA	12.0
202 D + L1 + TL1	706.8	0.00	-2.0	17.6	3.3	796.6	530.4	282707	16.1
203 D + L + TL2	706.8	0.00	-2.0	-23.6	-3.4	796.1	517.5	NA	12.0
204 D + L1 + TL2	706.8	0.00	-2.0	17.6	3.3	796.1	530.1	282706.4	16
205 D + 0.75XL + 0.75XTL1	706.4	0.00	-1.6	-17.7	-2.6	992.6	688.9	NA	16.0
206 D + 0.75XL1 + 0.75XTL1	706.4	0.00	-1.6	13.2	2.5	992.6	708.0	282572	21.5
207 D + 0.75XL + 0.75XTL2	706.4	0.00	-1.6	-17.7	-2.6	992.6	688.9	NA	16.0
208 D + 0.75XL1 + 0.75XTL2	706.4	0.00	-1.6	13.2	2.5	992.6	708.0	282571.6	21
209 D + WL(+Z)	705.4	0.00	-0.4	-21.4	-3.6	3801.6	485.0	NA	13.2
218 D + 0.75XL1 + 0.75XWL(+Z)	705.9	-1.75	-0.9	-1.5	0.6	1809.9	2774.6	161.4	192.2
219 D + 0.75XL + 0.75XWL(-Z)	704.8	0.82	-0.5	9.8	2.6	3042.2	667.8	344.2	28.8
220 D + 0.75XL1 + 0.75XWL(-Z)	707.7	0.66	-1.9	14.5	3.4	843.0	525.8	430.2	19.5
221 D + 0.75XL + 0.75XWL(+X)	705.1	-0.71	-0.6	-9.3	-0.6	2878.0	3021.0	397.2	30.4
222 D + 0.75XL1 + 0.75XWL(+X)	706.8	-2.20	-1.2	6.5	2.0	1291.5	881.0	128.5	43.3
223 D + 0.75XL + 0.75XWL(-X)	705.0	1.98	-0.7	-8.5	-0.5	2161.9	3706.8	142.1	33.1
224 D + 0.75XL1 + 0.75XWL(-X)	706.8	1.11	-1.5	6.5	2.0	1035.1	885.0	254.9	43.2
225 D + 0.75XL + 0.525XEL(+Z)	707.3	-0.52	-2.0	-26.0	-4.0	813.5	446.0	543.0	10.9
226 D + 0.75XL1 + 0.525XEL(+Z)	707.3	-0.52	-2.0	4.9	1.1	813.0	1614.9	544.1	58.0
227 D + 0.75XL + 0.525XEL(+X)	705.5	0.24	-1.2	-9.7	-1.2	1328.4	1462.5	1171.0	29.1
228 D + 0.75XL1 + 0.525XEL(+X)	705.5	0.24	-1.2	21.2	3.9	1328.4	457.9	1161.4	13.3
229 D + 0.75XL + 0.525XEL(-Z)	706.5	-1.34	-1.5	-17.4	-2.5	1054.4	699.2	211.2	16.2
230 D + 0.75XL1 + 0.525XEL(-Z)	706.5	-1.34	-1.5	13.4	2.5	1055.3	697.1	211.2	21.0
231 D + 0.75XL + 0.525XEL(-X)	706.4	1.62	-1.7	-17.7	-2.6	910.2	690.5	174.6	16.0
224 D + 0.75XL1 + 0.75XWL(-X)	706.8	1.11	-1.5	6.5	2.0	1035.1	885.0	254.9	43.2
225 D + 0.75XL + 0.525XEL(+Z)	707.3	-0.52	-2.0	-26.0	-4.0	813.5	446.0	543.0	10.9
226 D + 0.75XL1 + 0.525XEL(+Z)	706.4	0.00	-1.6	12.6	2.4	992.6	732.7	282571.6	22.4
227 D + 0.75XL + 0.525XEL(+X)	706.4	-0.56	-1.5	-17.7	-2.6	1041.4	688.9	504	16.0
228 D + 0.75XL1 + 0.525XEL(+X)	706.4	-0.56	-1.5	13.2	2.5	1041.4	708.0	504.6	21.5
229 D + 0.75XL + 0.525XEL(-Z)	706.4	0.00	-1.6	-17.1	-2.5	991.9	712.7	NA	16.5
230 D + 0.75XL1 + 0.525XEL(-Z)	706.4	0.00	-1.6	13.7	2.6	992.6	684.1	NA	21
231 D + 0.75XL + 0.525XEL(-X)	706.4	0.56	-1.7	-17.7	-2.6	946.8	688.9	502.8	16.0
232 D + 0.75XL1 + 0.525XEL(-X)	706.4	0.56	-1.7	13.2	2.5	946.8	708.0	503.7	21.5
233 0.6XD + WL(+Z)	699.7	0.00	-0.3	-21.4	-3.6	6297.6	481.9	NA	13
234 0.6XD + WL(+X)	699.7	-3.50	0.0	-0.1	0.0	54762.0	166604	80.0	5281.0
235 0.6XD + WL(-Z)	699.7	0.00	-0.3	21.3	3.6	6297.6	484.3	NA	13.1
236 0.6XD + WL(-X)	699.7	3.50	-0.5	0.0	0.0	2970.5	194367	80.0	6219.7
237 0.6XD + 0.7XEL(+Z)	699.7	0.00	-0.3	-0.8	-0.1	6266.2	14050.8	NA	351.6
238 0.6XD + 0.7XEL(+X)	699.7	-0.75	-0.1	-0.1	0.0	10584.1	166602	374.2	5597.8
239 0.6XD + 0.7XEL(-Z)	699.7	0.00	-0.3	0.7	0.1	6266.2	16425.6	NA	401.0
240 0.6XD + 0.7XEL(-X)	699.7	0.75	-0.4	-0.1	0.0	4482.2	166602	374.2	5597.8
					Min.=	796.1	446.0	80.0	10.9

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DESIGN CALCULATION OF PEDESTAL FOR RCO-1/ F3:

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 600 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.8

Y_b = (b-d')/b = 0.8

Unsupported length of column parallel to X-X axis, l_{ux} = 2 m kl_{ux}/r_x = 13.33 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2 m kl_{uz}/r_z = 13.33

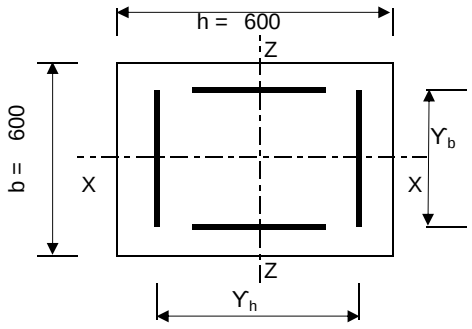
Radius of gyration r_x = 180 mm

Radius of gyration r_z = 180 mm

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 9746 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha = \log 0.5/\log\beta$	$(M_{nx}/M_{nx1})^{\alpha} + (M_{nz}/M_{nz1})^{\alpha}$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	180.6	13.8	25.5	278	21.2	39.2	0.022	0.090	680.4	0.03	0.090	680.4	0.06	1.609	0.01	safe
109	124.8	14.8	0.0	192	22.8	0.0	0.015	0.090	680.4	0.03	0.090	680.4	0.00		0.01	safe
110	26.2	0.2	0.4	40	0.3	0.6	0.003	0.090	680.4	0.00	0.090	680.4	0.00		0.00	safe
112	35.0	0.1	0.3	54	0.2	0.4	0.004	0.070	529.2	0.00	0.070	529.2	0.00		0.00	safe
111	109.0	15.1	29.2	168	23.3	45.0	0.013	0.065	491.4	0.05	0.065	491.4	0.09		0.03	safe
101	176.9	18.4	34.1	272	28.3	52.4	0.022	0.070	529.2	0.05	0.070	529.2	0.10		0.03	safe
100	34.7	0.3	0.6	53	0.5	0.9	0.004	0.060	453.6	0.00	0.060	453.6	0.00		0.00	safe
100	34.7	0.3	0.6	53	0.5	0.9	0.004	0.060	453.6	0.00	0.060	453.6	0.00		0.00	safe
101	176.9	18.4	34.1	272	28.3	52.4	0.022	0.070	529.2	0.05	0.070	529.2	0.10		0.03	safe
111	109.0	15.1	56.3	168	23.3	86.7	0.013	0.065	491.4	0.05	0.065	491.4	0.18		0.07	safe
110	26.2	0.2	0.4	40	0.3	0.6	0.003	0.090	680.4	0.00	0.090	680.4	0.00		0.00	safe
112	35.0	0.1	0.3	54	0.2	0.4	0.004	0.070	529.2	0.00	0.070	529.2	0.00		0.00	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3600	4	20 +	8	20	3772	1.05	10	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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- **DESIGN CALCULATION OF FOOTING RCO-2/ F4:**

Size of Columns at node	Along-X	Along-Z
1	0.60m	0.60m
2	0.60m	0.60m
3	0.60m	0.60m
4	0.60m	0.60m
Size of fdn. raft:	5.00m	4.00m

Thickness of raft:	0.50	m	RL (+) 913.60
Original Gr. Level:	0.00	m	RL (+) 913.95
Finished Gr. Level:	0.00	m	
U/S of B/Pl Above GL:	0.35	m	
U/S level of fdn:	-1.50	m	
Unit weight of conc:	24	kN/m^3	
Unit weight of soil:	20	kN/m^3	

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

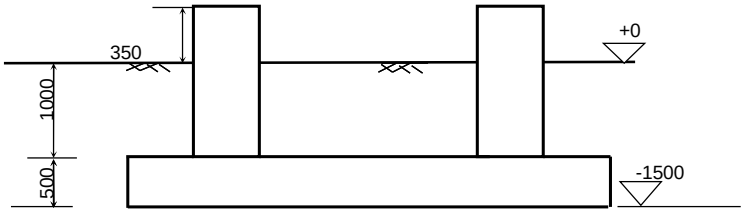
Calculation of Weight of Foundation :

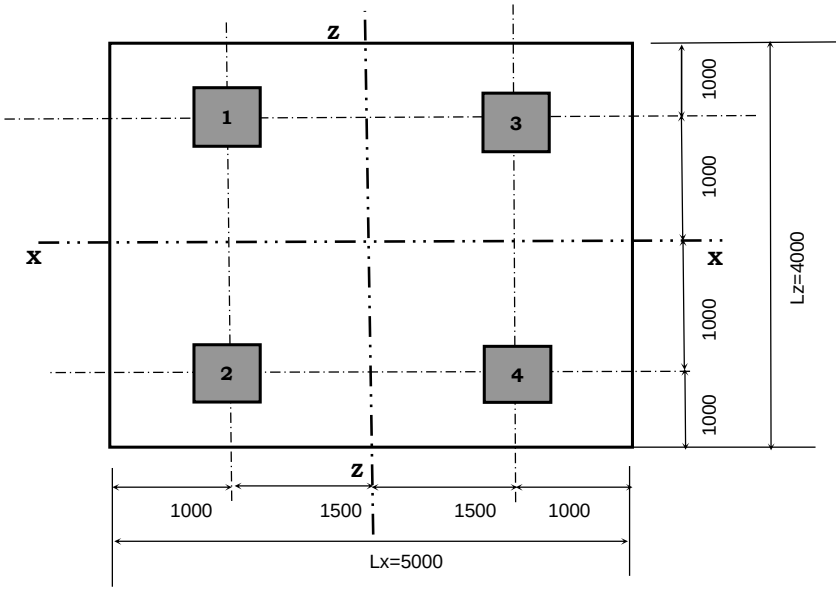
Weight of raft = (5*4*0.5*24) = 240.00KN

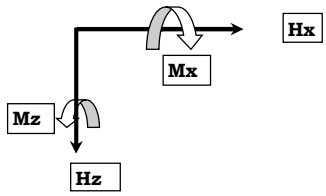
Weight of soil = ((5*4)-4*(0.6*0.6))*1*20 = 371.20KN

Total = 611.20KN

Submerged wt. = 611.2-5*4*1.5*10 = 311.20KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING RCO-2/ F4:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	102.652	0
Min Px	23	201	0	102.652	0
Max Py	1312	220	0	177.32	0
Min Py	1334	220	0	-68.766	0
Max Pz	23	201	0	102.652	0
Min Pz	23	201	0	102.652	0

Gross Maximum Base pressure, p_{max} = 177.3 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 480 KN/m^2
> 177.3 KN/m^2
Hence OK

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DESIGN CALCULATION OF FOOTING RCO-2/ F4:

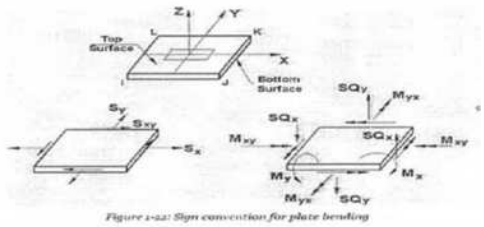
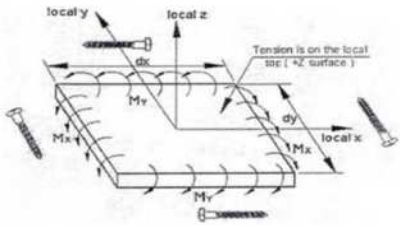


Figure 1-32: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; FDN_12_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
109 0.9X	876	0.383	0.028	0	0	0	43.088	42.062	-22.155	65.243	64.217	109 0.9XD + 1.6XWL(+Z)
109 0.9X	112	-0.436	0.023	0	0	0	-44.18	-48.878	0	-44.176	-48.878	109 0.9XD + 1.6XWL(+Z)
111 0.9X	883	-0.067	0.426	0	0	0	-66.63	-47.701	14.872	-81.501	-62.573	111 0.9XD + 1.6XWL(-Z)
109 0.9X	102	-0.067	-0.426	0	0	0	-66.67	-47.719	-14.896	-81.567	-62.615	109 0.9XD + 1.6XWL(+Z)
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
109 0.9X	884	-0.159	-0.335	0	0	0	60.472	43.067	-17.644	78.116	60.711	109 0.9XD + 1.6XWL(+Z)
109 0.9X	103	0.163	-0.378	0	0	0	-68.03	-46.276	-18.589	-86.62	-64.865	109 0.9XD + 1.6XWL(+Z)
109 0.9X	863	0.286	0.177	0	0	0	35.158	47.936	-31.883	67.041	79.819	109 0.9XD + 1.6XWL(+Z)
109 0.9X	111	-0.338	0.185	0	0	0	-36.23	-54.575	-32.989	-69.22	-87.564	109 0.9XD + 1.6XWL(+Z)
111 0.9X	864	-0.085	-0.171	0	0	0	-20.9	-12.966	40.143	-61.045	-53.109	111 0.9XD + 1.6XWL(-Z)
109 0.9X	110	-0.085	0.171	0	0	0	-20.9	-12.989	-40.207	-61.11	-53.196	109 0.9XD + 1.6XWL(+Z)

PLATE CENTER STRESS ; FDN_12_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
109 0.9X	1182	0.274	-0.027	0	0	0	14.282	1.667	-7.397	21.679	9.064	109 0.9XD + 1.6XWL(+Z)
109 0.9X	1206	-0.284	-0.029	0	0	0	-11.93	-1.841	0	-11.93	-1.841	109 0.9XD + 1.6XWL(+Z)
109 0.9X	130	-0.001	0.083	0	0	0	-5.214	0.813	-20.329	-25.543	21.142	109 0.9XD + 1.6XWL(+Z)
111 0.9X	855	-0.001	-0.082	0	0	0	-5.21	0.817	20.27	-25.48	21.087	111 0.9XD + 1.6XWL(-Z)
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
100 1.4X	1216	0	-0.001	0	0	0	-0.103	-0.134	0.083	-0.186	-0.217	100 1.4XD
109 0.9X	1184	0.155	-0.059	0	0	0	49.599	1.897	-7.32	56.919	9.217	109 0.9XD + 1.6XWL(+Z)
109 0.9X	1202	-0.16	-0.064	0	0	0	-53.53	-1.951	-6.92	-60.448	-8.871	109 0.9XD + 1.6XWL(+Z)
109 0.9X	1154	-0.021	0.04	0	0	0	0.098	26.544	-2.668	2.766	29.212	109 0.9XD + 1.6XWL(+Z)
109 0.9X	1143	0.026	0.039	0	0	0	-0.208	-31.29	-2.907	-3.115	-34.197	109 0.9XD + 1.6XWL(+Z)
111 0.9X	896	-0.014	-0.02	0	0	0	3.124	-1.298	37.835	40.959	-39.133	111 0.9XD + 1.6XWL(-Z)
109 0.9X	175	-0.014	0.02	0	0	0	3.124	-1.299	-37.896	41.02	-39.195	109 0.9XD + 1.6XWL(+Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

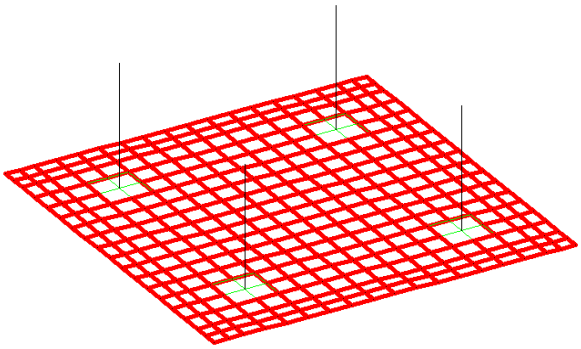
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	78.12	top face	50	442	78.116	0.53	0.0013	566.2	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-86.62	bottom face	50	442	86.62	0.59	0.0014	628.4		16	252.82	200	1006	14.18	17.73	0.072	tension control
111 0.9X	-61.05	top face	50	442	61.045	0.42	0.0010	441.6		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-61.11	top face	50	442	61.11	0.42	0.0010	442.0		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	109 0.9X	0.274	1	137	21.679	0.002	0.442	fc'466.1	fc'435.5	fc'758.3	435.48	1621	> 137	ok	326.6	φVc>Vu	not required
	109 0.9X	0.284	1	142	11.93			fc'508.3			435.48	1621	> 142	ok	326.6	φVc>Vu	not required

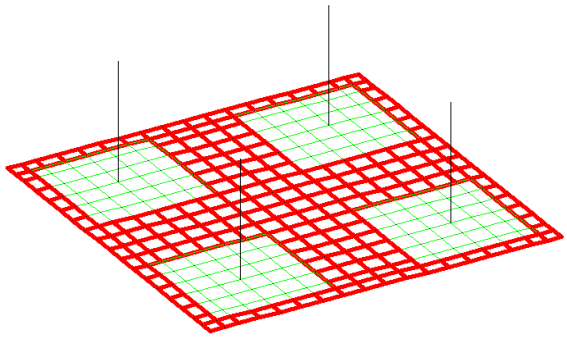
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	79.82	top face	50	442	79.819	0.54	0.0013	578.6	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-87.56	bottom face	50	442	87.564	0.60	0.0014	635.4		16	252.82	200	1006	14.18	17.73	0.072	tension control
111 0.9X	-53.11	top face	50	442	53.109	0.36	0.0009	383.8		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-53.20	top face	50	442	53.196	0.36	0.0009	384.4		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	kN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	109 0.9X	0.083	1	41.5	21.142	0.002	0.442	433.22	435.484	fc'758.3	433.22	454	> 41.5	ok	324.9	φVc>Vu	not required
	111 0.9X	0.082	1	41	21.087			433.08			433.08	454	> 41	ok	324.8	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING RCO-1/ F4:

Load at bottom of fooring	Node 1					Node 3				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	-0.8	-38.1	0.0	0.4	0.0	7.5	-38.6	0.0	-12.7	0.0
202 D + L1 + TL1	0.9	-18.5	0.0	0.2	0.0	9.2	-19.1	0.0	13.2	0.0
203 D + L + TL2	-0.8	-38.4	0.0	0.0	0.0	7.5	-38.9	0.0	-12.7	0.0
204 D + L1 + TL2	0.9	-18.9	0.0	-0.2	0.0	9.3	-19.4	0.0	13.2	0.0
205 D + 0.75XL + 0.75XTL1	0.3	-28.5	0.0	0.3	0.0	6.6	-29.0	0.0	-9.5	0.0
206 D + 0.75XL1 + 0.75XTL1	1.6	-13.8	0.0	0.2	0.0	7.9	-14.4	0.0	9.9	0.0
207 D + 0.75XL + 0.75XTL2	0.3	-28.7	0.0	0.0	0.0	6.6	-29.2	0.0	-9.5	0.0
208 D + 0.75XL1 + 0.75XTL2	1.5	-14.1	0.0	-0.1	0.0	7.9	-14.6	0.0	9.9	0.0
209 D + WL(+Z)	0.5	-25.9	0.0	-1.3	0.0	-0.1	-26.4	0.0	-43.0	0.0
210 D + WL(+X)	2.0	-13.1	0.0	0.1	0.0	5.5	-13.6	0.0	0.2	0.0
211 D + WL(-Z)	6.5	26.5	0.0	1.5	0.0	8.1	25.9	0.0	43.3	0.0
212 D + WL(-X)	5.0	13.7	0.0	0.1	0.0	2.5	13.1	0.0	0.1	0.0
213 D + 0.7XEL(+Z)	3.5	0.3	0.0	0.1	0.0	4.0	-0.3	0.0	-0.3	0.0
214 D + 0.7XEL(+X)	3.5	0.1	0.0	0.1	0.0	4.0	-0.5	0.0	0.1	0.0
215 D + 0.7XEL(-Z)	3.5	0.3	0.0	0.2	0.0	4.0	-0.2	0.0	0.6	0.0
216 D + 0.7XEL(-X)	3.5	0.5	0.0	0.1	0.0	4.0	0.0	0.0	0.1	0.0
217 D + 0.75XL + 0.75XWL(+Z)	-2.0	-48.2	0.0	-0.9	0.0	3.5	-48.7	0.0	-41.9	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	-0.7	-33.6	0.0	-1.0	0.0	4.8	-34.1	0.0	-22.4	0.0
219 D + 0.75XL + 0.75XWL(-Z)	2.6	-9.0	0.0	1.2	0.0	9.7	-9.5	0.0	22.9	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	3.8	5.7	0.0	1.1	0.0	11.0	5.1	0.0	42.3	0.0
221 D + 0.75XL + 0.75XWL(+X)	-0.8	-38.6	0.0	0.2	0.0	7.7	-39.2	0.0	-9.5	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	0.4	-24.0	0.0	0.0	0.0	9.0	-24.5	0.0	9.9	0.0
223 D + 0.75XL + 0.75XWL(-X)	1.5	-18.6	0.0	0.2	0.0	5.5	-19.1	0.0	-9.5	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	2.7	-3.9	0.0	0.1	0.0	6.8	-4.5	0.0	9.9	0.0
225 D + 0.75XL + 0.525XEL(+Z)	0.3	-28.6	0.0	0.1	0.0	6.6	-29.2	0.0	-9.8	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	1.5	-14.0	0.0	0.0	0.0	7.9	-14.5	0.0	9.6	0.0
227 D + 0.75XL + 0.525XEL(+X)	0.3	-28.8	0.0	0.2	0.0	6.6	-29.3	0.0	-9.5	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	1.5	-14.1	0.0	0.0	0.0	7.9	-14.7	0.0	9.9	0.0
229 D + 0.75XL + 0.525XEL(-Z)	0.3	-28.6	0.0	0.2	0.0	6.6	-29.1	0.0	-9.2	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	1.6	-13.9	0.0	0.1	0.0	8.0	-14.5	0.0	10.2	0.0
231 D + 0.75XL + 0.525XEL(-X)	0.3	-28.4	0.0	0.2	0.0	6.6	-29.0	0.0	-9.5	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	1.6	-13.8	0.0	0.0	0.0	7.9	-14.3	0.0	9.9	0.0
233 0.6XD + WL(+Z)	-0.9	-26.0	0.0	-1.3	0.0	-1.7	-26.3	0.0	-43.1	0.0
234 0.6XD + WL(+X)	0.6	-13.2	0.0	0.1	0.0	3.9	-13.5	0.0	0.1	0.0
235 0.6XD + WL(-Z)	5.1	26.4	0.0	1.5	0.0	6.5	26.0	0.0	43.3	0.0
236 0.6XD + WL(-X)	3.6	13.5	0.0	0.1	0.0	0.9	13.2	0.0	0.1	0.0
237 0.6XD + 0.7XEL(+Z)	2.1	0.1	0.0	0.0	0.0	2.4	-0.2	0.0	-0.3	0.0
238 0.6XD + 0.7XEL(+X)	2.1	0.0	0.0	0.1	0.0	2.4	-0.4	0.0	0.1	0.0
239 0.6XD + 0.7XEL(-Z)	2.1	0.2	0.0	0.1	0.0	2.4	-0.1	0.0	0.5	0.0
240 0.6XD + 0.7XEL(-X)	2.1	0.4	0.0	0.1	0.0	2.4	0.1	0.0	0.1	0.0

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Load at bottom of fooring	Node 2					Node 4				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	0.9	-19.1	0.0	0.0	0.0	9.2	-19.7	0.0	-13.3	0.0
202 D + L1 + TL1	-0.8	-38.7	0.0	-0.2	0.0	7.5	-39.3	0.0	12.7	0.0
203 D + L + TL2	0.9	-18.9	0.0	0.0	0.0	9.2	-19.4	0.0	-12.9	0.0
204 D + L1 + TL2	-0.8	-38.4	0.0	-0.2	0.0	7.5	-38.9	0.0	13.0	0.0
205 D + 0.75XL + 0.75XTL1	1.5	-14.3	0.0	-0.1	0.0	7.9	-14.9	0.0	-10.0	0.0
206 D + 0.75XL1 + 0.75XTL1	0.3	-29.0	0.0	-0.2	0.0	6.6	-29.5	0.0	9.5	0.0
207 D + 0.75XL + 0.75XTL2	1.5	-14.1	0.0	-0.1	0.0	7.9	-14.6	0.0	-9.7	0.0
208 D + 0.75XL1 + 0.75XTL2	0.3	-28.8	0.0	-0.2	0.0	6.6	-29.3	0.0	9.7	0.0
209 D + WL(+Z)	6.5	26.4	0.0	-1.5	0.0	8.1	25.8	0.0	-43.3	0.0
210 D + WL(+X)	2.0	-13.2	0.0	-0.1	0.0	5.5	-13.7	0.0	-0.1	0.0
211 D + WL(-Z)	0.5	-25.9	0.0	1.2	0.0	-0.1	-26.4	0.0	43.0	0.0
212 D + WL(-X)	5.0	13.6	0.0	-0.1	0.0	2.5	13.1	0.0	-0.1	0.0
213 D + 0.7XEL(+Z)	3.5	0.3	0.0	-0.2	0.0	4.0	-0.3	0.0	-0.5	0.0
214 D + 0.7XEL(+X)	3.5	0.0	0.0	-0.1	0.0	4.0	-0.5	0.0	-0.1	0.0
215 D + 0.7XEL(-Z)	3.5	0.2	0.0	-0.1	0.0	4.0	-0.3	0.0	0.3	0.0
216 D + 0.7XEL(-X)	3.5	0.5	0.0	-0.1	0.0	4.0	-0.1	0.0	-0.1	0.0
217 D + 0.75XL + 0.75XWL(+Z)	3.8	5.4	0.0	-1.1	0.0	11.0	4.9	0.0	-42.2	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	2.5	-9.2	0.0	-1.2	0.0	9.7	-9.8	0.0	-22.8	0.0
219 D + 0.75XL + 0.75XWL(-Z)	-0.7	-33.8	0.0	1.0	0.0	4.8	-34.3	0.0	22.5	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	-2.0	-48.5	0.0	0.9	0.0	3.5	-49.0	0.0	42.0	0.0
221 D + 0.75XL + 0.75XWL(+X)	0.4	-24.2	0.0	-0.1	0.0	9.0	-24.8	0.0	-9.9	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	-0.9	-38.9	0.0	-0.2	0.0	7.7	-39.4	0.0	9.6	0.0
223 D + 0.75XL + 0.75XWL(-X)	2.7	-4.2	0.0	-0.1	0.0	6.8	-4.7	0.0	-9.8	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	1.4	-18.8	0.0	-0.2	0.0	5.5	-19.4	0.0	9.6	0.0
225 D + 0.75XL + 0.525XEL(+Z)	1.6	-14.2	0.0	-0.1	0.0	7.9	-14.7	0.0	-10.2	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	0.3	-28.8	0.0	-0.2	0.0	6.6	-29.4	0.0	9.3	0.0
227 D + 0.75XL + 0.525XEL(+X)	1.5	-14.4	0.0	-0.1	0.0	7.9	-14.9	0.0	-9.8	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	0.3	-29.0	0.0	-0.2	0.0	6.6	-29.6	0.0	9.6	0.0
229 D + 0.75XL + 0.525XEL(-Z)	1.5	-14.2	0.0	0.0	0.0	7.9	-14.8	0.0	-9.5	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	0.3	-28.9	0.0	-0.1	0.0	6.6	-29.4	0.0	9.9	0.0
231 D + 0.75XL + 0.525XEL(-X)	1.6	-14.0	0.0	-0.1	0.0	7.9	-14.6	0.0	-9.8	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	0.3	-28.7	0.0	-0.2	0.0	6.6	-29.2	0.0	9.6	0.0
233 0.6XD + WL(+Z)	5.1	26.3	0.0	-1.5	0.0	6.5	26.0	0.0	-43.2	0.0
234 0.6XD + WL(+X)	0.6	-13.3	0.0	-0.1	0.0	3.9	-13.6	0.0	-0.1	0.0
235 0.6XD + WL(-Z)	-0.9	-26.0	0.0	1.3	0.0	-1.7	-26.3	0.0	43.1	0.0
236 0.6XD + WL(-X)	3.6	13.5	0.0	-0.1	0.0	0.9	13.2	0.0	-0.1	0.0
237 0.6XD + 0.7XEL(+Z)	2.1	0.2	0.0	-0.1	0.0	2.4	-0.1	0.0	-0.5	0.0
238 0.6XD + 0.7XEL(+X)	2.1	-0.1	0.0	-0.1	0.0	2.4	-0.4	0.0	-0.1	0.0
239 0.6XD + 0.7XEL(-Z)	2.1	0.1	0.0	0.0	0.0	2.4	-0.2	0.0	0.3	0.0
240 0.6XD + 0.7XEL(-X)	2.1	0.4	0.0	-0.1	0.0	2.4	0.1	0.0	-0.1	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + L + TL1	628.0	-115.60	24.9	-25.7	-3.4	63.0	371.6	2	9.8
202 D + L1 + TL1	628.0	-115.60	24.9	25.9	3.4	63.0	364.3	2	9.7
203 D + L + TL2	628.0	-115.60	24.9	-25.7	-3.4	63.1	371.7	2.2	9.8
204 D + L1 + TL2	628.0	-115.60	24.9	25.9	3.4	63.1	364.2	2.2	10
205 D + 0.75XL + 0.75XTL1	627.6	-86.70	19.0	-19.2	-2.5	82.4	495.3	2.9	13.0
206 D + 0.75XL1 + 0.75XTL1	627.6	-86.70	19.0	19.4	2.6	82.4	485.4	3	12.9
207 D + 0.75XL + 0.75XTL2	627.6	-86.70	19.0	-19.2	-2.5	82.4	495.3	2.9	13.0
208 D + 0.75XL1 + 0.75XTL2	627.6	-86.70	19.0	19.4	2.6	82.4	485.4	2.9	13
209 D + WL(+Z)	626.2	0.00	1.5	-89.1	-14.3	1068.2	87.5	NA	2.8
218 D + 0.75XL1 + 0.75XWL(+Z)	622.8	-94.52	13.6	-23.7	-3.3	114.9	374.8	2.6	10.5
219 D + 0.75XL + 0.75XWL(-Z)	623.8	-70.90	9.8	68.3	11.9	159.4	104.5	3.5	3.7
220 D + 0.75XL1 + 0.75XWL(-Z)	633.6	37.52	7.1	43.1	7.3	224.5	174.0	6.8	5.9
221 D + 0.75XL + 0.75XWL(+X)	625.6	-77.80	13.6	-10.1	-0.7	114.8	1834.7	3.2	24.8
222 D + 0.75XL1 + 0.75XWL(+X)	628.1	-49.03	13.8	9.7	1.9	114.0	648.9	5.1	25.9
223 D + 0.75XL + 0.75XWL(-X)	625.6	-37.85	6.8	-9.1	-0.5	230.4	2612.1	6.6	27.4
224 D + 0.75XL1 + 0.75XWL(-X)	628.2	-7.97	6.8	9.7	2.0	229.9	621.1	31.5	25.9
225 D + 0.75XL + 0.525XEL(+Z)	632.9	-47.52	20.3	-53.0	-7.9	78.0	159.4	5.3	4.8
226 D + 0.75XL1 + 0.525XEL(+Z)	632.9	-47.52	20.3	-14.4	-2.8	78.0	448.5	5.3	17.6
227 D + 0.75XL + 0.525XEL(+X)	622.2	-126.27	17.8	14.2	2.8	87.2	439.7	2.0	17.6
228 D + 0.75XL1 + 0.525XEL(+X)	622.2	-126.27	17.8	52.8	8.0	87.2	156.5	2.0	4.7
229 D + 0.75XL + 0.525XEL(-Z)	627.6	-106.73	22.5	-18.9	-2.4	69.9	512.9	2.4	13.3
230 D + 0.75XL1 + 0.525XEL(-Z)	627.6	-106.73	22.5	19.7	2.7	69.9	469.2	2.4	12.7
231 D + 0.75XL + 0.525XEL(-X)	627.6	-66.28	15.6	-19.2	-2.6	100.6	486.2	3.8	13.0
224 D + 0.75XL1 + 0.75XWL(-X)	628.2	-7.97	6.8	9.7	2.0	229.9	621.1	31.5	25.9
225 D + 0.75XL + 0.525XEL(+Z)	632.9	-47.52	20.3	-53.0	-7.9	78.0	159.4	5.3	4.8
226 D + 0.75XL1 + 0.525XEL(+Z)	627.6	-86.70	19.0	18.7	2.5	82.4	502.3	2.9	13.4
227 D + 0.75XL + 0.525XEL(+X)	627.6	-87.39	19.2	-19.2	-2.5	81.9	495.1	3	13.0
228 D + 0.75XL1 + 0.525XEL(+X)	627.6	-87.39	19.2	19.4	2.6	81.9	485.4	2.9	12.9
229 D + 0.75XL + 0.525XEL(-Z)	627.6	-86.70	19.0	-18.6	-2.5	82.4	512.3	2.9	13.5
230 D + 0.75XL1 + 0.525XEL(-Z)	627.6	-86.70	19.0	20.1	2.7	82.4	469.7	2.9	12
231 D + 0.75XL + 0.525XEL(-X)	627.6	-86.01	18.9	-19.2	-2.5	82.9	494.9	2.9	13.0
232 D + 0.75XL1 + 0.525XEL(-X)	627.6	-86.01	18.9	19.4	2.6	82.9	485.4	2.9	12.9
233 0.6XD + WL(+Z)	620.2	0.00	0.9	-89.1	-14.3	1760.9	86.7	NA	3
234 0.6XD + WL(+X)	620.1	-53.52	9.9	0.0	0.0	156.2	620080	4.6	41338.7
235 0.6XD + WL(-Z)	620.2	0.00	0.9	89.1	14.3	1760.9	86.7	NA	2.8
236 0.6XD + WL(-X)	620.3	53.52	-8.2	0.0	0.0	189.9	413547	4.6	124064.2
237 0.6XD + 0.7XEL(+Z)	620.2	0.00	0.9	-0.9	-0.1	1760.9	10786.1	NA	269.1
238 0.6XD + 0.7XEL(+X)	620.2	-0.93	1.0	0.0	0.0	1491.6	1240402	268.2	62020.1
239 0.6XD + 0.7XEL(-Z)	620.2	0.00	0.9	0.9	0.1	1763.9	10880.7	248080.0	267.3
240 0.6XD + 0.7XEL(-X)	620.2	0.93	0.7	0.0	0.0	2149.0	1240402	267.9	82693.5
					Min.=	63.0	86.7	2.0	2.8

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DESIGN CALCULATION OF PEDESTAL FOR RCO-1/ F4:

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 600 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.8

Y_b = (b-d')/b = 0.8

Unsupported length of column parallel to X-X axis, l_{ux} = 2 m kl_{ux}/r_x = 13.33 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2 m kl_{uz}/r_z = 13.33

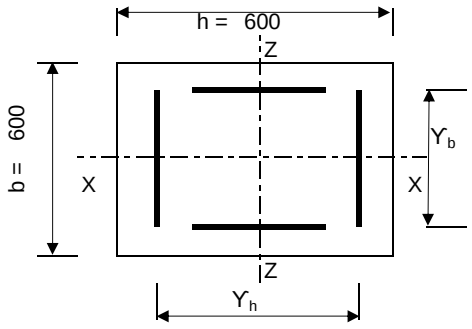
Radius of gyration r_x = 180 mm

Radius of gyration r_z = 180 mm

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 9746 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha = \log 0.5/\log\beta$	$(M_{nx}/M_{nx1})^{\alpha} + (M_{nz}/M_{nz1})^{\alpha}$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	180.6	13.8	25.5	278	21.2	39.2	0.022	0.090	680.4	0.03	0.090	680.4	0.06	1.609	0.01	safe
109	124.8	14.8	0.0	192	22.8	0.0	0.015	0.090	680.4	0.03	0.090	680.4	0.00		0.01	safe
110	26.2	0.2	0.4	40	0.3	0.6	0.003	0.090	680.4	0.00	0.090	680.4	0.00		0.00	safe
112	35.0	0.1	0.3	54	0.2	0.4	0.004	0.070	529.2	0.00	0.070	529.2	0.00		0.00	safe
111	109.0	15.1	29.2	168	23.3	45.0	0.013	0.065	491.4	0.05	0.065	491.4	0.09		0.03	safe
101	176.9	18.4	34.1	272	28.3	52.4	0.022	0.070	529.2	0.05	0.070	529.2	0.10		0.03	safe
100	34.7	0.3	0.6	53	0.5	0.9	0.004	0.060	453.6	0.00	0.060	453.6	0.00		0.00	safe
100	34.7	0.3	0.6	53	0.5	0.9	0.004	0.060	453.6	0.00	0.060	453.6	0.00		0.00	safe
101	176.9	18.4	34.1	272	28.3	52.4	0.022	0.070	529.2	0.05	0.070	529.2	0.10		0.03	safe
111	109.0	15.1	56.3	168	23.3	86.7	0.013	0.065	491.4	0.05	0.065	491.4	0.18		0.07	safe
110	26.2	0.2	0.4	40	0.3	0.6	0.003	0.090	680.4	0.00	0.090	680.4	0.00		0.00	safe
112	35.0	0.1	0.3	54	0.2	0.4	0.004	0.070	529.2	0.00	0.070	529.2	0.00		0.00	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars, db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3600	4	20	+	8	20	3772	1.05	10	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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- **DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F5:**

Size of Columns at node	Along-X	Along-Z
Col.1	0.60m	0.85m
Col.2	0.60m	0.85m
Size of fdn. raft:	4.00m	2.50m
Thickness of raft:	0.50 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.35 m	
U/S level of fdn:	-1.50 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	20 kN/m^3	

Calculation of Weight of Foundation:

Weight of raft = (4*2.5*0.5*25) = 125.00KN

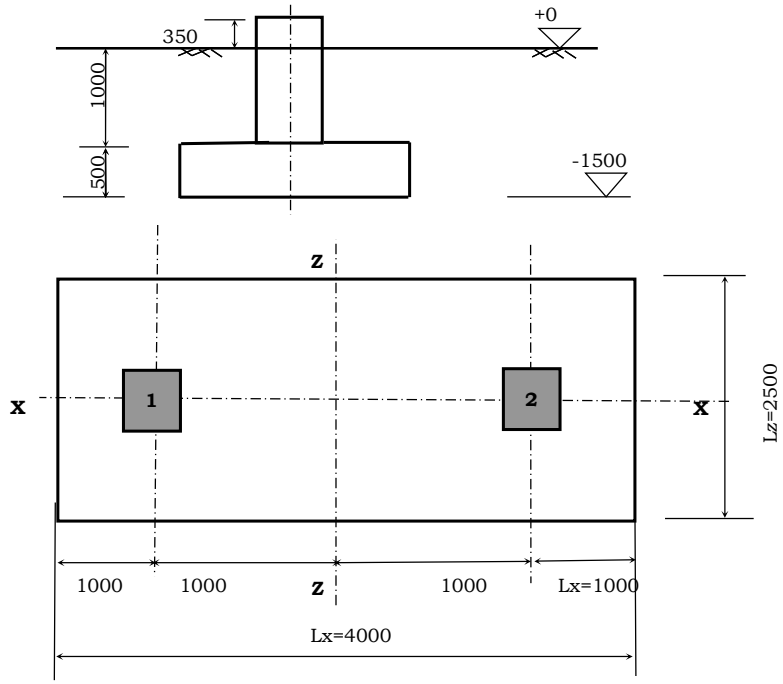
Weight of soil = ((4*2.5)-2*(0.6*0.85))*1*20 = 179.60KN

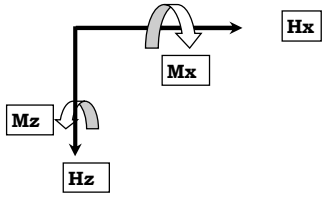
Total = 304.60KN

Submerged wt. = 304.6-4*2.5*1.5*10 = 154.60KN

Design Input:

Characteristic strength of concrete, f '$_c$ =	35 MPa
Yield stress of steel reinforcement, f_y =	420 MPa
Clear cover for main reinforcement bars =	50mm





Hx=Force along X-X direction

Hz=Force along Z-Z direction

Mx=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F5:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	68.76	0
Min Px	23	201	0	68.76	0
Max Py	489	209	0	125.90	0
Min Py	489	235	0	-51.98	0
Max Pz	23	201	0	68.76	0
Min Pz	23	201	0	68.76	0

Gross Maximum Base pressure, p_{max} = 125.9 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

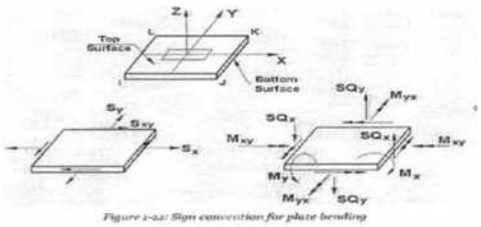
Gross Safe Bearing Capacity = 480 KN/m^2
> 125.9 KN/m^2
Hence OK

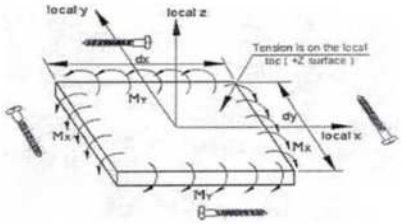
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DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F5:





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F5_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	59	0.607	0.07	0	0	0	-87.7	-64.806	7.881	-95.58	-72.687	111 0.9XD + 1.6XWL(-Z)
Min Qx	111	-0.607	0.07	0	0	0	-87.42	-64.742	0	-87.419	-64.742	109 0.9XD + 1.6XWL(+Z)
Max Qy	58	0.435	0.381	0	0	0	-77.24	-44.472	16.472	-93.713	-60.944	111 0.9XD + 1.6XWL(-Z)
Min Qy	61	0.435	-0.381	0	0	0	-77.23	-44.473	-16.516	-93.747	-60.989	111 0.9XD + 1.6XWL(-Z)
Max Sx	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Min Sx	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Max Sy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Min Sy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Max Sxy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Min Sxy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Max Mx	59	-0.527	-0.059	0	0	0	76.415	48.162	-6.793	83.208	54.955	109 0.9XD + 1.6XWL(+Z)
Min Mx	59	0.607	0.07	0	0	0	-87.7	-64.806	7.881	-95.58	-72.687	111 0.9XD + 1.6XWL(-Z)
Max My	60	-0.527	0.059	0	0	0	76.408	48.163	6.874	83.282	55.037	109 0.9XD + 1.6XWL(+Z)
Min My	60	0.607	-0.07	0	0	0	-87.7	-64.806	-7.926	-95.622	-72.732	111 0.9XD + 1.6XWL(-Z)
Max Mxy	65	-0.178	-0.047	0	0	0	-16.71	-29.39	25.837	-42.545	-55.227	111 0.9XD + 1.6XWL(-Z)
Min Mxy	62	-0.178	0.047	0	0	0	-16.71	-29.395	-25.863	-42.577	-55.258	111 0.9XD + 1.6XWL(-Z)

PLATE CENTER STRESS : FDN_F5_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	147	0.345	0.014	0	0	0	-12.92	-15.048	7.738	-20.655	-22.786	111 0.9XD + 1.6XWL(-Z)
Min Qx	171	-0.345	0.014	0	0	0	-12.68	-15.004	0	-12.677	-15.004	109 0.9XD + 1.6XWL(+Z)
Max Qy	150	0.062	0.047	0	0	0	-23.15	-11.067	6.534	-29.679	-17.601	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qy	176	-0.061	-0.046	0	0	0	-23.2	-10.886	6.532	-29.729	-17.418	101 1.2XD + 1.2XTL1 + 1.6XL
Max Sx	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Min Sx	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Max Sy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Min Sy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Max Sxy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Min Sxy	475	-0.002	-0.002	0	0	0	-0.056	-1.655	0.328	-0.384	-1.983	100 1.4XD
Max Mx	147	-0.335	-0.007	0	0	0	9.077	2.964	-7.511	16.588	10.475	109 0.9XD + 1.6XWL(+Z)
Min Mx	468	0.037	-0.032	0	0	0	-23.45	-3.662	-5.546	-28.992	-9.208	102 1.2XD + 1.2XTL1 + 1.6XL1
Max My	227	-0.038	-0.006	0	0	0	0.042	23.11	-2.098	2.14	25.208	111 0.9XD + 1.6XWL(-Z)
Min My	227	0.049	0.01	0	0	0	-0.043	-32.326	2.652	-2.695	-34.978	109 0.9XD + 1.6XWL(+Z)
Max Mxy	176	-0.223	-0.038	0	0	0	-12.37	-6.288	24.642	-37.013	-30.93	109 0.9XD + 1.6XWL(+Z)
Min Mxy	152	0.223	-0.038	0	0	0	-12.59	-6.333	-24.62	-37.207	-30.953	111 0.9XD + 1.6XWL(-Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

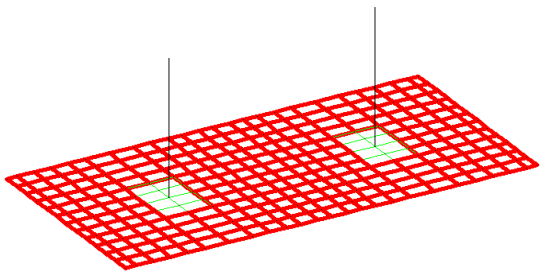
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	83.21	top face	50	442	83.208	0.57	0.0014	603.4	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
111 0.9X	-95.58	bottom face	50	442	95.58	0.65	0.0016	694.2		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
111 0.9X	-42.55	top face	50	442	42.545	0.29	0.0007	307.1		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
111 0.9X	-42.58	top face	50	442	42.577	0.29	0.0007	307.3		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.345	1	172.5	20.655	0.002	0.442	481.502	435.484	758.323	435.48	1621	> 172.5	ok	326.6	φVc>Vu	not required
	Min Qx	0.345	1	172.5	12.677			521.224			435.48	1621	> 172.5	ok	326.6	φVc>Vu	not required

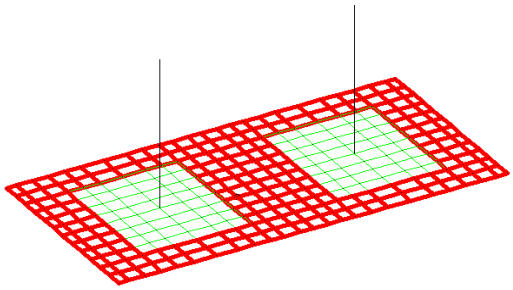
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	55.04	top face	50	442	55.037	0.38	0.0009	397.8	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
111 0.9X	-72.73	bottom face	50	442	72.732	0.50	0.0012	526.8		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
111 0.9X	-55.23	top face	50	442	55.227	0.38	0.0009	399.2		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
111 0.9X	-55.26	top face	50	442	55.258	0.38	0.0009	399.4		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	102 1.2X	0.047	1	23.5	17.601	0.002	0.442	428.476	435.484	fc'758.3	428.48	451	> 23.5	ok	321.4	φVc>Vu	not required
	101 1.2X	0.046	1	23	17.418			428.365			428.36	451	> 23	ok	321.3	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F5:

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	3.7	0.0	0.0	8.0	0.0	5.2	1.0	0.0	-30.1	0.0
202 D + L1 + TL1	5.2	0.0	0.0	30.0	0.0	3.7	0.9	0.0	-8.2	0.0
203 D + L + TL2	3.7	0.0	0.0	-12.8	0.0	5.1	1.0	0.0	-9.3	0.0
204 D + L1 + TL2	5.1	0.0	0.0	9.2	0.0	3.7	0.9	0.0	12.6	0.0
205 D + 0.75XL + 0.75XTL1	3.7	0.0	0.0	6.9	0.0	4.8	0.7	0.0	-23.5	0.0
206 D + 0.75XL1 + 0.75XTL1	4.8	0.0	0.0	23.4	0.0	3.7	0.7	0.0	-7.0	0.0
207 D + 0.75XL + 0.75XTL2	3.7	0.0	0.0	-8.7	0.0	4.8	0.7	0.0	-7.9	0.0
208 D + 0.75XL1 + 0.75XTL2	4.8	0.0	0.0	7.8	0.0	3.7	0.7	0.0	8.6	0.0
209 D + WL(+Z)	0.1	0.0	0.0	-32.4	0.0	7.1	0.2	0.0	-39.2	0.0
210 D + WL(+X)	3.7	0.0	0.0	3.8	0.0	3.7	0.5	0.0	-3.8	0.0
211 D + WL(-Z)	7.1	0.0	0.0	39.3	0.0	0.1	-0.1	0.0	32.1	0.0
212 D + WL(-X)	3.6	0.0	0.0	3.2	0.0	3.6	-0.4	0.0	-3.2	0.0
213 D + 0.7XEL(+Z)	3.6	0.0	0.0	3.2	0.0	3.6	0.1	0.0	-3.8	0.0
214 D + 0.7XEL(+X)	3.6	-0.3	0.0	3.5	0.0	3.6	-0.2	0.0	-3.5	0.0
215 D + 0.7XEL(-Z)	3.6	0.0	0.0	3.7	0.0	3.6	0.1	0.0	-3.3	0.0
216 D + 0.7XEL(-X)	3.6	0.3	0.0	3.5	0.0	3.6	0.3	0.0	-3.5	0.0
217 D + 0.75XL + 0.75XWL(+Z)	1.1	0.0	0.0	-27.8	0.0	7.4	0.8	0.0	-42.4	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	2.1	0.0	0.0	-11.3	0.0	6.3	0.8	0.0	-26.0	0.0
219 D + 0.75XL + 0.75XWL(-Z)	6.3	0.0	0.0	25.9	0.0	2.1	0.6	0.0	11.1	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	7.4	0.0	0.0	42.4	0.0	1.1	0.6	0.0	27.5	0.0
221 D + 0.75XL + 0.75XWL(+X)	3.7	0.0	0.0	-0.7	0.0	4.8	1.1	0.0	-15.9	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	4.8	0.0	0.0	15.8	0.0	3.7	1.1	0.0	0.5	0.0
223 D + 0.75XL + 0.75XWL(-X)	3.7	0.0	0.0	-1.2	0.0	4.7	0.4	0.0	-15.4	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	4.7	0.0	0.0	15.3	0.0	3.6	0.3	0.0	1.0	0.0
225 D + 0.75XL + 0.525XEL(+Z)	3.7	0.0	0.0	-1.1	0.0	4.8	0.7	0.0	-15.9	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	4.8	0.0	0.0	15.4	0.0	3.7	0.7	0.0	0.6	0.0
227 D + 0.75XL + 0.525XEL(+X)	3.7	-0.2	0.0	-0.9	0.0	4.8	0.5	0.0	-15.7	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	4.8	-0.2	0.0	15.6	0.0	3.7	0.5	0.0	0.8	0.0
229 D + 0.75XL + 0.525XEL(-Z)	3.7	0.0	0.0	-0.7	0.0	4.8	0.7	0.0	-15.5	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	4.8	0.0	0.0	15.8	0.0	3.7	0.7	0.0	1.0	0.0
231 D + 0.75XL + 0.525XEL(-X)	3.7	0.2	0.0	-0.9	0.0	4.8	0.9	0.0	-15.7	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	4.8	0.2	0.0	15.6	0.0	3.7	0.9	0.0	0.8	0.0
233 0.6XD + WL(+Z)	-1.3	0.0	0.0	-33.7	0.0	5.7	0.2	0.0	-37.8	0.0
234 0.6XD + WL(+X)	2.2	0.0	0.0	2.4	0.0	2.2	0.5	0.0	-2.4	0.0
235 0.6XD + WL(-Z)	5.7	0.0	0.0	37.9	0.0	-1.3	-0.1	0.0	33.6	0.0
236 0.6XD + WL(-X)	2.1	0.0	0.0	1.8	0.0	2.1	-0.4	0.0	-1.8	0.0
237 0.6XD + 0.7XEL(+Z)	2.2	0.0	0.0	1.8	0.0	2.2	0.0	0.0	-2.4	0.0
238 0.6XD + 0.7XEL(+X)	2.2	-0.3	0.0	2.1	0.0	2.2	-0.2	0.0	-2.1	0.0
239 0.6XD + 0.7XEL(-Z)	2.2	0.0	0.0	2.3	0.0	2.2	0.0	0.0	-1.9	0.0
240 0.6XD + 0.7XEL(-X)	2.2	0.3	0.0	2.1	0.0	2.2	0.3	0.0	-2.1	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + L + TL1	313.5	0.96	1.4	-22.1	0.0	437.5	NA	130.3	5.7
202 D + L1 + TL1	313.5	0.92	-1.4	21.8	0.0	432.7	NA	137.0	5.7
203 D + L + TL2	313.4	0.96	1.4	-22.1	0.0	437.2	NA	130.3	5.7
204 D + L1 + TL2	313.4	0.92	-1.5	21.8	0.0	432.3	NA	137.0	5.7
205 D + 0.75XL + 0.75XTL1	313.1	0.74	1.1	-16.6	0.0	583.0	NA	169.9	7.5
206 D + 0.75XL1 + 0.75XTL1	313.1	0.70	-1.1	16.4	0.0	575.5	NA	178.6	7.7
207 D + 0.75XL + 0.75XTL2	313.0	0.74	1.1	-16.6	0.0	582.4	NA	169.9	7.5
208 D + 0.75XL1 + 0.75XTL2	313.0	0.70	-1.1	16.4	0.0	576.0	NA	178.6	7.7
209 D + WL(+Z)	311.8	0.17	7.0	-71.5	0.0	89.1	NA	716.8	1.7
210 D + WL(+X)	311.9	0.53	0.0	0.0	0.0	311920.0	NA	235.4	3780.8
211 D + WL(-Z)	311.8	-0.06	-7.0	71.5	0.0	89.1	NA	2227.3	1.7
212 D + WL(-X)	311.7	-0.41	0.0	0.0	0.0	207812.7	NA	302.6	4299.6
213 D + 0.7XEL(+Z)	311.8	0.06	0.0	-0.5	0.0	20788.0	NA	2114.0	227.6
214 D + 0.7XEL(+X)	311.8	-0.46	0.0	0.0	0.0	NA	NA	272.3	4023.5
215 D + 0.7XEL(-Z)	311.8	0.06	0.0	0.5	0.0	20788.0	NA	2114.0	256.6
216 D + 0.7XEL(-X)	311.8	0.58	0.0	0.0	0.0	NA	NA	216.5	4023.5
217 D + 0.75XL + 0.75XWL(+Z)	313.1	0.82	6.3	-70.2	0.0	99.0	NA	152.2	1.8
218 D + 0.75XL1 + 0.75XWL(+Z)	313.1	0.79	4.2	-37.3	0.0	150.4	NA	158.9	3.4
219 D + 0.75XL + 0.75XWL(-Z)	313.1	0.65	-4.2	37.0	0.0	149.9	NA	192.6	3.4
220 D + 0.75XL1 + 0.75XWL(-Z)	313.1	0.62	-6.3	70.0	0.0	98.8	NA	203.6	1.8
221 D + 0.75XL + 0.75XWL(+X)	313.1	1.09	1.1	-16.6	0.0	582.0	NA	114.9	7.5
222 D + 0.75XL1 + 0.75XWL(+X)	313.1	1.06	-1.1	16.3	0.0	576.7	NA	118.7	7.7
223 D + 0.75XL + 0.75XWL(-X)	313.0	0.38	1.1	-16.6	0.0	582.8	NA	326.9	7.5
224 D + 0.75XL1 + 0.75XWL(-X)	313.0	0.35	-1.1	16.4	0.0	574.8	NA	359.7	7.7
225 D + 0.75XL + 0.525XEL(+Z)	313.1	0.74	1.1	-17.0	0.0	570.7	NA	169.9	7.4
226 D + 0.75XL1 + 0.525XEL(+Z)	313.1	0.70	-1.1	16.0	0.0	587.9	NA	178.6	7.8
227 D + 0.75XL + 0.525XEL(+X)	313.1	0.35	1.1	-16.6	0.0	582.4	NA	359.8	7.5
228 D + 0.75XL1 + 0.525XEL(+X)	313.1	0.31	-1.1	16.4	0.0	576.0	NA	400.1	7.7
229 D + 0.75XL + 0.525XEL(-Z)	313.1	0.74	1.1	-16.2	0.0	594.6	NA	169.9	7.7
230 D + 0.75XL1 + 0.525XEL(-Z)	313.1	0.70	-1.1	16.7	0.0	564.1	NA	178.6	7.5
231 D + 0.75XL + 0.525XEL(-X)	313.1	1.12	1.1	-16.6	0.0	582.4	NA	111.4	7.5
232 D + 0.75XL1 + 0.525XEL(-X)	313.1	1.09	-1.1	16.4	0.0	576.0	NA	115.0	7.7
233 0.6XD + WL(+Z)	308.9	0.15	7.0	-71.5	0.0	88.3	NA	823.8	1.7
234 0.6XD + WL(+X)	309.0	0.51	0.0	0.0	0.0	309032.0	NA	244.3	5886.3
235 0.6XD + WL(-Z)	308.9	-0.08	-7.0	71.5	0.0	88.3	NA	1544.7	1.7
236 0.6XD + WL(-X)	308.8	-0.44	0.0	0.0	0.0	308832.0	NA	283.3	7266.6
237 0.6XD + 0.7XEL(+Z)	308.9	0.04	0.0	-0.5	0.0	20595.5	NA	3530.7	230.5
238 0.6XD + 0.7XEL(+X)	308.9	-0.48	0.0	0.0	0.0	NA	NA	256.4	6503.8
239 0.6XD + 0.7XEL(-Z)	308.9	0.04	0.0	0.5	0.0	20595.5	NA	3530.7	248.1
240 0.6XD + 0.7XEL(-X)	308.9	0.55	0.0	0.0	0.0	NA	NA	223.5	6503.8
					Min.=	88.3	NA	111.4	1.7

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DESIGN CALCULATION OF PEDESTAL OF RCO-2/ F5:

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 850 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.77

Y_b = (b-d')/b = 0.84

Unsupported length of column parallel to X-X axis, l_{ux} = 1.85 m kl_{ux}/r_x = 7.255 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.85 m kl_{uz}/r_z = 10.28

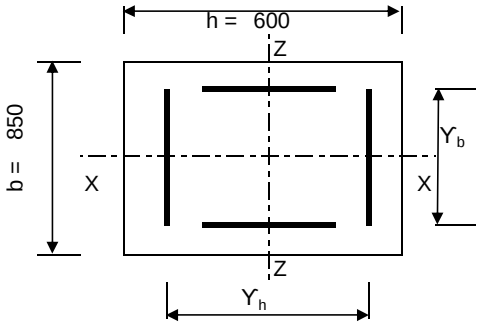
Radius of gyration r_x = 255 mm

Radius of gyration r_z = 180 mm

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 13708.05 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
109	636.2	141.5	0.6	979	217.7	0.9	0.055	0.075	1137.94	0.19	0.075	803.3	0.00	1.609	0.07	safe
215	185.2	360.3	32.0	285	554.3	49.3	0.016	0.060	910.35	0.61	0.060	642.6	0.08		0.467	safe
112	55.2	6.2	1.6	85	9.6	2.5	0.005	0.075	1137.94	0.01	0.075	803.3	0.00		0.001	safe
101	292.1	100.2	3.6	449	154.2	5.5	0.025	0.070	1062.08	0.15	0.070	749.7	0.01		0.046	safe
111	636.0	142.1	0.0	979	218.6	0.0	0.055	0.070	1062.08	0.21	0.070	749.7	0.00		0.079	safe
109	636.2	141.5	0.6	979	217.7	0.9	0.055	0.060	910.35	0.24	0.060	642.6	0.00		0.101	safe
100	93.8	11.6	0.2	144	17.8	0.3	0.008	0.060	910.35	0.02	0.060	642.6	0.00		0.002	safe
100	93.8	11.6	0.2	144	17.8	0.3	0.008	0.060	910.35	0.02	0.060	642.6	0.00		0.002	safe
109	636.2	141.5	0.6	979	217.7	0.9	0.055	0.060	910.35	0.24	0.060	642.6	0.00		0.101	safe
111	636.0	142.1	0.0	979	218.6	0.0	0.055	0.060	910.35	0.24	0.060	642.6	0.00		0.101	safe
112	55.2	6.2	1.6	85	9.6	2.5	0.005	0.075	1137.94	0.01	0.075	803.3	0.00		0.001	safe
101	292.1	100.2	3.6	449	154.2	5.5	0.025	0.070	1062.08	0.15	0.070	749.7	0.01		0.046	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5100	4	20	+	12	20	5029	0.99	10	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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- **DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F6:**

Size of Columns at node	Along-X	Along-Z
Col.1	0.60m	0.85m
Col.2	0.60m	0.85m
Size of fdn. raft:	4.00m	1.50m
Thickness of raft:	0.50 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.35 m	
U/S level of fdn:	-1.50 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	20 kN/m^3	

Calculation of Weight of Foundation:

Weight of raft = (4*1.5*0.5*25) = 75.00KN

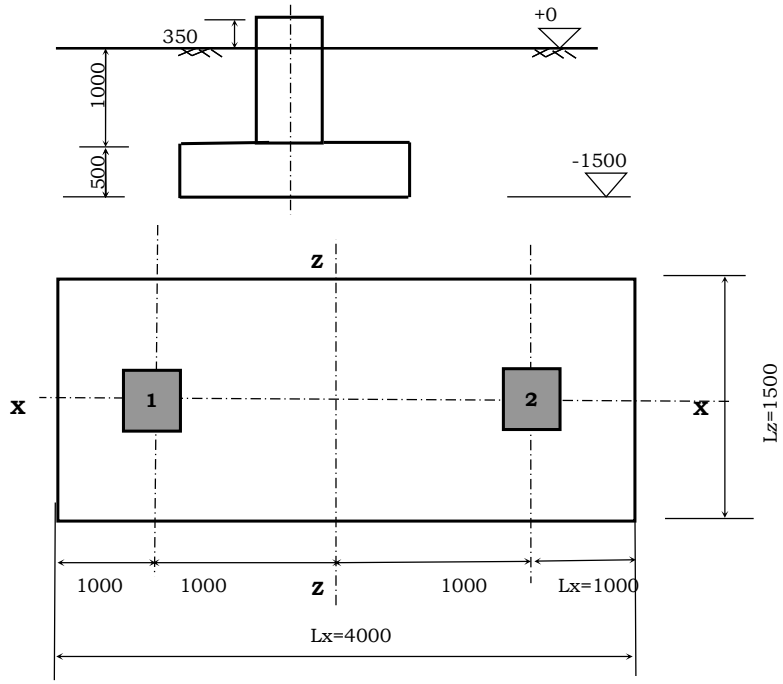
Weight of soil = ((4*1.5)-2*(0.6*0.85))*1*20 = 99.60KN

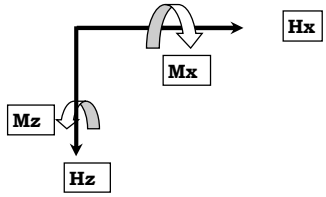
Total = 174.60KN

Submerged wt. = 174.6-4*1.5*1.5*10 = 84.60KN

Design Input:

Characteristic strength of concrete, f'_c =	35 MPa
Yield stress of steel reinforcement, f_y =	420 MPa
Clear cover for main reinforcement bars =	50mm





Hx=Force along X-X direction

Hz=Force along Z-Z direction

Mx=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F6:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	75.26	0
Min Px	23	201	0	75.26	0
Max Py	489	209	0	131.98	0
Min Py	502	233	0	-49.69	0
Max Pz	23	201	0	75.26	0
Min Pz	23	201	0	75.26	0

Gross Maximum Base pressure, p_{max} = 132.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

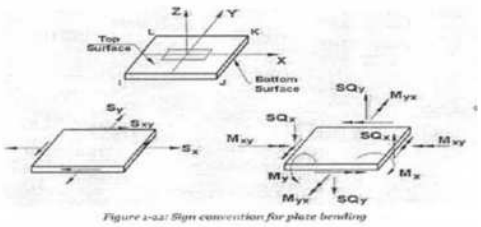
Gross Safe Bearing Capacity = 480 KN/m^2
> 132.0 KN/m^2
Hence OK

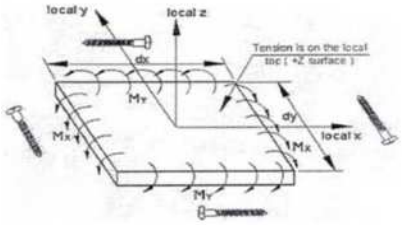
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DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F6:





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F5_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	60	0.393	-0.032	0	0	0	-71.96	-30.426	-3.524	-75.479	-33.95	111 0.9XD + 1.6XWL(-Z)
Min Qx	112	-0.393	-0.032	0	0	0	-71.81	-30.469	0	-71.814	-30.469	109 0.9XD + 1.6XWL(+Z)
Max Qy	58	0.291	0.207	0	0	0	-66.81	-18.725	6.607	-73.42	-25.332	111 0.9XD + 1.6XWL(-Z)
Min Qy	61	0.291	-0.207	0	0	0	-66.81	-18.725	-6.607	-73.42	-25.332	111 0.9XD + 1.6XWL(-Z)
Max Sx	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Min Sx	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Max Sy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Min Sy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Max Sxy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Min Sxy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Max Mx	60	-0.324	0.027	0	0	0	59.744	22.729	2.593	62.337	25.322	109 0.9XD + 1.6XWL(+Z)
Min Mx	136	0.016	-0.096	0	0	0	-73.36	-2.302	1.705	-75.067	-4.007	109 0.9XD + 1.6XWL(+Z)
Max My	60	-0.324	0.027	0	0	0	59.744	22.729	2.593	62.337	25.322	109 0.9XD + 1.6XWL(+Z)
Min My	112	-0.393	-0.032	0	0	0	-71.81	-30.469	3.54	-75.354	-34.009	109 0.9XD + 1.6XWL(+Z)
Max Mxy	106	0.155	0.003	0	0	0	-21.21	-6.55	14.71	-35.915	-21.26	109 0.9XD + 1.6XWL(+Z)
Min Mxy	109	0.155	-0.003	0	0	0	-21.21	-6.55	-14.71	-35.915	-21.26	109 0.9XD + 1.6XWL(+Z)

PLATE CENTER STRESS : FDN_F5_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	148	0.24	-0.006	0	0	0	-12.78	-5.835	-2.454	-15.237	-8.289	111 0.9XD + 1.6XWL(-Z)
Min Qx	172	-0.24	-0.006	0	0	0	-12.58	-5.822	0	-12.578	-5.822	109 0.9XD + 1.6XWL(+Z)
Max Qy	170	-0.069	0.019	0	0	0	-31.02	-5.028	-1.88	-32.896	-6.908	101 1.2XD + 1.2XTL1 + 1.6XL
Min Qy	173	-0.069	-0.019	0	0	0	-31.02	-5.028	1.88	-32.896	-6.908	101 1.2XD + 1.2XTL1 + 1.6XL
Max Sx	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Min Sx	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Max Sy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Min Sy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Max Sxy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Min Sxy	395	-0.002	-0.001	0	0	0	-0.08	-0.635	0.103	-0.183	-0.738	100 1.4XD
Max Mx	148	-0.054	-0.001	0	0	0	9.494	-2.116	0.437	9.931	-2.553	103 1.2XD + 1.2XTL2 + 1.6XL
Min Mx	176	-0.05	-0.012	0	0	0	-31.47	-1.927	1.306	-32.78	-3.233	101 1.2XD + 1.2XTL1 + 1.6XL
Max My	212	0.037	0.001	0	0	0	1.097	4.956	-0.82	1.917	5.776	109 0.9XD + 1.6XWL(+Z)
Min My	227	0.05	0.002	0	0	0	-1.417	-7.715	1.14	-2.557	-8.855	109 0.9XD + 1.6XWL(+Z)
Max Mxy	173	-0.22	-0.015	0	0	0	-12.91	-3.791	5.483	-18.395	-9.274	109 0.9XD + 1.6XWL(+Z)
Min Mxy	170	-0.22	0.015	0	0	0	-12.91	-3.791	-5.483	-18.395	-9.274	109 0.9XD + 1.6XWL(+Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

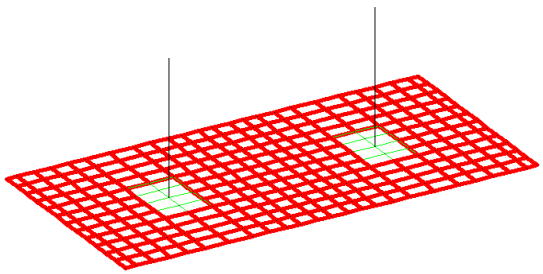
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	62.34	top face	50	442	62.337	0.43	0.0010	451.0	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-75.07	bottom face	50	442	75.067	0.51	0.0012	543.9		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-35.92	top face	50	442	35.915	0.25	0.0006	259.0		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-35.92	top face	50	442	35.915	0.25	0.0006	259.0		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	Max Qx	0.24	1	120	15.237	0.002	0.442	477.905	435.484	758.323	435.48	1621	> 120	ok	326.6	φVc>Vu	not required
	Min Qx	0.24	1	120	12.578			490.488			435.48	1621	> 120	ok	326.6	φVc>Vu	not required

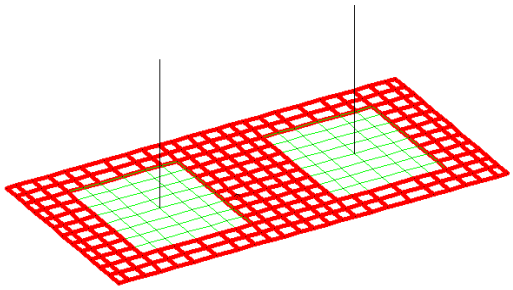
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109 0.9X	25.32	top face	50	442	25.322	0.17	0.0004	182.4	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-34.01	bottom face	50	442	34.009	0.23	0.0006	245.2		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-21.26	top face	50	442	21.26	0.15	0.0003	153.1		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
109 0.9X	-21.26	top face	50	442	21.26	0.15	0.0003	153.1		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema	φV_c	Remar	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		rks	KN	ks	
Load condition	101 1.2X	0.019	1	9.5	6.908	0.002	0.442	428.778	435.484	fc'758.3	428.78	451	> 9.5	ok	321.6	φVc>Vu	not required
	101 1.2X	0.019	1	9.5	6.908			428.778			428.78	451	> 9.5	ok	321.6	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING MKD. RCO-2/ F6:

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + L + TL1	4.1	0.0	0.0	9.1	0.0	5.6	0.0	0.0	-23.4	0.0
202 D + L1 + TL1	5.6	0.0	0.0	23.3	0.0	4.2	0.0	0.0	-9.2	0.0
203 D + L + TL2	4.1	0.0	0.0	-11.7	0.0	5.6	0.0	0.0	-2.6	0.0
204 D + L1 + TL2	5.6	0.0	0.0	2.5	0.0	4.1	0.0	0.0	11.6	0.0
205 D + 0.75XL + 0.75XTL1	4.1	0.0	0.0	7.5	0.0	5.2	0.0	0.0	-18.2	0.0
206 D + 0.75XL1 + 0.75XTL1	5.2	0.0	0.0	18.1	0.0	4.1	0.0	0.0	-7.6	0.0
207 D + 0.75XL + 0.75XTL2	4.1	0.0	0.0	-8.1	0.0	5.2	0.0	0.0	-2.6	0.0
208 D + 0.75XL1 + 0.75XTL2	5.2	0.0	0.0	2.5	0.0	4.1	0.0	0.0	8.0	0.0
209 D + WL(+Z)	0.6	0.0	0.0	-20.1	0.0	7.5	0.0	0.0	-25.2	0.0
210 D + WL(+X)	4.1	0.0	0.0	2.9	0.0	4.1	0.0	0.0	-2.9	0.0
211 D + WL(-Z)	7.4	0.0	0.0	25.2	0.0	0.6	0.0	0.0	20.0	0.0
212 D + WL(-X)	3.9	0.0	0.0	2.3	0.0	4.0	0.0	0.0	-2.3	0.0
213 D + 0.7XEL(+Z)	4.0	0.0	0.0	2.3	0.0	4.1	0.0	0.0	-2.8	0.0
214 D + 0.7XEL(+X)	4.0	-0.3	0.0	2.6	0.0	4.0	-0.3	0.0	-2.6	0.0
215 D + 0.7XEL(-Z)	4.0	0.0	0.0	2.8	0.0	4.0	0.0	0.0	-2.3	0.0
216 D + 0.7XEL(-X)	4.0	0.3	0.0	2.6	0.0	4.0	0.3	0.0	-2.6	0.0
217 D + 0.75XL + 0.75XWL(+Z)	1.5	0.0	0.0	-17.3	0.0	7.8	0.0	0.0	-27.3	0.0
218 D + 0.75XL1 + 0.75XWL(+Z)	2.6	0.0	0.0	-6.7	0.0	6.7	0.0	0.0	-16.7	0.0
219 D + 0.75XL + 0.75XWL(-Z)	6.6	0.0	0.0	16.7	0.0	2.7	0.0	0.0	6.6	0.0
220 D + 0.75XL1 + 0.75XWL(-Z)	7.7	0.0	0.0	27.3	0.0	1.6	0.0	0.0	17.2	0.0
221 D + 0.75XL + 0.75XWL(+X)	4.1	0.0	0.0	-0.1	0.0	5.3	0.0	0.0	-10.6	0.0
222 D + 0.75XL1 + 0.75XWL(+X)	5.2	0.0	0.0	10.5	0.0	4.2	0.0	0.0	0.0	0.0
223 D + 0.75XL + 0.75XWL(-X)	4.0	0.0	0.0	-0.6	0.0	5.2	0.0	0.0	-10.1	0.0
224 D + 0.75XL1 + 0.75XWL(-X)	5.1	0.0	0.0	10.1	0.0	4.1	0.0	0.0	0.5	0.0
225 D + 0.75XL + 0.525XEL(+Z)	4.1	0.0	0.0	-0.5	0.0	5.2	0.0	0.0	-10.6	0.0
226 D + 0.75XL1 + 0.525XEL(+Z)	5.2	0.0	0.0	10.1	0.0	4.1	0.0	0.0	0.0	0.0
227 D + 0.75XL + 0.525XEL(+X)	4.1	-0.2	0.0	-0.3	0.0	5.2	-0.2	0.0	-10.4	0.0
228 D + 0.75XL1 + 0.525XEL(+X)	5.2	-0.2	0.0	10.3	0.0	4.1	-0.2	0.0	0.2	0.0
229 D + 0.75XL + 0.525XEL(-Z)	4.1	0.0	0.0	-0.2	0.0	5.2	0.0	0.0	-10.2	0.0
230 D + 0.75XL1 + 0.525XEL(-Z)	5.2	0.0	0.0	10.5	0.0	4.1	0.0	0.0	0.4	0.0
231 D + 0.75XL + 0.525XEL(-X)	4.1	0.2	0.0	-0.3	0.0	5.2	0.2	0.0	-10.4	0.0
232 D + 0.75XL1 + 0.525XEL(-X)	5.2	0.2	0.0	10.3	0.0	4.1	0.2	0.0	0.2	0.0
233 0.6XD + WL(+Z)	-1.0	0.0	0.0	-21.1	0.0	5.8	0.0	0.0	-24.1	0.0
234 0.6XD + WL(+X)	2.5	0.0	0.0	1.9	0.0	2.5	0.0	0.0	-1.9	0.0
235 0.6XD + WL(-Z)	5.8	0.0	0.0	24.2	0.0	-1.0	0.0	0.0	21.0	0.0
236 0.6XD + WL(-X)	2.3	0.0	0.0	1.2	0.0	2.3	0.0	0.0	-1.2	0.0
237 0.6XD + 0.7XEL(+Z)	2.4	0.0	0.0	1.3	0.0	2.4	0.0	0.0	-1.8	0.0
238 0.6XD + 0.7XEL(+X)	2.4	-0.3	0.0	1.5	0.0	2.4	-0.3	0.0	-1.6	0.0
239 0.6XD + 0.7XEL(-Z)	2.4	0.0	0.0	1.8	0.0	2.4	0.0	0.0	-1.3	0.0
240 0.6XD + 0.7XEL(-X)	2.4	0.3	0.0	1.5	0.0	2.4	0.3	0.0	-1.6	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + L + TL1	314.4	0.00	1.5	-14.3	0.0	410.7	NA	NA	8.8
202 D + L1 + TL1	314.4	0.00	-1.4	14.1	0.0	441.2	NA	NA	8.9
203 D + L + TL2	314.3	0.00	1.5	-14.3	0.0	410.3	NA	NA	8.8
204 D + L1 + TL2	314.3	0.00	-1.4	14.1	0.0	440.8	NA	NA	8.9
205 D + 0.75XL + 0.75XTL1	313.9	0.00	1.2	-10.7	0.0	544.1	NA	NA	11.7
206 D + 0.75XL1 + 0.75XTL1	313.9	0.00	-1.1	10.5	0.0	590.7	NA	NA	11.9
207 D + 0.75XL + 0.75XTL2	313.9	0.00	1.2	-10.7	0.0	544.0	NA	NA	11.7
208 D + 0.75XL1 + 0.75XTL2	313.9	0.00	-1.1	10.5	0.0	590.5	NA	NA	11.9
209 D + WL(+Z)	312.7	0.00	6.8	-45.2	0.0	91.3	NA	NA	2.8
210 D + WL(+X)	312.8	0.00	0.0	0.0	0.0	23171.9	NA	NA	7820.5
211 D + WL(-Z)	312.7	0.00	-6.8	45.2	0.0	91.9	NA	NA	2.8
212 D + WL(-X)	312.5	0.00	0.0	0.0	0.0	34722.0	NA	NA	7812.5
213 D + 0.7XEL(+Z)	312.7	0.00	0.1	-0.5	0.0	8933.1	NA	NA	234.6
214 D + 0.7XEL(+X)	312.7	-0.52	0.0	0.0	0.0	28423.6	NA	241.9	7816.5
215 D + 0.7XEL(-Z)	312.7	0.00	0.0	0.5	0.0	25012.7	NA	NA	249.6
216 D + 0.7XEL(-X)	312.7	0.52	0.0	0.0	0.0	28423.6	NA	241.9	7816.5
217 D + 0.75XL + 0.75XWL(+Z)	313.9	0.00	6.3	-44.6	0.0	100.1	NA	NA	2.8
218 D + 0.75XL1 + 0.75XWL(+Z)	313.9	0.00	4.1	-23.4	0.0	154.8	NA	NA	5.4
219 D + 0.75XL + 0.75XWL(-Z)	313.9	0.00	-4.0	23.2	0.0	158.4	NA	NA	5.4
220 D + 0.75XL1 + 0.75XWL(-Z)	313.9	0.00	-6.2	44.5	0.0	101.6	NA	NA	2.8
221 D + 0.75XL + 0.75XWL(+X)	314.0	0.00	1.2	-10.7	0.0	542.8	NA	NA	11.7
222 D + 0.75XL1 + 0.75XWL(+X)	314.0	0.00	-1.1	10.5	0.0	592.5	NA	NA	11.9
223 D + 0.75XL + 0.75XWL(-X)	313.8	0.00	1.2	-10.7	0.0	545.2	NA	NA	11.7
224 D + 0.75XL1 + 0.75XWL(-X)	313.8	0.00	-1.1	10.5	0.0	588.2	NA	NA	11.9
225 D + 0.75XL + 0.525XEL(+Z)	313.9	0.00	1.2	-11.1	0.0	528.0	NA	NA	11.3
226 D + 0.75XL1 + 0.525XEL(+Z)	313.9	0.00	-1.0	10.2	0.0	611.3	NA	NA	12.4
227 D + 0.75XL + 0.525XEL(+X)	313.9	-0.39	1.2	-10.7	0.0	543.6	NA	323.6	11.7
228 D + 0.75XL1 + 0.525XEL(+X)	313.9	-0.39	-1.1	10.5	0.0	590.6	NA	323.6	11.9
229 D + 0.75XL + 0.525XEL(-Z)	313.9	0.00	1.1	-10.3	0.0	561.0	NA	NA	12.2
230 D + 0.75XL1 + 0.525XEL(-Z)	313.9	0.00	-1.1	10.9	0.0	571.3	NA	NA	11.5
231 D + 0.75XL + 0.525XEL(-X)	313.9	0.39	1.2	-10.7	0.0	543.6	NA	323.6	11.7
232 D + 0.75XL1 + 0.525XEL(-X)	313.9	0.39	-1.1	10.5	0.0	590.6	NA	323.6	11.9
233 0.6XD + WL(+Z)	309.4	0.00	6.8	-45.2	0.0	90.5	NA	NA	2.7
234 0.6XD + WL(+X)	309.6	0.00	0.0	0.0	0.0	34399.8	NA	NA	12383.9
235 0.6XD + WL(-Z)	309.4	0.00	-6.8	45.2	0.0	90.9	NA	NA	2.7
236 0.6XD + WL(-X)	309.3	0.00	0.0	0.0	0.0	68727.3	NA	NA	12370.9
237 0.6XD + 0.7XEL(+Z)	309.4	0.00	0.1	-0.5	0.0	10145.4	NA	NA	234.9
238 0.6XD + 0.7XEL(+X)	309.4	-0.52	0.0	0.0	0.0	44205.1	NA	239.4	12377.4
239 0.6XD + 0.7XEL(-Z)	309.4	0.00	0.0	0.5	0.0	18202.1	NA	NA	244.1
240 0.6XD + 0.7XEL(-X)	309.4	0.52	0.0	0.0	0.0	44205.1	NA	239.4	12377.4
					Min.=	90.5	NA	239.4	2.7

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DESIGN CALCULATION OF PEDESTAL OF RCO-2/ F6:

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 850 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.77

Y_b = (b-d')/b = 0.84

Unsupported length of column parallel to X-X axis, l_{ux} = 1.85 m kl_{ux}/r_x = 7.255 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.85 m kl_{uz}/r_z = 10.28

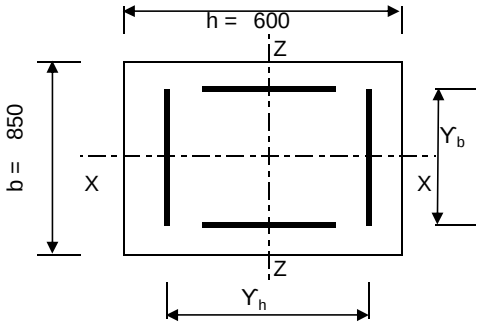
Radius of gyration r_x = 255 mm

Radius of gyration r_z = 180 mm

$P_{n,\,Max} = 0.8[0.85 f_c'(A_g - A_{st}) + f_y A_{st}]$ = 13708.05 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
109	407.8	90.3	0.0	627	138.9	0.0	0.035	0.075	1137.94	0.12	0.075	803.3	0.00	1.609	0.034	safe
215	185.2	360.3	32.0	285	554.3	49.3	0.016	0.060	910.35	0.61	0.060	642.6	0.08		0.467	safe
130	52.7	5.5	0.9	81	8.4	1.3	0.005	0.075	1137.94	0.01	0.075	803.3	0.00		0.001	safe
132	52.7	5.5	0.9	81	8.4	1.3	0.005	0.070	1062.08	0.01	0.070	749.7	0.00		0.001	safe
111	406.8	90.7	0.0	626	139.5	0.0	0.035	0.070	1062.08	0.13	0.070	749.7	0.00		0.039	safe
109	407.8	90.3	0.0	627	138.9	0.0	0.035	0.060	910.35	0.15	0.060	642.6	0.00		0.049	safe
100	81.9	8.5	0.0	126	13.1	0.0	0.007	0.060	910.35	0.01	0.060	642.6	0.00		0.002	safe
100	81.9	8.5	0.0	126	13.1	0.0	0.007	0.060	910.35	0.01	0.060	642.6	0.00		0.002	safe
109	407.8	90.3	0.0	627	138.9	0.0	0.035	0.060	910.35	0.15	0.060	642.6	0.00		0.049	safe
111	406.8	90.7	0.0	626	139.5	0.0	0.035	0.060	910.35	0.15	0.060	642.6	0.00		0.049	safe
130	52.7	5.5	0.9	81	8.4	1.3	0.005	0.075	1137.94	0.01	0.075	803.3	0.00		0.001	safe
132	52.7	5.5	0.9	81	8.4	1.3	0.005	0.070	1062.08	0.01	0.070	749.7	0.00		0.001	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5100	4	20	+	12	20	5029	0.99	10	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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REFERENCES

[1] Saudi concrete structures Code; SBC 304-2018
[2] Building Code Requirements for Structural Concrete; ACI 318-14
[3] Saudi Building Code Requirements : Structural - Loading and Forces; SBC 301-2007

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ATTACHMENT 1 STAAD OUTPUT

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*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUL 17, 2020                 *
*           Time=    10:19:53                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
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     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\RCO1 _RCO2\CIVIL\Working\F1\F1_TW1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-JULY-20
     4. *FOUNDATION F12 (TW-27)
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. SET NL 100
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 0 0.35 -2.5; 2 0 0.35 0.5; 3 3 0.35 -2.5; 4 3 0.35 0.5; 23 3 -1.5 0.5
    11. 121 0 -1.5 0.5; 122 0.3 -1.5 0.5; 123 0.3 -1.5 0.8; 124 0 -1.5 0.8
    12. 125 0 -1.5 0.2; 126 0.3 -1.5 0.2; 127 -0.3 -1.5 0.5; 128 -0.3 -1.5 0.8
    13. 129 -0.3 -1.5 0.2; 130 0.3 -1.5 1.1; 131 0 -1.5 1.1; 132 -0.3 -1.5 1.1
    14. 133 0 -1.5 -0.1; 134 0.3 -1.5 -0.1; 135 -0.3 -1.5 -0.1; 136 0.6 -1.5 0.5
    15. 137 0.6 -1.5 0.8; 138 0.6 -1.5 0.2; 139 0.6 -1.5 1.1; 140 0.6 -1.5 -0.1
    16. 141 -0.6 -1.5 -0.1; 142 -0.6 -1.5 0.2; 143 -0.6 -1.5 0.5; 144 -0.6 -1.5 0.8
    17. 145 -0.6 -1.5 1.1; 146 0.9 -1.5 -0.1; 147 0.9 -1.5 0.2; 148 0.9 -1.5 0.5
    18. 149 0.9 -1.5 0.8; 150 0.9 -1.5 1.1; 156 0 -1.5 -0.4; 157 0.3 -1.5 -0.4
    19. 158 -0.3 -1.5 -0.4; 159 0.6 -1.5 -0.4; 160 -0.6 -1.5 -0.4; 161 0.9 -1.5 -0.4
    20. 164 -0.6 -1.5 -0.7; 165 -0.3 -1.5 -0.7; 166 0 -1.5 -0.7; 167 0.3 -1.5 -0.7
    21. 168 0.6 -1.5 -0.7; 169 0.9 -1.5 -0.7; 184 3.3 -1.5 0.5; 185 3.3 -1.5 0.8
    22. 186 3 -1.5 0.8; 187 3 -1.5 0.2; 188 3.3 -1.5 0.2; 189 2.7 -1.5 0.5
    23. 190 2.7 -1.5 0.8; 191 2.7 -1.5 0.2; 192 3 -1.5 1.1; 193 2.7 -1.5 1.1
    24. 194 3.3 -1.5 1.1; 195 2.7 -1.5 -0.1; 196 3 -1.5 -0.1; 197 3.3 -1.5 -0.1
    25. 198 3.6 -1.5 -0.1; 199 3.6 -1.5 0.2; 200 3.6 -1.5 0.5; 201 3.6 -1.5 0.8
    26. 202 3.6 -1.5 1.1; 203 2.4 -1.5 -0.1; 204 2.4 -1.5 0.2; 205 2.4 -1.5 0.5
    27. 206 2.4 -1.5 0.8; 207 2.4 -1.5 1.1; 213 2.1 -1.5 -0.1; 214 2.1 -1.5 0.2
    28. 215 2.1 -1.5 0.5; 216 2.1 -1.5 0.8; 217 2.1 -1.5 1.1; 218 2.1 -1.5 -0.4
    29. 219 2.4 -1.5 -0.4; 220 2.7 -1.5 -0.4; 221 3 -1.5 -0.4; 222 3.3 -1.5 -0.4
    30. 223 3.6 -1.5 -0.4; 225 2.1 -1.5 -0.7; 226 2.4 -1.5 -0.7; 227 2.7 -1.5 -0.7
    31. 228 3 -1.5 -0.7; 229 3.3 -1.5 -0.7; 230 3.6 -1.5 -0.7; 246 1.2 -1.5 -0.1
    32. 247 1.2 -1.5 0.2; 248 1.2 -1.5 0.5; 249 1.2 -1.5 0.8; 250 1.2 -1.5 1.1
    33. 251 1.2 -1.5 -0.4; 252 1.2 -1.5 -0.7; 273 1.8 -1.5 -0.1; 274 1.8 -1.5 0.2
    34. 275 1.8 -1.5 0.5; 276 1.8 -1.5 0.8; 277 1.8 -1.5 1.1; 278 1.8 -1.5 -0.4
    35. 279 1.8 -1.5 -0.7; 527 1.5 -1.5 0.2; 528 1.5 -1.5 -0.1; 531 1.5 -1.5 0.5
    36. 533 1.5 -1.5 0.8; 535 1.5 -1.5 1.1; 537 1.5 -1.5 -0.4; 539 1.5 -1.5 -0.7
    37. 808 -0.6 -1.5 -1; 810 -0.3 -1.5 -1; 811 0 -1.5 -1; 812 0.3 -1.5 -1.
    38. 813 0.6 -1.5 -1; 814 0.9 -1.5 -1; 815 2.4 -1.5 -1; 816 2.1 -1.5 -1.
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39. 817 2.7 -1.5 -1; 818 3 -1.5 -1; 819 3.3 -1.5 -1; 820 3.6 -1.5 -1.
40. 822 1.2 -1.5 -1; 823 1.8 -1.5 -1; 845 1.5 -1.5 -1; 927 0 -1.5 -1.6
41. 928 0.3 -1.5 -1.6; 929 0.3 -1.5 -1.9; 930 0 -1.5 -1.9; 931 0.3 -1.5 -2.2
42. 932 0 -1.5 -2.2; 933 -0.3 -1.5 -1.6; 934 -0.3 -1.5 -1.9; 935 -0.3 -1.5 -2.2
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51. 975 0.9 -1.5 -3.1; 976 3 -1.5 -1.6; 977 3.3 -1.5 -1.6; 978 3.3 -1.5 -1.9
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 97. 1311 -0.8 -1.5 -3.5; 1312 4 -1.5 -3.5; 1313 -1 -1.5 -3.5; 1314 -0.3 -1.5 1.5
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102. 1333 4 -1.5 1.5; 1334 -1 -1.5 1.5
103. MEMBER INCIDENCES
104. 48 23 4; 49 121 2; 1011 988 3; 1012 939 1
105. ELEMENT INCIDENCES SHELL
106. 50 124 123 122 121; 51 121 122 126 125; 52 128 124 121 127; 53 127 121 125 129
107. 54 132 131 124 128; 55 131 130 123 124; 56 129 125 133 135; 57 125 126 134 133
108. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
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133. 257 279 225 816 823; 258 539 279 823 845; 415 247 527 528 246
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135. 427 246 528 537 251; 430 251 537 539 252; 439 252 539 845 822
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161. 902 1087 1036 1035 1085; 903 1080 1033 1037 1089; 904 1089 1037 1038 1091
162. 934 1025 1079 1080 1026; 936 1027 1083 1079 1025; 938 1028 1085 1083 1027
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164. 1014 808 810 936 951; 1015 810 811 937 936; 1016 811 812 938 937
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169. 1042 200 1153 1151 199; 1043 201 1154 1153 200; 1044 202 1155 1154 201
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183. 1114 1200 969 1202 1227; 1115 1228 1229 131 132; 1116 1229 1230 130 131
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185. 1120 1234 1235 192 193; 1121 1235 1236 194 192; 1122 1236 1237 202 194
186. 1123 1238 1234 193 207; 1124 1239 1238 207 217; 1125 1233 1240 250 150
187. 1126 1241 1239 217 277; 1127 1243 1241 277 535; 1128 1240 1243 535 250
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198. 1173 1289 1199 1200 1290; 1174 1283 1193 1198 1288; 1175 1290 1200 1227 1291
199. 1176 1292 1254 1180 1278; 1177 1202 1201 1293 1294; 1178 1201 1203 1295 1293
200. 1179 1203 1204 1296 1295; 1180 1204 1205 1297 1296; 1181 1205 1206 1298 1297
201. 1182 1208 1207 1299 1300; 1183 1207 1209 1301 1299; 1184 1209 1210 1302 1301
202. 1185 1210 1211 1303 1302; 1186 1211 1212 1304 1303; 1187 1206 1213 1305 1298
203. 1188 1214 1208 1300 1306; 1189 1216 1214 1306 1308; 1190 1213 1216 1308 1305
204. 1193 1212 1226 1310 1304; 1194 1227 1202 1294 1311; 1195 1226 1272 1312 1310
205. 1196 1291 1227 1311 1313; 1197 1314 1315 1229 1228; 1198 1315 1316 1230 1229
206. 1199 1316 1317 1231 1230; 1200 1318 1314 1228 1232; 1201 1317 1319 1233 1231

```
207. 1202 1320 1321 1235 1234; 1203 1321 1322 1236 1235; 1204 1322 1323 1237 1236
208. 1205 1324 1320 1234 1238; 1206 1325 1324 1238 1239; 1207 1319 1326 1240 1233
209. 1208 1327 1325 1239 1241; 1209 1329 1327 1241 1243; 1210 1326 1329 1243 1240
210. 1213 1323 1331 1253 1237; 1214 1332 1318 1232 1254; 1215 1331 1333 1273 1253
211. 1216 1334 1332 1254 1292
212. START GROUP DEFINITION
213. FLOOR
214. ELEMENT
215. _FDN_12_MOMENT 54 TO 69 75 TO 79 81 TO 85 102 TO 113 118 TO 126 128 TO 132 -
216. 146 TO 151 170 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 -
217. 430 439 815 TO 820 823 TO 834 840 843 844 846 TO 856 859 TO 866 871 TO 876 -
218. 879 881 TO 885 887 TO 904 934 936 938 940 942 944 1014 TO 1019 1021 1031 -
219. 1032 TO 1035 1037 1038 1041 TO 1047 1058 TO 1071 1082 TO 1102 1113 TO 1128 -
220. 1139 TO 1147 1150 TO 1165 1168 TO 1190 1193 TO 1210 1213 TO 1216
221. _FDN_12_SHEAR 81 TO 85 128 TO 132 146 TO 151 170 TO 175 178 TO 183 -
222. 243 TO 252 254 257 258 415 418 421 424 427 430 439 819 820 826 830 834 855 -
223. 856 862 866 874 887 TO 904 934 936 938 940 942 944 1014 TO 1019 1021 1031 -
224. 1032 TO 1035 1037 1038 1046 1047 1061 1064 1070 1071 1085 1088 1099 TO 1102 -
225. 1125 TO 1128 1141 TO 1147 1150 TO 1165 1168 TO 1190 1193 TO 1210 -
226. 1213 TO 1216
227. END GROUP DEFINITION
228. ELEMENT PROPERTY
229. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
230. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
231. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
232. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
233. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
234. 1193 TO 1210 1213 TO 1216 THICKNESS 0.5
235. DEFINE MATERIAL START
236. ISOTROPIC CONCRETE
237. E 2.17185E+007
238. POISSON 0.17
239. DENSITY 25
240. ALPHA 1E-005
241. DAMP 0.05
242. TYPE CONCRETE
243. STRENGTH FCU 27579
244. END DEFINE MATERIAL
245. MEMBER PROPERTY AMERICAN
246. 48 49 1011 1012 PRIS YD 0.6 ZD 0.6
247. CONSTANTS
248. MATERIAL CONCRETE ALL
249. SUPPORTS
250. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
251. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
252. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
253. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
254. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
255. 1193 TO 1210 1213 TO 1216 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
```

JOINT	INFLUENCE AREA
23	0.090000
121	0.090000
122	0.090000

123	0.090000
124	0.090000
125	0.090000
126	0.090000
127	0.090000
128	0.090000
129	0.090000
130	0.075000
131	0.075000
132	0.075000
133	0.090000
134	0.090000
135	0.090000
136	0.090000
137	0.090000
138	0.090000
139	0.075000
140	0.090000
141	0.075000
142	0.075000
143	0.075000
144	0.075000
145	0.062500
146	0.090000
147	0.090000
148	0.090000
149	0.090000
150	0.075000
156	0.090000
157	0.090000
158	0.090000
159	0.090000
160	0.075000
161	0.090000
164	0.075000
165	0.090000
166	0.090000
167	0.090000
168	0.090000
169	0.090000
184	0.090000
185	0.090000
186	0.090000
187	0.090000
188	0.090000
189	0.090000
190	0.090000
191	0.090000
192	0.075000
193	0.075000
194	0.075000
195	0.090000
196	0.090000
197	0.090000
198	0.075000
199	0.075000

200	0.075000
201	0.075000
202	0.062500
203	0.090000
204	0.090000
205	0.090000
206	0.090000
207	0.075000
213	0.090000
214	0.090000
215	0.090000
216	0.090000
217	0.075000
218	0.090000
219	0.090000
220	0.090000
221	0.090000
222	0.090000
223	0.075000
225	0.090000
226	0.090000
227	0.090000
228	0.090000
229	0.090000
230	0.075000
246	0.090000
247	0.090000
248	0.090000
249	0.090000
250	0.075000
251	0.090000
252	0.090000
273	0.090000
274	0.090000
275	0.090000
276	0.090000
277	0.075000
278	0.090000
279	0.090000
527	0.090000
528	0.090000
531	0.090000
533	0.090000
535	0.075000
537	0.090000
539	0.090000
808	0.075000
810	0.090000
811	0.090000
812	0.090000
813	0.090000
814	0.090000
815	0.090000
816	0.090000
817	0.090000
818	0.090000

819	0.090000
820	0.075000
822	0.090000
823	0.090000
845	0.090000
927	0.090000
928	0.090000
929	0.090000
930	0.090000
931	0.090000
932	0.090000
933	0.090000
934	0.090000
935	0.090000
936	0.090000
937	0.090000
938	0.090000
939	0.090000
940	0.090000
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942	0.090000
943	0.090000
944	0.090000
945	0.090000
946	0.090000
947	0.075000
948	0.075000
949	0.075000
950	0.075000
951	0.075000
952	0.090000
953	0.090000
954	0.090000
955	0.090000
956	0.090000
962	0.075000
964	0.090000
965	0.090000
966	0.090000
967	0.090000
968	0.090000
969	0.062500
971	0.075000
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973	0.075000
974	0.075000
975	0.075000
976	0.090000
977	0.090000
978	0.090000
979	0.090000
980	0.090000
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983	0.090000
984	0.090000

985	0.090000
986	0.090000
987	0.090000
988	0.090000
989	0.090000
990	0.090000
991	0.075000
992	0.075000
993	0.075000
994	0.075000
995	0.075000
996	0.090000
997	0.090000
998	0.090000
999	0.090000
1000	0.090000
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1007	0.090000
1008	0.090000
1009	0.090000
1010	0.090000
1011	0.090000
1012	0.090000
1013	0.090000
1014	0.090000
1015	0.090000
1016	0.075000
1018	0.075000
1019	0.075000
1020	0.075000
1021	0.075000
1022	0.075000
1023	0.062500
1025	0.090000
1026	0.090000
1027	0.090000
1028	0.090000
1029	0.090000
1030	0.090000
1031	0.075000
1032	0.090000
1033	0.090000
1034	0.090000
1035	0.090000
1036	0.090000
1037	0.090000
1038	0.075000
1079	0.090000
1080	0.090000
1083	0.090000
1085	0.090000
1087	0.090000
1089	0.090000
1091	0.075000
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1152	0.060000

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1155	0.050000
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1175	0.050000
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1179	0.060000
1180	0.050000
1181	0.060000
1182	0.060000
1193	0.060000
1194	0.060000
1195	0.060000
1196	0.060000
1197	0.060000
1198	0.060000
1199	0.060000
1200	0.050000
1201	0.060000
1202	0.050000
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1207	0.060000
1208	0.060000
1209	0.060000
1210	0.060000
1211	0.060000
1212	0.050000
1213	0.060000
1214	0.060000
1216	0.060000
1226	0.040000
1227	0.040000
1228	0.060000
1229	0.060000
1230	0.060000
1231	0.060000
1232	0.050000
1233	0.060000
1234	0.060000
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1236	0.060000
1237	0.050000
1238	0.060000

1239	0.060000
1240	0.060000
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1243	0.060000
1253	0.040000
1254	0.040000
1255	0.030000
1256	0.030000
1257	0.030000
1258	0.030000
1259	0.025000
1260	0.030000
1261	0.030000
1264	0.030000
1265	0.030000
1266	0.030000
1267	0.030000
1268	0.030000
1269	0.030000
1270	0.030000
1271	0.025000
1272	0.020000
1273	0.020000
1274	0.030000
1275	0.030000
1276	0.030000
1277	0.030000
1278	0.025000
1279	0.030000
1280	0.030000
1283	0.030000
1284	0.030000
1285	0.030000
1286	0.030000
1287	0.030000
1288	0.030000
1289	0.030000
1290	0.025000
1291	0.020000
1292	0.020000
1293	0.030000
1294	0.025000
1295	0.030000
1296	0.030000
1297	0.030000
1298	0.030000
1299	0.030000
1300	0.030000
1301	0.030000
1302	0.030000
1303	0.030000
1304	0.025000
1305	0.030000
1306	0.030000
1308	0.030000
1310	0.020000

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     1311          0.020000
     1312          0.010000
     1313          0.010000
     1314          0.030000
     1315          0.030000
     1316          0.030000
     1317          0.030000
     1318          0.025000
     1319          0.030000
     1320          0.030000
     1321          0.030000
     1322          0.030000
     1323          0.025000
     1324          0.030000
     1325          0.030000
     1326          0.030000
     1327          0.030000
     1329          0.030000
     1331          0.020000
     1332          0.020000
     1333          0.010000
     1334          0.010000
   TOTAL          25.000027
256. ***
257. ************************************************
258. ***LOAD COMBINATION FOR STRESS DESIGN
259. ************************************************
260. ***
261. ***
262. LOAD 1 LOADTYPE DEAD  TITLE DL
263. ***
264. SELFWEIGHT Y -1.1
265. ***FROM GALLERY-1
266. JOINT LOAD
267. 1 FX -0.3 FY -6.647 FZ -0.192
268. 2 FX -0.178 FY -6.41 FZ 0.285
269. 3 FX 0.178 FY -20.587 FZ -0.309
270. 4 FX 0.3 FY -20.158 FZ 0.173
271. ELEMENT LOAD
272. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
273. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
274. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
275. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
276. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
277. 1193 TO 1210 1213 TO 1216 PR GY -20
278. LOAD 2 LOADTYPE LIVE  TITLE LL
279. JOINT LOAD
280. 1 FX 18.996 FY 93.722 FZ 4.133
281. 2 FX 10.785 FY 22.894 FZ 4.126
282. 3 FX 19 FY -97.187 FZ 12.735
283. 4 FX 10.797 FY -157.982 FZ 12.759
284. *
285. LOAD 3 LOADTYPE LIVE  TITLE LL1
286. JOINT LOAD
287. 1 FX 8.52 FY 9.675 FZ -3.488
288. 2 FX 21.26 FY 106.94 FZ -3.495
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289. 3 FX 8.539 FY -189.641 FZ -15.853
290. 4 FX 21.259 FY -65.527 FZ -15.83
291. *
292. LOAD 11 LOADTYPE WIND  TITLE WL+Z
293. JOINT LOAD
294. 1 FX 4.303 FY 38.74 FZ 4.155
295. 2 FX -4.303 FY -38.74 FZ 4.155
296. 3 FX 4.296 FY 49.245 FZ 12.549
297. 4 FX -4.296 FY -49.245 FZ 12.549
298. LOAD 12 LOADTYPE WIND  TITLE WL-Z
299. JOINT LOAD
300. 1 FX -4.303 FY -38.74 FZ -4.155
301. 2 FX 4.303 FY 38.74 FZ -4.155
302. 3 FX -4.296 FY -49.245 FZ -12.549
303. 4 FX 4.296 FY 49.245 FZ -12.549
304. LOAD 13 LOADTYPE WIND  TITLE WL+X
305. JOINT LOAD
306. 1 FX 2.073 FY 7.641 FZ 0.002
307. 2 FX 2.077 FY 7.643 FZ 0.001
308. 3 FX 2.073 FY -7.408 FZ -0.003
309. 4 FX 2.077 FY -7.411 FZ -0.002
310. LOAD 14 LOADTYPE WIND  TITLE WL-X
311. JOINT LOAD
312. 1 FX -2.073 FY -7.641 FZ -0.002
313. 2 FX -2.077 FY -7.643 FZ -0.001
314. 3 FX -2.073 FY 7.408 FZ 0.003
315. 4 FX -2.077 FY 7.411 FZ 0.002
316. *
317. LOAD 20 LOADTYPE NONE  TITLE TL1
318. JOINT LOAD
319. 1 FX -0.016 FY 0 FZ -0.129
320. 2 FX -0.016 FY 0 FZ 0.129
321. 3 FX 0.016 FY 0 FZ -0.129
322. 4 FX 0.016 FY 0 FZ 0.129
323. *
324. LOAD 21 LOADTYPE NONE  TITLE TL2
325. JOINT LOAD
326. 1 FX 0.016 FY 0 FZ 0.129
327. 2 FX 0.016 FY 0 FZ -0.129
328. 3 FX -0.016 FY 0 FZ 0.129
329. 4 FX -0.016 FY 0 FZ -0.129
330. *
331. LOAD 30 EARTHQUAKE LOAD (+Z)
332. JOINT LOAD
333. 1 FX 0.109 FY 1.458 FZ 0.22
334. 2 FX -0.109 FY -1.458 FZ 0.22
335. 3 FX 0.109 FY 1.541 FZ 0.442
336. 4 FX -0.109 FY -1.541 FZ 0.442
337. *
338. LOAD 31 EARTHQUAKE LOAD (-Z)
339. JOINT LOAD
340. 1 FX -0.109 FY -1.458 FZ -0.22
341. 2 FX 0.109 FY 1.458 FZ -0.22
342. 3 FX -0.109 FY -1.541 FZ -0.442
343. 4 FX 0.109 FY 1.541 FZ -0.442
344. *
```

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345. LOAD 32 EARTHQUAKE LOAD (+X)
346. JOINT LOAD
347. 1 FX 0.351 FY 1.507 FZ -0.02
348. 2 FX 0.311 FY 1.492 FZ -0.02
349. 3 FX 0.351 FY -1.507 FZ 0.02
350. 4 FX 0.311 FY -1.492 FZ 0.02
351. *
352. LOAD 33 EARTHQUAKE LOAD (-X)
353. JOINT LOAD
354. 1 FX -0.351 FY -1.507 FZ 0.02
355. 2 FX -0.311 FY -1.492 FZ 0.02
356. 3 FX -0.351 FY 1.507 FZ -0.02
357. 4 FX -0.311 FY 1.492 FZ -0.02
358. *
359. ****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
360. *
361. LOAD COMB 201 D + L + TL1
362. 1 1.0 2 1.0 20 1.0
363. *
364. LOAD COMB 202 D + L1 + TL1
365. 1 1.0 3 1.0 20 1.0
366. *
367. LOAD COMB 203 D + L + TL2
368. 1 1.0 2 1.0 21 1.0
369. *
370. LOAD COMB 204 D + L1 + TL2
371. 1 1.0 3 1.0 21 1.0
372. *
373. LOAD COMB 205 D + 0.75XL + 0.75XTL1
374. 1 1.0 2 0.75 20 0.75
375. *
376. LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
377. 1 1.0 3 0.75 20 0.75
378. *
379. LOAD COMB 207 D + 0.75XL + 0.75XTL2
380. 1 1.0 2 0.75 21 0.75
381. *
382. LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
383. 1 1.0 3 0.75 21 0.75
384. *
385. LOAD COMB 209 D + WL(+Z)
386. 1 1.0 11 1.0
387. *
388. LOAD COMB 210 D + WL(+X)
389. 1 1.0 13 1.0
390. *
391. LOAD COMB 211 D + WL(-Z)
392. 1 1.0 12 1.0
393. *
394. LOAD COMB 212 D + WL(-X)
395. 1 1.0 14 1.0
396. *
397. LOAD COMB 213 D + 0.7XEL(+Z)
398. 1 1.0 30 0.7
399. *
400. LOAD COMB 214 D + 0.7XEL(+X)
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401. 1 1.0 32 0.7
402. *
403. LOAD COMB 215 D + 0.7XEL(-Z)
404. 1 1.0 31 0.7
405. *
406. LOAD COMB 216 D + 0.7XEL(-X)
407. 1 1.0 33 0.7
408. *
409. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
410. 1 1.0 2 0.75 11 0.75
411. *
412. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
413. 1 1.0 3 0.75 11 0.75
414. *
415. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
416. 1 1.0 2 0.75 12 0.75
417. *
418. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
419. 1 1.0 3 0.75 12 0.75
420. *
421. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
422. 1 1.0 2 0.75 13 0.75
423. *
424. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
425. 1 1.0 3 0.75 13 0.75
426. *
427. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
428. 1 1.0 2 0.75 14 0.75
429. *
430. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
431. 1 1.0 3 0.75 14 0.75
432. *
433. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
434. 1 1.0 2 0.75 30 0.525
435. *
436. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
437. 1 1.0 3 0.75 30 0.525
438. *
439. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
440. 1 1.0 2 0.75 32 0.525
441. *
442. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
443. 1 1.0 3 0.75 32 0.525
444. *
445. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
446. 1 1.0 2 0.75 31 0.525
447. *
448. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
449. 1 1.0 3 0.75 31 0.525
450. *
451. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
452. 1 1.0 2 0.75 33 0.525
453. *
454. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
455. 1 1.0 3 0.75 33 0.525
456. *
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457. LOAD COMB 233 0.6XD + WL(+Z)
458. 1 0.6 11 1.0
459. *
460. LOAD COMB 234 0.6XD + WL(+X)
461. 1 0.6 13 1.0
462. *
463. LOAD COMB 235 0.6XD + WL(-Z)
464. 1 0.6 12 1.0
465. *
466. LOAD COMB 236 0.6XD + WL(-X)
467. 1 0.6 14 1.0
468. *
469. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
470. 1 0.6 30 0.7
471. *
472. LOAD COMB 238 0.6XD + 0.7XEL(+X)
473. 1 0.6 32 0.7
474. *
475. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
476. 1 0.6 31 0.7
477. *
478. LOAD COMB 240 0.6XD + 0.7XEL(-X)
479. 1 0.6 33 0.7
480. *
481. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**365**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**324**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**361**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 265/ 32/ 99 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 1107
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 110 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.1/ 74039.5 MB

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482. CHANGE
483. SUPPORTS
484. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
485. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
486. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
487. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
488. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
```

489. 1193 TO 1210 1213 TO 1215 -
490. 1216 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION

JOINT	INFLUENCE AREA
23	0.090000
121	0.090000
122	0.090000
123	0.090000
124	0.090000
125	0.090000
126	0.090000
127	0.090000
128	0.090000
129	0.090000
130	0.075000
131	0.075000
132	0.075000
133	0.090000
134	0.090000
135	0.090000
136	0.090000
137	0.090000
138	0.090000
139	0.075000
140	0.090000
141	0.075000
142	0.075000
143	0.075000
144	0.075000
145	0.062500
146	0.090000
147	0.090000
148	0.090000
149	0.090000
150	0.075000
156	0.090000
157	0.090000
158	0.090000
159	0.090000
160	0.075000
161	0.090000
164	0.075000
165	0.090000
166	0.090000
167	0.090000
168	0.090000
169	0.090000
184	0.090000
185	0.090000
186	0.090000
187	0.090000
188	0.090000
189	0.090000
190	0.090000

191	0.090000
192	0.075000
193	0.075000
194	0.075000
195	0.090000
196	0.090000
197	0.090000
198	0.075000
199	0.075000
200	0.075000
201	0.075000
202	0.062500
203	0.090000
204	0.090000
205	0.090000
206	0.090000
207	0.075000
213	0.090000
214	0.090000
215	0.090000
216	0.090000
217	0.075000
218	0.090000
219	0.090000
220	0.090000
221	0.090000
222	0.090000
223	0.075000
225	0.090000
226	0.090000
227	0.090000
228	0.090000
229	0.090000
230	0.075000
246	0.090000
247	0.090000
248	0.090000
249	0.090000
250	0.075000
251	0.090000
252	0.090000
273	0.090000
274	0.090000
275	0.090000
276	0.090000
277	0.075000
278	0.090000
279	0.090000
527	0.090000
528	0.090000
531	0.090000
533	0.090000
535	0.075000
537	0.090000
539	0.090000
808	0.075000

810	0.090000
811	0.090000
812	0.090000
813	0.090000
814	0.090000
815	0.090000
816	0.090000
817	0.090000
818	0.090000
819	0.090000
820	0.075000
822	0.090000
823	0.090000
845	0.090000
927	0.090000
928	0.090000
929	0.090000
930	0.090000
931	0.090000
932	0.090000
933	0.090000
934	0.090000
935	0.090000
936	0.090000
937	0.090000
938	0.090000
939	0.090000
940	0.090000
941	0.090000
942	0.090000
943	0.090000
944	0.090000
945	0.090000
946	0.090000
947	0.075000
948	0.075000
949	0.075000
950	0.075000
951	0.075000
952	0.090000
953	0.090000
954	0.090000
955	0.090000
956	0.090000
962	0.075000
964	0.090000
965	0.090000
966	0.090000
967	0.090000
968	0.090000
969	0.062500
971	0.075000
972	0.075000
973	0.075000
974	0.075000
975	0.075000

976	0.090000
977	0.090000
978	0.090000
979	0.090000
980	0.090000
981	0.090000
982	0.090000
983	0.090000
984	0.090000
985	0.090000
986	0.090000
987	0.090000
988	0.090000
989	0.090000
990	0.090000
991	0.075000
992	0.075000
993	0.075000
994	0.075000
995	0.075000
996	0.090000
997	0.090000
998	0.090000
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1009	0.090000
1010	0.090000
1011	0.090000
1012	0.090000
1013	0.090000
1014	0.090000
1015	0.090000
1016	0.075000
1018	0.075000
1019	0.075000
1020	0.075000
1021	0.075000
1022	0.075000
1023	0.062500
1025	0.090000
1026	0.090000
1027	0.090000
1028	0.090000
1029	0.090000
1030	0.090000
1031	0.075000
1032	0.090000
1033	0.090000
1034	0.090000
1035	0.090000
1036	0.090000
1037	0.090000
1038	0.075000

1079	0.090000
1080	0.090000
1083	0.090000
1085	0.090000
1087	0.090000
1089	0.090000
1091	0.075000
1151	0.060000
1152	0.060000
1153	0.060000
1154	0.060000
1155	0.050000
1156	0.060000
1157	0.060000
1168	0.060000
1169	0.060000
1170	0.060000
1171	0.060000
1172	0.060000
1173	0.060000
1174	0.060000
1175	0.050000
1176	0.060000
1177	0.060000
1178	0.060000
1179	0.060000
1180	0.050000
1181	0.060000
1182	0.060000
1193	0.060000
1194	0.060000
1195	0.060000
1196	0.060000
1197	0.060000
1198	0.060000
1199	0.060000
1200	0.050000
1201	0.060000
1202	0.050000
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1204	0.060000
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1208	0.060000
1209	0.060000
1210	0.060000
1211	0.060000
1212	0.050000
1213	0.060000
1214	0.060000
1216	0.060000
1226	0.040000
1227	0.040000
1228	0.060000
1229	0.060000

1230	0.060000
1231	0.060000
1232	0.050000
1233	0.060000
1234	0.060000
1235	0.060000
1236	0.060000
1237	0.050000
1238	0.060000
1239	0.060000
1240	0.060000
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1243	0.060000
1253	0.040000
1254	0.040000
1255	0.030000
1256	0.030000
1257	0.030000
1258	0.030000
1259	0.025000
1260	0.030000
1261	0.030000
1264	0.030000
1265	0.030000
1266	0.030000
1267	0.030000
1268	0.030000
1269	0.030000
1270	0.030000
1271	0.025000
1272	0.020000
1273	0.020000
1274	0.030000
1275	0.030000
1276	0.030000
1277	0.030000
1278	0.025000
1279	0.030000
1280	0.030000
1283	0.030000
1284	0.030000
1285	0.030000
1286	0.030000
1287	0.030000
1288	0.030000
1289	0.030000
1290	0.025000
1291	0.020000
1292	0.020000
1293	0.030000
1294	0.025000
1295	0.030000
1296	0.030000
1297	0.030000
1298	0.030000
1299	0.030000

1300	0.030000
1301	0.030000
1302	0.030000
1303	0.030000
1304	0.025000
1305	0.030000
1306	0.030000
1308	0.030000
1310	0.020000
1311	0.020000
1312	0.010000
1313	0.010000
1314	0.030000
1315	0.030000
1316	0.030000
1317	0.030000
1318	0.025000
1319	0.030000
1320	0.030000
1321	0.030000
1322	0.030000
1323	0.025000
1324	0.030000
1325	0.030000
1326	0.030000
1327	0.030000
1329	0.030000
1331	0.020000
1332	0.020000
1333	0.010000
1334	0.010000
TOTAL	25.000027

```
491. *
492. *
493. *
494. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
495. LOAD COMB 100 1.4XD
496. 1 1.4
497. *
498. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
499. 1 1.2 20 1.2 2 1.6
500. *
501. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
502. 1 1.2 20 1.2 3 1.6
503. *
504. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
505. 1 1.2 21 1.2 2 1.6
506. *
507. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
508. 1 1.2 21 1.2 3 1.6
509. *
510. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
511. 1 1.2 11 0.8
512. *
513. LOAD COMB 106 1.2XD + 0.8XWL(+X)
514. 1 1.2 13 0.8
```

```
515. *
516. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
517. 1 1.2 12 0.8
518. *
519. LOAD COMB 108 1.2XD + 0.8XWL(-X)
520. 1 1.2 14 0.8
521. *
522. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
523. 1 0.9 11 1.6
524. *
525. LOAD COMB 110 0.9XD + 1.6XWL(+X)
526. 1 0.9 13 1.6
527. *
528. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
529. 1 0.9 12 1.6
530. *
531. LOAD COMB 112 0.9XD + 1.6XWL(-X)
532. 1 0.9 14 1.6
533. *
534. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
535. 1 1.2 2 0.5 1.6
536. *
537. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
538. 1 1.2 2 0.5 1.6
539. *
540. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
541. 1 1.2 2 0.5 1.6
542. *
543. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
544. 1 1.2 2 0.5 1.6
545. *
546. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
547. 1 1.2 3 0.5 1.6
548. *
549. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
550. 1 1.2 3 0.5 1.6
551. *
552. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
553. 1 1.2 3 0.5 1.6
554. *
555. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
556. 1 1.2 3 0.5 1.6
557. *
558. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
559. 1 1.2 2 0.5 1
560. *
561. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
562. 1 1.2 2 0.5 1
563. *
564. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
565. 1 1.2 2 0.5 1
566. *
567. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
568. 1 1.2 2 0.5 1
569. *
570. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
```

```
571. 1 1.2 3 0.5 1
572. *
573. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
574. 1 1.2 3 0.5 1
575. *
576. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
577. 1 1.2 3 0.5 1
578. *
579. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
580. 1 1.2 3 0.5 1
581. *
582. LOAD COMB 129 0.9XD + EL(+Z)
583. 1 0.9 30 1.0
584. *
585. LOAD COMB 130 0.9XD + EL(+X)
586. 1 0.9 32 1.0
587. *
588. LOAD COMB 131 0.9XD + EL(-Z)
589. 1 0.9 31 1.0
590. *
591. LOAD COMB 132 0.9XD + EL(-X)
592. 1 0.9 33 1.0
593. *
594. PERFORM ANALYSIS
595. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 17,2020 TIME= 10:20: 1 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUL 17, 2020                 *
*           Time=    14:10: 8                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\RCO1 _RCO2\CIVIL\Working\F2\F2.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18-MARCH-20
     4. *FOUNDATION F5
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. SET NL 100
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 -0.25 0.35 0; 2 -0.25 0.35 -3; 23 -0.25 -2 -3; 121 -0.25 -2 0
    11. 122 -0.25 -2 -0.35; 123 0 -2 -0.35; 124 0 -2 0; 125 -0.5 -2 0
    12. 126 -0.5 -2 -0.35; 127 -0.25 -2 0.35; 128 0 -2 0.35; 129 -0.5 -2 0.35
    13. 130 0.25 -2 -0.35; 131 0.25 -2 0; 132 0.25 -2 0.35; 133 -0.75 -2 0
    14. 134 -0.75 -2 -0.35; 135 -0.75 -2 0.35; 136 -0.25 -2 -0.6; 137 0 -2 -0.6
    15. 138 -0.5 -2 -0.6; 139 0.25 -2 -0.6; 140 -0.75 -2 -0.6; 141 -0.75 -2 0.6
    16. 142 -0.5 -2 0.6; 143 -0.25 -2 0.6; 144 0 -2 0.6; 145 0.25 -2 0.6
    17. 146 -0.75 -2 -0.85; 147 -0.5 -2 -0.85; 148 -0.25 -2 -0.85; 149 0 -2 -0.85
    18. 150 0.25 -2 -0.85; 151 -0.75 -2 0.85; 152 -0.5 -2 0.85; 153 -0.25 -2 0.85
    19. 154 0 -2 0.85; 155 0.25 -2 0.85; 156 -1 -2 0; 157 -1 -2 -0.35; 158 -1 -2 0.35
    20. 159 -1 -2 -0.6; 160 -1 -2 0.6; 161 -1 -2 -0.85; 162 -1 -2 0.85
    21. 163 -1.25 -2 0.85; 164 -1.25 -2 0.6; 165 -1.25 -2 0.35; 166 -1.25 -2 0
    22. 167 -1.25 -2 -0.35; 168 -1.25 -2 -0.6; 169 -1.25 -2 -0.85; 170 0.5 -2 0.6
    23. 171 0.5 -2 0.85; 172 0.5 -2 0.35; 173 0.5 -2 0; 174 0.5 -2 -0.35
    24. 175 0.5 -2 -0.6; 176 0.5 -2 -0.85; 177 0.75 -2 0.6; 178 0.75 -2 0.85
    25. 179 0.75 -2 0.35; 180 0.75 -2 0; 181 0.75 -2 -0.35; 182 0.75 -2 -0.6
    26. 183 0.75 -2 -0.85; 184 -0.25 -2 -3.35; 185 0 -2 -3.35; 186 0 -2 -3
    27. 187 -0.5 -2 -3; 188 -0.5 -2 -3.35; 189 -0.25 -2 -2.65; 190 0 -2 -2.65
    28. 191 -0.5 -2 -2.65; 192 0.25 -2 -3; 193 0.25 -2 -2.65; 194 0.25 -2 -3.35
    29. 195 -0.75 -2 -2.65; 196 -0.75 -2 -3; 197 -0.75 -2 -3.35; 198 -0.75 -2 -3.6
    30. 199 -0.5 -2 -3.6; 200 -0.25 -2 -3.6; 201 0 -2 -3.6; 202 0.25 -2 -3.6
    31. 203 -0.75 -2 -2.4; 204 -0.5 -2 -2.4; 205 -0.25 -2 -2.4; 206 0 -2 -2.4
    32. 207 0.25 -2 -2.4; 208 -0.75 -2 -3.85; 209 -0.5 -2 -3.85; 210 -0.25 -2 -3.85
    33. 211 0 -2 -3.85; 212 0.25 -2 -3.85; 213 -0.75 -2 -2.15; 214 -0.5 -2 -2.15
    34. 215 -0.25 -2 -2.15; 216 0 -2 -2.15; 217 0.25 -2 -2.15; 218 -1 -2 -2.15
    35. 219 -1 -2 -2.4; 220 -1 -2 -2.65; 221 -1 -2 -3; 222 -1 -2 -3.35; 223 -1 -2 -3.6
    36. 224 -1 -2 -3.85; 225 -1.25 -2 -2.15; 226 -1.25 -2 -2.4; 227 -1.25 -2 -2.65
    37. 228 -1.25 -2 -3; 229 -1.25 -2 -3.35; 230 -1.25 -2 -3.6; 231 -1.25 -2 -3.85
    38. 232 0.5 -2 -2.4; 233 0.5 -2 -2.15; 234 0.5 -2 -2.65; 235 0.5 -2 -3
```

39. 236 0.5 -2 -3.35; 237 0.5 -2 -3.6; 238 0.5 -2 -3.85; 239 0.75 -2 -2.4
40. 240 0.75 -2 -2.15; 241 0.75 -2 -2.65; 242 0.75 -2 -3; 243 0.75 -2 -3.35
41. 244 0.75 -2 -3.6; 245 0.75 -2 -3.85; 246 -0.75 -2 -1.05; 247 -0.5 -2 -1.05
42. 248 -0.25 -2 -1.05; 249 0 -2 -1.05; 250 0.25 -2 -1.05; 251 -1 -2 -1.05
43. 252 -1.25 -2 -1.05; 253 0.5 -2 -1.05; 254 0.75 -2 -1.05; 255 -0.75 -2 -1.3
44. 256 -0.5 -2 -1.3; 257 -0.25 -2 -1.3; 258 0 -2 -1.3; 259 0.25 -2 -1.3
45. 260 -1 -2 -1.3; 261 -1.25 -2 -1.3; 262 0.5 -2 -1.3; 263 0.75 -2 -1.3
46. 273 -0.75 -2 -1.95; 274 -0.5 -2 -1.95; 275 -0.25 -2 -1.95; 276 0 -2 -1.95
47. 277 0.25 -2 -1.95; 278 -1 -2 -1.95; 279 -1.25 -2 -1.95; 280 0.5 -2 -1.95
48. 281 0.75 -2 -1.95; 282 -0.75 -2 -1.75; 283 -0.5 -2 -1.75; 284 -0.25 -2 -1.75
49. 285 0 -2 -1.75; 286 0.25 -2 -1.75; 287 -1 -2 -1.75; 288 -1.25 -2 -1.75
50. 289 0.5 -2 -1.75; 290 0.75 -2 -1.75; 291 -0.75 -2 -1.5; 292 -0.5 -2 -1.5
51. 293 -0.25 -2 -1.5; 294 0 -2 -1.5; 295 0.25 -2 -1.5; 296 -1 -2 -1.5
52. 297 -1.25 -2 -1.5; 298 0.5 -2 -1.5; 299 0.75 -2 -1.5; 309 -0.75 -2 0.9
53. 310 -0.5 -2 0.9; 311 -0.25 -2 0.9; 312 0 -2 0.9; 313 0.25 -2 0.9
54. 314 -1 -2 0.9; 315 -1.25 -2 0.9; 316 0.5 -2 0.9; 317 0.75 -2 0.9
55. 327 -0.75 -2 -3.9; 328 -0.5 -2 -3.9; 329 -0.25 -2 -3.9; 330 0 -2 -3.9
56. 331 0.25 -2 -3.9; 332 -1 -2 -3.9; 333 -1.25 -2 -3.9; 334 0.5 -2 -3.9
57. 335 0.75 -2 -3.9; 345 -1.375 -2 0.85; 346 -1.375 -2 0.6; 347 -1.375 -2 0.35
58. 348 -1.375 -2 0; 349 -1.375 -2 -0.35; 350 -1.375 -2 -0.6; 351 -1.375 -2 -0.85
59. 352 -1.375 -2 -2.15; 353 -1.375 -2 -2.4; 354 -1.375 -2 -2.65; 355 -1.375 -2 -3
60. 356 -1.375 -2 -3.35; 357 -1.375 -2 -3.6; 358 -1.375 -2 -3.85
61. 359 -1.375 -2 -1.05; 360 -1.375 -2 -1.3; 362 -1.375 -2 -1.95
62. 363 -1.375 -2 -1.75; 364 -1.375 -2 -1.5; 366 -1.375 -2 0.9; 368 -1.375 -2 -3.9
63. 395 0.875 -2 0.6; 396 0.875 -2 0.85; 397 0.875 -2 0.35; 398 0.875 -2 0
64. 399 0.875 -2 -0.35; 400 0.875 -2 -0.6; 401 0.875 -2 -0.85; 402 0.875 -2 -2.4
65. 403 0.875 -2 -2.15; 404 0.875 -2 -2.65; 405 0.875 -2 -3; 406 0.875 -2 -3.35
66. 407 0.875 -2 -3.6; 408 0.875 -2 -3.85; 409 0.875 -2 -1.05; 410 0.875 -2 -1.3
67. 412 0.875 -2 -1.95; 413 0.875 -2 -1.75; 414 0.875 -2 -1.5; 416 0.875 -2 0.9
68. 418 0.875 -2 -3.9; 419 -1.5 -2 0.6; 420 -1.5 -2 0.85; 421 -1.5 -2 0.35
69. 422 -1.5 -2 0; 423 -1.5 -2 -0.35; 424 -1.5 -2 -0.6; 425 -1.5 -2 -0.85
70. 426 -1.5 -2 -2.4; 427 -1.5 -2 -2.15; 428 -1.5 -2 -2.65; 429 -1.5 -2 -3
71. 430 -1.5 -2 -3.35; 431 -1.5 -2 -3.6; 432 -1.5 -2 -3.85; 433 -1.5 -2 -1.05
72. 434 -1.5 -2 -1.3; 435 -1.5 -2 -1.5; 436 -1.5 -2 -1.95; 437 -1.5 -2 -1.75
73. 438 -1.5 -2 0.9; 439 -1.5 -2 -3.9; 440 1 -2 0.85; 441 1 -2 0.6; 442 1 -2 0.35
74. 443 1 -2 0; 444 1 -2 -0.35; 445 1 -2 -0.6; 446 1 -2 -0.85; 447 1 -2 -2.15
75. 448 1 -2 -2.4; 449 1 -2 -2.65; 450 1 -2 -3; 451 1 -2 -3.35; 452 1 -2 -3.6
76. 453 1 -2 -3.85; 454 1 -2 -1.05; 455 1 -2 -1.3; 456 1 -2 -1.5; 457 1 -2 -1.95
77. 458 1 -2 -1.75; 459 1 -2 0.9; 460 1 -2 -3.9; 461 -0.5 -2 -3.95
78. 462 -0.75 -2 -3.95; 463 -0.25 -2 -3.95; 464 0 -2 -3.95; 465 0.25 -2 -3.95
79. 466 -1 -2 -3.95; 467 -1.25 -2 -3.95; 468 0.5 -2 -3.95; 469 0.75 -2 -3.95
80. 470 -1.375 -2 -3.95; 471 0.875 -2 -3.95; 472 -1.5 -2 -3.95; 473 1 -2 -3.95
81. 474 -0.5 -2 0.95; 475 -0.75 -2 0.95; 476 -0.25 -2 0.95; 477 0 -2 0.95
82. 478 0.25 -2 0.95; 479 -1 -2 0.95; 480 -1.25 -2 0.95; 481 0.5 -2 0.95
83. 482 0.75 -2 0.95; 483 -1.375 -2 0.95; 484 0.875 -2 0.95; 485 -1.5 -2 0.95
84. 486 1 -2 0.95; 487 -0.5 -2 -4; 488 -0.75 -2 -4; 489 -0.25 -2 -4; 490 0 -2 -4
85. 491 0.25 -2 -4; 492 -1 -2 -4; 493 -1.25 -2 -4; 494 0.5 -2 -4; 495 0.75 -2 -4
86. 496 -1.375 -2 -4; 497 0.875 -2 -4; 498 -1.5 -2 -4; 499 1 -2 -4; 500 -0.5 -2 1
87. 501 -0.75 -2 1; 502 -0.25 -2 1; 503 0 -2 1; 504 0.25 -2 1; 505 -1 -2 1
88. 506 -1.25 -2 1; 507 0.5 -2 1; 508 0.75 -2 1; 509 -1.375 -2 1; 510 0.875 -2 1
89. 511 -1.5 -2 1; 512 1 -2 1
90. MEMBER INCIDENCES
91. 48 23 2; 49 121 1
92. ELEMENT INCIDENCES SHELL
93. 50 124 123 122 121; 51 121 122 126 125; 52 128 124 121 127; 53 127 121 125 129
94. 54 132 131 124 128; 55 131 130 123 124; 56 129 125 133 135; 57 125 126 134 133

95. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
96. 62 142 129 135 141; 63 143 127 129 142; 64 144 128 127 143; 65 145 132 128 144
97. 66 138 147 146 140; 67 136 148 147 138; 68 137 149 148 136; 69 139 150 149 137
98. 70 152 142 141 151; 71 153 143 142 152; 72 154 144 143 153; 73 155 145 144 154
99. 74 151 141 160 162; 75 141 135 158 160; 76 135 133 156 158; 77 133 134 157 156
100. 78 134 140 159 157; 79 140 146 161 159; 80 162 160 164 163; 81 160 158 165 164
101. 82 158 156 166 165; 83 156 157 167 166; 84 157 159 168 167; 85 159 161 169 168
102. 86 171 170 145 155; 87 170 172 132 145; 88 172 173 131 132; 89 173 174 130 131
103. 90 174 175 139 130; 91 175 176 150 139; 92 178 177 170 171; 93 177 179 172 170
104. 94 179 180 173 172; 95 180 181 174 173; 96 181 182 175 174; 97 182 183 176 175
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118. 138 236 237 202 194; 139 237 238 212 202; 140 240 239 232 233
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121. 147 148 248 247 147; 148 149 249 248 148; 149 150 250 249 149
122. 150 146 246 251 161; 151 161 251 252 169; 152 176 253 250 150
123. 153 183 254 253 176; 154 247 256 255 246; 155 248 257 256 247
124. 156 249 258 257 248; 157 250 259 258 249; 158 246 255 260 251
125. 159 251 260 261 252; 160 253 262 259 250; 161 254 263 262 253
126. 162 256 292 291 255; 163 257 293 292 256; 164 258 294 293 257
127. 165 259 295 294 258; 166 255 291 296 260; 167 260 296 297 261
128. 168 262 298 295 259; 169 263 299 298 262; 170 274 214 213 273
129. 171 275 215 214 274; 172 276 216 215 275; 173 277 217 216 276
130. 174 273 213 218 278; 175 278 218 225 279; 176 280 233 217 277
131. 177 281 240 233 280; 178 283 274 273 282; 179 284 275 274 283
132. 180 285 276 275 284; 181 286 277 276 285; 182 282 273 278 287
133. 183 287 278 279 288; 184 289 280 277 286; 185 290 281 280 289
134. 186 292 283 282 291; 187 293 284 283 292; 188 294 285 284 293
135. 189 295 286 285 294; 190 291 282 287 296; 191 296 287 288 297
136. 192 298 289 286 295; 193 299 290 289 298; 210 310 152 151 309
137. 211 311 153 152 310; 212 312 154 153 311; 213 313 155 154 312
138. 214 309 151 162 314; 215 314 162 163 315; 216 316 171 155 313
139. 217 317 178 171 316; 226 209 328 327 208; 227 210 329 328 209
140. 228 211 330 329 210; 229 212 331 330 211; 230 208 327 332 224
141. 231 224 332 333 231; 232 238 334 331 212; 233 245 335 334 238
142. 242 163 164 346 345; 243 164 165 347 346; 244 165 166 348 347
143. 245 166 167 349 348; 246 167 168 350 349; 247 168 169 351 350
144. 248 225 226 353 352; 249 226 227 354 353; 250 227 228 355 354
145. 251 228 229 356 355; 252 229 230 357 356; 253 230 231 358 357
146. 254 169 252 359 351; 255 252 261 360 359; 256 261 297 364 360
147. 257 279 225 352 362; 258 288 279 362 363; 259 297 288 363 364
148. 262 315 163 345 366; 264 231 333 368 358; 290 396 395 177 178
149. 291 395 397 179 177; 292 397 398 180 179; 293 398 399 181 180
150. 294 399 400 182 181; 295 400 401 183 182; 296 403 402 239 240

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151. 297 402 404 241 239; 298 404 405 242 241; 299 405 406 243 242
152. 300 406 407 244 243; 301 407 408 245 244; 302 401 409 254 183
153. 303 409 410 263 254; 304 410 414 299 263; 305 412 403 240 281
154. 306 413 412 281 290; 307 414 413 290 299; 310 416 396 178 317
155. 312 408 418 335 245; 313 345 346 419 420; 314 346 347 421 419
156. 315 347 348 422 421; 316 348 349 423 422; 317 349 350 424 423
157. 318 350 351 425 424; 319 352 353 426 427; 320 353 354 428 426
158. 321 354 355 429 428; 322 355 356 430 429; 323 356 357 431 430
159. 324 357 358 432 431; 325 351 359 433 425; 326 359 360 434 433
160. 327 360 364 435 434; 328 362 352 427 436; 329 363 362 436 437
161. 330 364 363 437 435; 331 366 345 420 438; 332 358 368 439 432
162. 333 440 441 395 396; 334 441 442 397 395; 335 442 443 398 397
163. 336 443 444 399 398; 337 444 445 400 399; 338 445 446 401 400
164. 339 447 448 402 403; 340 448 449 404 402; 341 449 450 405 404
165. 342 450 451 406 405; 343 451 452 407 406; 344 452 453 408 407
166. 345 446 454 409 401; 346 454 455 410 409; 347 455 456 414 410
167. 348 457 447 403 412; 349 458 457 412 413; 350 456 458 413 414
168. 351 459 440 396 416; 352 453 460 418 408; 353 328 461 462 327
169. 354 329 463 461 328; 355 330 464 463 329; 356 331 465 464 330
170. 357 327 462 466 332; 358 332 466 467 333; 359 334 468 465 331
171. 360 335 469 468 334; 361 333 467 470 368; 362 418 471 469 335
172. 363 368 470 472 439; 364 460 473 471 418; 365 474 310 309 475
173. 366 476 311 310 474; 367 477 312 311 476; 368 478 313 312 477
174. 369 475 309 314 479; 370 479 314 315 480; 371 481 316 313 478
175. 372 482 317 316 481; 373 480 315 366 483; 374 484 416 317 482
176. 375 483 366 438 485; 376 486 459 416 484; 377 461 487 488 462
177. 378 463 489 487 461; 379 464 490 489 463; 380 465 491 490 464
178. 381 462 488 492 466; 382 466 492 493 467; 383 468 494 491 465
179. 384 469 495 494 468; 385 467 493 496 470; 386 471 497 495 469
180. 387 470 496 498 472; 388 473 499 497 471; 389 500 474 475 501
181. 390 502 476 474 500; 391 503 477 476 502; 392 504 478 477 503
182. 393 501 475 479 505; 394 505 479 480 506; 395 507 481 478 504
183. 396 508 482 481 507; 397 506 480 483 509; 398 510 484 482 508
184. 399 509 483 485 511; 400 512 486 484 510
185. START GROUP DEFINITION
186. FLOOR
187. ELEMENT
188. _FDN_5_MOMENT 58 TO 97 106 TO 193 210 TO 217 226 TO 233 242 TO 259 262 264 -
189. 290 TO 307 310 312 TO 400
190. _FDN_5_SHEAR 146 TO 193 210 TO 217 226 TO 233 242 TO 259 262 264 290 TO 307 -
191. 310 312 TO 400
192. END GROUP DEFINITION
193. ELEMENT PROPERTY
194. 50 TO 193 210 TO 217 226 TO 233 242 TO 259 262 264 290 TO 307 310 312 TO 399 -
195. 400 THICKNESS 0.5
196. DEFINE MATERIAL START
197. ISOTROPIC CONCRETE
198. E 2.17185E+007
199. POISSON 0.17
200. DENSITY 25
201. ALPHA 1E-005
202. DAMP 0.05
203. TYPE CONCRETE
204. STRENGTH FCU 27579
205. END DEFINE MATERIAL
206. MEMBER PROPERTY AMERICAN
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207. 48 49 PRIS YD 0.85 ZD 0.6
208. CONSTANTS
209. MATERIAL CONCRETE ALL
210. SUPPORTS
211. 50 TO 193 210 TO 217 226 TO 233 242 TO 259 262 264 290 TO 307 310 312 TO 399 -
212. 400 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
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JOINT	INFLUENCE AREA
23	0.087500
121	0.087500
122	0.075000
123	0.075000
124	0.087500
125	0.087500
126	0.075000
127	0.075000
128	0.075000
129	0.075000
130	0.075000
131	0.087500
132	0.075000
133	0.087500
134	0.075000
135	0.075000
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062500
142	0.062500
143	0.062500
144	0.062500
145	0.062500
146	0.056250
147	0.056250
148	0.056250
149	0.056250
150	0.056250
151	0.037500
152	0.037500
153	0.037500
154	0.037500
155	0.037500
156	0.087500
157	0.075000
158	0.075000
159	0.062500
160	0.062500
161	0.056250
162	0.037500
163	0.028125
164	0.046875
165	0.056250

166	0.065625
167	0.056250
168	0.046875
169	0.042188
170	0.062500
171	0.037500
172	0.075000
173	0.087500
174	0.075000
175	0.062500
176	0.056250
177	0.046875
178	0.028125
179	0.056250
180	0.065625
181	0.056250
182	0.046875
183	0.042188
184	0.075000
185	0.075000
186	0.087500
187	0.087500
188	0.075000
189	0.075000
190	0.075000
191	0.075000
192	0.087500
193	0.075000
194	0.075000
195	0.075000
196	0.087500
197	0.075000
198	0.062500
199	0.062500
200	0.062500
201	0.062500
202	0.062500
203	0.062500
204	0.062500
205	0.062500
206	0.062500
207	0.062500
208	0.037500
209	0.037500
210	0.037500
211	0.037500
212	0.037500
213	0.056250
214	0.056250
215	0.056250
216	0.056250
217	0.056250
218	0.056250
219	0.062500
220	0.075000
221	0.087500

222	0.075000
223	0.062500
224	0.037500
225	0.042188
226	0.046875
227	0.056250
228	0.065625
229	0.056250
230	0.046875
231	0.028125
232	0.062500
233	0.056250
234	0.075000
235	0.087500
236	0.075000
237	0.062500
238	0.037500
239	0.046875
240	0.042188
241	0.056250
242	0.065625
243	0.056250
244	0.046875
245	0.028125
246	0.056250
247	0.056250
248	0.056250
249	0.056250
250	0.056250
251	0.056250
252	0.042187
253	0.056250
254	0.042187
255	0.056250
256	0.056250
257	0.056250
258	0.056250
259	0.056250
260	0.056250
261	0.042187
262	0.056250
263	0.042187
273	0.050000
274	0.050000
275	0.050000
276	0.050000
277	0.050000
278	0.050000
279	0.037500
280	0.050000
281	0.037500
282	0.056250
283	0.056250
284	0.056250
285	0.056250
286	0.056250

287	0.056250
288	0.042187
289	0.056250
290	0.042187
291	0.056250
292	0.056250
293	0.056250
294	0.056250
295	0.056250
296	0.056250
297	0.042187
298	0.056250
299	0.042187
309	0.012500
310	0.012500
311	0.012500
312	0.012500
313	0.012500
314	0.012500
315	0.009375
316	0.012500
317	0.009375
327	0.012500
328	0.012500
329	0.012500
330	0.012500
331	0.012500
332	0.012500
333	0.009375
334	0.012500
335	0.009375
345	0.018750
346	0.031250
347	0.037500
348	0.043750
349	0.037500
350	0.031250
351	0.028125
352	0.028125
353	0.031250
354	0.037500
355	0.043750
356	0.037500
357	0.031250
358	0.018750
359	0.028125
360	0.028125
362	0.025000
363	0.028125
364	0.028125
366	0.006250
368	0.006250
395	0.031250
396	0.018750
397	0.037500
398	0.043750

399	0.037500
400	0.031250
401	0.028125
402	0.031250
403	0.028125
404	0.037500
405	0.043750
406	0.037500
407	0.031250
408	0.018750
409	0.028125
410	0.028125
412	0.025000
413	0.028125
414	0.028125
416	0.006250
418	0.006250
419	0.015625
420	0.009375
421	0.018750
422	0.021875
423	0.018750
424	0.015625
425	0.014063
426	0.015625
427	0.014063
428	0.018750
429	0.021875
430	0.018750
431	0.015625
432	0.009375
433	0.014063
434	0.014063
435	0.014063
436	0.012500
437	0.014063
438	0.003125
439	0.003125
440	0.009375
441	0.015625
442	0.018750
443	0.021875
444	0.018750
445	0.015625
446	0.014063
447	0.014063
448	0.015625
449	0.018750
450	0.021875
451	0.018750
452	0.015625
453	0.009375
454	0.014063
455	0.014063
456	0.014063
457	0.012500

458	0.014063
459	0.003125
460	0.003125
461	0.012500
462	0.012500
463	0.012500
464	0.012500
465	0.012500
466	0.012500
467	0.009375
468	0.012500
469	0.009375
470	0.006250
471	0.006250
472	0.003125
473	0.003125
474	0.012500
475	0.012500
476	0.012500
477	0.012500
478	0.012500
479	0.012500
480	0.009375
481	0.012500
482	0.009375
483	0.006250
484	0.006250
485	0.003125
486	0.003125
487	0.006250
488	0.006250
489	0.006250
490	0.006250
491	0.006250
492	0.006250
493	0.004688
494	0.006250
495	0.004688
496	0.003125
497	0.003125
498	0.001563
499	0.001563
500	0.006250
501	0.006250
502	0.006250
503	0.006250
504	0.006250
505	0.006250
506	0.004688
507	0.006250
508	0.004688
509	0.003125
510	0.003125
511	0.001563
512	0.001563
TOTAL	12.500000

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213. ***
214. ************************************************
215. ***LOAD COMBINATION FOR STRESS DESIGN
216. ************************************************
217. ***
218. ***
219. LOAD 1 LOADTYPE DEAD  TITLE DL
220. ***
221. SELFWEIGHT Y -1.1
222. ***FROM GALLERY-1
223. JOINT LOAD
224. 1 FX -0.276 FY -35.143 FZ -3.686
225. 2 FX 0.276 FY -35.349 FZ 3.646
228. ELEMENT LOAD
229. 50 TO 193 210 TO 217 226 TO 233 242 TO 259 262 264 290 TO 307 310 312 TO 399 -
230. 400 PR GY -20
231. *
232. LOAD 2 LOADTYPE LIVE  TITLE LL
233. JOINT LOAD
234. 1 FX 27.108 FY 12.258 FZ 17.017
235. 2 FX 26.144 FY -205.908 FZ 35.948
237. *
238. LOAD 3 LOADTYPE LIVE  TITLE LL1
239. JOINT LOAD
240. 1 FX 24.241 FY -254.309 FZ -36.447
241. 2 FX 29.011 FY 60.658 FZ -17.43
243. *
244. LOAD 11 LOADTYPE WIND  TITLE WL+Z
245. JOINT LOAD
246. 1 FX 0.524 FY 135.576 FZ 23.644
247. 2 FX -0.524 FY -135.576 FZ 23.606
249. *
250. LOAD 12 LOADTYPE WIND  TITLE WL-Z
251. JOINT LOAD
252. 1 FX -0.524 FY -135.576 FZ -23.644
253. 2 FX 0.524 FY 135.576 FZ -23.606
255. *
256. LOAD 13 LOADTYPE WIND  TITLE WL+X
257. JOINT LOAD
258. 1 FX 1.589 FY -0.007 FZ -0.001
259. 2 FX 1.611 FY 0.007 FZ -0.001
261. *
262. LOAD 14 LOADTYPE WIND  TITLE WL-X
263. JOINT LOAD
264. 1 FX -1.589 FY 0.007 FZ 0.001
265. 2 FX -1.611 FY -0.007 FZ 0.001
267. *
268. LOAD 20 LOADTYPE NONE  TITLE TL1
269. JOINT LOAD
270. 1 FX 0 FY 0 FZ -23.554
271. 2 FX 0 FY 0 FZ 23.554
273. *
274. LOAD 21 LOADTYPE NONE  TITLE TL2
275. JOINT LOAD
276. 1 FX 0 FY 0 FZ 23.554
277. 2 FX 0 FY 0 FZ -23.554
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279. *
280. LOAD 30 EARTHQUAKE LOAD (+Z)
281. JOINT LOAD
282. 1 FX 0 FY 1.719 FZ 0.379
283. 2 FX 0 FY -1.719 FZ 0.379
285. *
286. LOAD 31 EARTHQUAKE LOAD (-Z)
287. JOINT LOAD
288. 1 FX 0 FY -1.719 FZ -0.379
289. 2 FX 0 FY 1.719 FZ -0.379
291. *
292. LOAD 32 EARTHQUAKE LOAD (+X)
293. JOINT LOAD
294. 1 FX 0.377 FY 0 FZ 0
295. 2 FX 0.382 FY 0 FZ 0
297. *
298. LOAD 33 EARTHQUAKE LOAD (-X)
299. JOINT LOAD
300. 1 FX -0.377 FY 0 FZ 0
301. 2 FX -0.382 FY 0 FZ 0
302. *
303. ****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
304. *
305. LOAD COMB 201 D + L + TL1
306. 1 1.0 2 1.0 20 1.0
307. *
308. LOAD COMB 202 D + L1 + TL1
309. 1 1.0 3 1.0 20 1.0
310. *
311. LOAD COMB 203 D + L + TL2
312. 1 1.0 2 1.0 21 1.0
313. *
314. LOAD COMB 204 D + L1 + TL2
315. 1 1.0 3 1.0 21 1.0
316. *
317. LOAD COMB 205 D + 0.75XL + 0.75XTL1
318. 1 1.0 2 0.75 20 0.75
319. *
320. LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
321. 1 1.0 3 0.75 20 0.75
322. *
323. LOAD COMB 207 D + 0.75XL + 0.75XTL2
324. 1 1.0 2 0.75 21 0.75
325. *
326. LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
327. 1 1.0 3 0.75 21 0.75
328. *
329. LOAD COMB 209 D + WL(+Z)
330. 1 1.0 11 1.0
331. *
332. LOAD COMB 210 D + WL(+X)
333. 1 1.0 13 1.0
334. *
335. LOAD COMB 211 D + WL(-Z)
336. 1 1.0 12 1.0
337. *
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338. LOAD COMB 212 D + WL(-X)
339. 1 1.0 14 1.0
340. *
341. LOAD COMB 213 D + 0.7XEL(+Z)
342. 1 1.0 30 0.7
343. *
344. LOAD COMB 214 D + 0.7XEL(+X)
345. 1 1.0 32 0.7
346. *
347. LOAD COMB 215 D + 0.7XEL(-Z)
348. 1 1.0 31 0.7
349. *
350. LOAD COMB 216 D + 0.7XEL(-X)
351. 1 1.0 33 0.7
352. *
353. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
354. 1 1.0 2 0.75 11 0.75
355. *
356. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
357. 1 1.0 3 0.75 11 0.75
358. *
359. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
360. 1 1.0 2 0.75 12 0.75
361. *
362. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
363. 1 1.0 3 0.75 12 0.75
364. *
365. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
366. 1 1.0 2 0.75 13 0.75
367. *
368. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
369. 1 1.0 3 0.75 13 0.75
370. *
371. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
372. 1 1.0 2 0.75 14 0.75
373. *
374. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
375. 1 1.0 3 0.75 14 0.75
376. *
377. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
378. 1 1.0 2 0.75 30 0.525
379. *
380. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
381. 1 1.0 3 0.75 30 0.525
382. *
383. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
384. 1 1.0 2 0.75 32 0.525
385. *
386. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
387. 1 1.0 3 0.75 32 0.525
388. *
389. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
390. 1 1.0 2 0.75 31 0.525
391. *
392. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
393. 1 1.0 3 0.75 31 0.525
```

```
394. *
395. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
396. 1 1.0 2 0.75 33 0.525
397. *
398. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
399. 1 1.0 3 0.75 33 0.525
400. *
401. LOAD COMB 233 0.6XD + WL(+Z)
402. 1 0.6 11 1.0
403. *
404. LOAD COMB 234 0.6XD + WL(+X)
405. 1 0.6 13 1.0
406. *
407. LOAD COMB 235 0.6XD + WL(-Z)
408. 1 0.6 12 1.0
409. *
410. LOAD COMB 236 0.6XD + WL(-X)
411. 1 0.6 14 1.0
412. *
413. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
414. 1 0.6 30 0.7
415. *
416. LOAD COMB 238 0.6XD + 0.7XEL(+X)
417. 1 0.6 32 0.7
418. *
419. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
420. 1 0.6 31 0.7
421. *
422. LOAD COMB 240 0.6XD + 0.7XEL(-X)
423. 1 0.6 33 0.7
424. *
425. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**327**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**288**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**325**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 171/ 22/ 69 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 987
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 69 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.4/ 73949.2 MB

```
426. CHANGE
427. SUPPORTS
428. 50 TO 193 210 TO 217 226 TO 233 242 TO 259 262 264 290 TO 307 310 312 TO 399 -
429. 400 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION
```

JOINT	INFLUENCE AREA
23	0.087500
121	0.087500
122	0.075000
123	0.075000
124	0.087500
125	0.087500
126	0.075000
127	0.075000
128	0.075000
129	0.075000
130	0.075000
131	0.087500
132	0.075000
133	0.087500
134	0.075000
135	0.075000
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062500
142	0.062500
143	0.062500
144	0.062500
145	0.062500
146	0.056250
147	0.056250
148	0.056250
149	0.056250
150	0.056250
151	0.037500
152	0.037500
153	0.037500
154	0.037500
155	0.037500
156	0.087500
157	0.075000
158	0.075000
159	0.062500
160	0.062500
161	0.056250
162	0.037500
163	0.028125
164	0.046875
165	0.056250
166	0.065625
167	0.056250

168	0.046875
169	0.042188
170	0.062500
171	0.037500
172	0.075000
173	0.087500
174	0.075000
175	0.062500
176	0.056250
177	0.046875
178	0.028125
179	0.056250
180	0.065625
181	0.056250
182	0.046875
183	0.042188
184	0.075000
185	0.075000
186	0.087500
187	0.087500
188	0.075000
189	0.075000
190	0.075000
191	0.075000
192	0.087500
193	0.075000
194	0.075000
195	0.075000
196	0.087500
197	0.075000
198	0.062500
199	0.062500
200	0.062500
201	0.062500
202	0.062500
203	0.062500
204	0.062500
205	0.062500
206	0.062500
207	0.062500
208	0.037500
209	0.037500
210	0.037500
211	0.037500
212	0.037500
213	0.056250
214	0.056250
215	0.056250
216	0.056250
217	0.056250
218	0.056250
219	0.062500
220	0.075000
221	0.087500
222	0.075000
223	0.062500

224	0.037500
225	0.042188
226	0.046875
227	0.056250
228	0.065625
229	0.056250
230	0.046875
231	0.028125
232	0.062500
233	0.056250
234	0.075000
235	0.087500
236	0.075000
237	0.062500
238	0.037500
239	0.046875
240	0.042188
241	0.056250
242	0.065625
243	0.056250
244	0.046875
245	0.028125
246	0.056250
247	0.056250
248	0.056250
249	0.056250
250	0.056250
251	0.056250
252	0.042187
253	0.056250
254	0.042187
255	0.056250
256	0.056250
257	0.056250
258	0.056250
259	0.056250
260	0.056250
261	0.042187
262	0.056250
263	0.042187
273	0.050000
274	0.050000
275	0.050000
276	0.050000
277	0.050000
278	0.050000
279	0.037500
280	0.050000
281	0.037500
282	0.056250
283	0.056250
284	0.056250
285	0.056250
286	0.056250
287	0.056250
288	0.042187

289	0.056250
290	0.042187
291	0.056250
292	0.056250
293	0.056250
294	0.056250
295	0.056250
296	0.056250
297	0.042187
298	0.056250
299	0.042187
309	0.012500
310	0.012500
311	0.012500
312	0.012500
313	0.012500
314	0.012500
315	0.009375
316	0.012500
317	0.009375
327	0.012500
328	0.012500
329	0.012500
330	0.012500
331	0.012500
332	0.012500
333	0.009375
334	0.012500
335	0.009375
345	0.018750
346	0.031250
347	0.037500
348	0.043750
349	0.037500
350	0.031250
351	0.028125
352	0.028125
353	0.031250
354	0.037500
355	0.043750
356	0.037500
357	0.031250
358	0.018750
359	0.028125
360	0.028125
362	0.025000
363	0.028125
364	0.028125
366	0.006250
368	0.006250
395	0.031250
396	0.018750
397	0.037500
398	0.043750
399	0.037500
400	0.031250

401	0.028125
402	0.031250
403	0.028125
404	0.037500
405	0.043750
406	0.037500
407	0.031250
408	0.018750
409	0.028125
410	0.028125
412	0.025000
413	0.028125
414	0.028125
416	0.006250
418	0.006250
419	0.015625
420	0.009375
421	0.018750
422	0.021875
423	0.018750
424	0.015625
425	0.014063
426	0.015625
427	0.014063
428	0.018750
429	0.021875
430	0.018750
431	0.015625
432	0.009375
433	0.014063
434	0.014063
435	0.014063
436	0.012500
437	0.014063
438	0.003125
439	0.003125
440	0.009375
441	0.015625
442	0.018750
443	0.021875
444	0.018750
445	0.015625
446	0.014063
447	0.014063
448	0.015625
449	0.018750
450	0.021875
451	0.018750
452	0.015625
453	0.009375
454	0.014063
455	0.014063
456	0.014063
457	0.012500
458	0.014063
459	0.003125

460	0.003125
461	0.012500
462	0.012500
463	0.012500
464	0.012500
465	0.012500
466	0.012500
467	0.009375
468	0.012500
469	0.009375
470	0.006250
471	0.006250
472	0.003125
473	0.003125
474	0.012500
475	0.012500
476	0.012500
477	0.012500
478	0.012500
479	0.012500
480	0.009375
481	0.012500
482	0.009375
483	0.006250
484	0.006250
485	0.003125
486	0.003125
487	0.006250
488	0.006250
489	0.006250
490	0.006250
491	0.006250
492	0.006250
493	0.004688
494	0.006250
495	0.004688
496	0.003125
497	0.003125
498	0.001563
499	0.001563
500	0.006250
501	0.006250
502	0.006250
503	0.006250
504	0.006250
505	0.006250
506	0.004688
507	0.006250
508	0.004688
509	0.003125
510	0.003125
511	0.001563
512	0.001563
TOTAL	12.500000

430. *
431. *

```
*
****** LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 100 1.4XD
1 1.4
*
LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
1 1.2 20 1.2 2 1.6
*
LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
1 1.2 20 1.2 3 1.6
*
LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
1 1.2 21 1.2 2 1.6
*
LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
1 1.2 21 1.2 3 1.6
*
LOAD COMB 105 1.2XD + 0.8XWL(+Z)
1 1.2 11 0.8
*
LOAD COMB 106 1.2XD + 0.8XWL(+X)
1 1.2 13 0.8
*
LOAD COMB 107 1.2XD + 0.8XWL(-Z)
1 1.2 12 0.8
*
LOAD COMB 108 1.2XD + 0.8XWL(-X)
1 1.2 14 0.8
*
LOAD COMB 109 0.9XD + 1.6XWL(+Z)
1 0.9 11 1.6
*
LOAD COMB 110 0.9XD + 1.6XWL(+X)
1 0.9 13 1.6
*
LOAD COMB 111 0.9XD + 1.6XWL(-Z)
1 0.9 12 1.6
*
LOAD COMB 112 0.9XD + 1.6XWL(-X)
1 0.9 14 1.6
*
LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
1 1.2 2 0.5 1.6
*
LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
1 1.2 2 0.5 1.6
*
LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
1 1.2 2 0.5 1.6
*
LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
1 1.2 2 0.5 1.6
*
LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
1 1.2 3 0.5 1.6
*
```

```
488. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
489. 1 1.2 3 0.5 1.6
490. *
491. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
492. 1 1.2 3 0.5 1.6
493. *
494. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
495. 1 1.2 3 0.5 1.6
496. *
497. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
498. 1 1.2 2 0.5 1
499. *
500. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
501. 1 1.2 2 0.5 1
502. *
503. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
504. 1 1.2 2 0.5 1
505. *
506. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
507. 1 1.2 2 0.5 1
508. *
509. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
510. 1 1.2 3 0.5 1
511. *
512. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
513. 1 1.2 3 0.5 1
514. *
515. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
516. 1 1.2 3 0.5 1
517. *
518. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
519. 1 1.2 3 0.5 1
520. *
521. LOAD COMB 129 0.9XD + EL(+Z)
522. 1 0.9 30 1.0
523. *
524. LOAD COMB 130 0.9XD + EL(+X)
525. 1 0.9 32 1.0
526. *
527. LOAD COMB 131 0.9XD + EL(-Z)
528. 1 0.9 31 1.0
529. *
530. LOAD COMB 132 0.9XD + EL(-X)
531. 1 0.9 33 1.0
532. PERFORM ANALYSIS
533. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 17,2020 TIME= 14:10:15 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUL 17, 2020                 *
*           Time=    11:23: 7                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\RCO1 _RCO2\CIVIL\Working\F3\F3.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-JULY-20
     4. *FOUNDATION F12 (TW-27)
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. SET NL 100
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 0 0.35 -2.5; 2 0 0.35 0.5; 3 2.5 0.35 -2.5; 4 2.5 0.35 0.5; 23 2.5 -1.5 0.5
    11. 121 0 -1.5 0.5; 122 0.3 -1.5 0.5; 123 0.3 -1.5 0.8; 124 0 -1.5 0.8
    12. 125 0 -1.5 0.2; 126 0.3 -1.5 0.2; 127 -0.3 -1.5 0.5; 128 -0.3 -1.5 0.8
    13. 129 -0.3 -1.5 0.2; 130 0.3 -1.5 1.1; 131 0 -1.5 1.1; 132 -0.3 -1.5 1.1
    14. 133 0 -1.5 -0.1; 134 0.3 -1.5 -0.1; 135 -0.3 -1.5 -0.1; 136 0.6 -1.5 0.5
    15. 137 0.6 -1.5 0.8; 138 0.6 -1.5 0.2; 139 0.6 -1.5 1.1; 140 0.6 -1.5 -0.1
    16. 141 -0.6 -1.5 -0.1; 142 -0.6 -1.5 0.2; 143 -0.6 -1.5 0.5; 144 -0.6 -1.5 0.8
    17. 145 -0.6 -1.5 1.1; 146 0.9 -1.5 -0.1; 147 0.9 -1.5 0.2; 148 0.9 -1.5 0.5
    18. 149 0.9 -1.5 0.8; 150 0.9 -1.5 1.1; 156 0 -1.5 -0.4; 157 0.3 -1.5 -0.4
    19. 158 -0.3 -1.5 -0.4; 159 0.6 -1.5 -0.4; 160 -0.6 -1.5 -0.4; 161 0.9 -1.5 -0.4
    20. 164 -0.6 -1.5 -0.7; 165 -0.3 -1.5 -0.7; 166 0 -1.5 -0.7; 167 0.3 -1.5 -0.7
    21. 168 0.6 -1.5 -0.7; 169 0.9 -1.5 -0.7; 184 2.8 -1.5 0.5; 185 2.8 -1.5 0.8
    22. 186 2.5 -1.5 0.8; 187 2.5 -1.5 0.2; 188 2.8 -1.5 0.2; 189 2.2 -1.5 0.5
    23. 190 2.2 -1.5 0.8; 191 2.2 -1.5 0.2; 192 2.5 -1.5 1.1; 193 2.2 -1.5 1.1
    24. 194 2.8 -1.5 1.1; 195 2.2 -1.5 -0.1; 196 2.5 -1.5 -0.1; 197 2.8 -1.5 -0.1
    25. 198 3.1 -1.5 -0.1; 199 3.1 -1.5 0.2; 200 3.1 -1.5 0.5; 201 3.1 -1.5 0.8
    26. 202 3.1 -1.5 1.1; 203 1.9 -1.5 -0.1; 204 1.9 -1.5 0.2; 205 1.9 -1.5 0.5
    27. 206 1.9 -1.5 0.8; 207 1.9 -1.5 1.1; 213 1.6 -1.5 -0.1; 214 1.6 -1.5 0.2
    28. 215 1.6 -1.5 0.5; 216 1.6 -1.5 0.8; 217 1.6 -1.5 1.1; 218 1.6 -1.5 -0.4
    29. 219 1.9 -1.5 -0.4; 220 2.2 -1.5 -0.4; 221 2.5 -1.5 -0.4; 222 2.8 -1.5 -0.4
    30. 223 3.1 -1.5 -0.4; 225 1.6 -1.5 -0.7; 226 1.9 -1.5 -0.7; 227 2.2 -1.5 -0.7
    31. 228 2.5 -1.5 -0.7; 229 2.8 -1.5 -0.7; 230 3.1 -1.5 -0.7; 246 1.1 -1.5 -0.1
    32. 247 1.1 -1.5 0.2; 248 1.1 -1.5 0.5; 249 1.1 -1.5 0.8; 250 1.1 -1.5 1.1
    33. 251 1.1 -1.5 -0.4; 252 1.1 -1.5 -0.7; 273 1.4 -1.5 -0.1; 274 1.4 -1.5 0.2
    34. 275 1.4 -1.5 0.5; 276 1.4 -1.5 0.8; 277 1.4 -1.5 1.1; 278 1.4 -1.5 -0.4
    35. 279 1.4 -1.5 -0.7; 527 1.25 -1.5 0.2; 528 1.25 -1.5 -0.1; 531 1.25 -1.5 0.5
    36. 533 1.25 -1.5 0.8; 535 1.25 -1.5 1.1; 537 1.25 -1.5 -0.4; 539 1.25 -1.5 -0.7
    37. 808 -0.6 -1.5 -1; 810 -0.3 -1.5 -1; 811 0 -1.5 -1; 812 0.3 -1.5 -1.
    38. 813 0.6 -1.5 -1; 814 0.9 -1.5 -1; 815 1.9 -1.5 -1; 816 1.6 -1.5 -1.
```

39. 817 2.2 -1.5 -1; 818 2.5 -1.5 -1; 819 2.8 -1.5 -1; 820 3.1 -1.5 -1.
40. 822 1.1 -1.5 -1; 823 1.4 -1.5 -1; 845 1.25 -1.5 -1; 927 0 -1.5 -1.6
41. 928 0.3 -1.5 -1.6; 929 0.3 -1.5 -1.9; 930 0 -1.5 -1.9; 931 0.3 -1.5 -2.2
42. 932 0 -1.5 -2.2; 933 -0.3 -1.5 -1.6; 934 -0.3 -1.5 -1.9; 935 -0.3 -1.5 -2.2
43. 936 -0.3 -1.5 -1.3; 937 0 -1.5 -1.3; 938 0.3 -1.5 -1.3; 939 0 -1.5 -2.5
44. 940 -0.3 -1.5 -2.5; 941 0.3 -1.5 -2.5; 942 0.6 -1.5 -2.2; 943 0.6 -1.5 -2.5
45. 944 0.6 -1.5 -1.9; 945 0.6 -1.5 -1.6; 946 0.6 -1.5 -1.3; 947 -0.6 -1.5 -2.2
46. 948 -0.6 -1.5 -2.5; 949 -0.6 -1.5 -1.9; 950 -0.6 -1.5 -1.6; 951 -0.6 -1.5 -1.3
47. 952 0.9 -1.5 -2.2; 953 0.9 -1.5 -2.5; 954 0.9 -1.5 -1.9; 955 0.9 -1.5 -1.6
48. 956 0.9 -1.5 -1.3; 962 -0.6 -1.5 -2.8; 964 -0.3 -1.5 -2.8; 965 0 -1.5 -2.8
49. 966 0.3 -1.5 -2.8; 967 0.6 -1.5 -2.8; 968 0.9 -1.5 -2.8; 969 -0.6 -1.5 -3.1
50. 971 -0.3 -1.5 -3.1; 972 0 -1.5 -3.1; 973 0.3 -1.5 -3.1; 974 0.6 -1.5 -3.1
51. 975 0.9 -1.5 -3.1; 976 2.5 -1.5 -1.6; 977 2.8 -1.5 -1.6; 978 2.8 -1.5 -1.9
52. 979 2.5 -1.5 -1.9; 980 2.8 -1.5 -2.2; 981 2.5 -1.5 -2.2; 982 2.2 -1.5 -1.6
53. 983 2.2 -1.5 -1.9; 984 2.2 -1.5 -2.2; 985 2.2 -1.5 -1.3; 986 2.5 -1.5 -1.3
54. 987 2.8 -1.5 -1.3; 988 2.5 -1.5 -2.5; 989 2.2 -1.5 -2.5; 990 2.8 -1.5 -2.5
55. 991 3.1 -1.5 -2.2; 992 3.1 -1.5 -2.5; 993 3.1 -1.5 -1.9; 994 3.1 -1.5 -1.6
56. 995 3.1 -1.5 -1.3; 996 1.9 -1.5 -2.2; 997 1.9 -1.5 -2.5; 998 1.9 -1.5 -1.9
57. 999 1.9 -1.5 -1.6; 1000 1.9 -1.5 -1.3; 1006 1.6 -1.5 -2.2; 1007 1.6 -1.5 -2.5
58. 1008 1.6 -1.5 -1.9; 1009 1.6 -1.5 -1.6; 1010 1.6 -1.5 -1.3; 1011 1.9 -1.5 -2.8
59. 1012 1.6 -1.5 -2.8; 1013 2.2 -1.5 -2.8; 1014 2.5 -1.5 -2.8; 1015 2.8 -1.5 -2.8
60. 1016 3.1 -1.5 -2.8; 1018 1.9 -1.5 -3.1; 1019 1.6 -1.5 -3.1; 1020 2.2 -1.5 -3.1
61. 1021 2.5 -1.5 -3.1; 1022 2.8 -1.5 -3.1; 1023 3.1 -1.5 -3.1; 1025 1.1 -1.5 -2.2
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212. 1216 1334 1332 1254 1292
213. START GROUP DEFINITION
214. FLOOR
215. ELEMENT
216. _FDN_12_MOMENT 54 TO 69 75 TO 79 81 TO 85 102 TO 113 118 TO 126 128 TO 132 -
217. 146 TO 151 170 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 -
218. 430 439 815 TO 820 823 TO 834 840 843 844 846 TO 856 859 TO 866 871 TO 876 -
219. 879 881 TO 885 887 TO 904 934 936 938 940 942 944 1014 TO 1019 1021 1031 -
220. 1032 TO 1035 1037 1038 1041 TO 1047 1058 TO 1071 1082 TO 1102 1113 TO 1128 -
221. 1139 TO 1147 1150 TO 1165 1168 TO 1190 1193 TO 1210 1213 TO 1216
222. _FDN_12_SHEAR 81 TO 85 128 TO 132 146 TO 151 170 TO 175 178 TO 183 -
223. 243 TO 252 254 257 258 415 418 421 424 427 430 439 819 820 826 830 834 855 -
224. 856 862 866 874 887 TO 904 934 936 938 940 942 944 1014 TO 1019 1021 1031 -
225. 1032 TO 1035 1037 1038 1046 1047 1061 1064 1070 1071 1085 1088 1099 TO 1102 -
226. 1125 TO 1128 1141 TO 1147 1150 TO 1165 1168 TO 1190 1193 TO 1210 -
227. 1213 TO 1216
228. END GROUP DEFINITION
229. ELEMENT PROPERTY
230. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
231. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
232. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
233. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
234. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
235. 1193 TO 1210 1213 TO 1216 THICKNESS 0.5
236. DEFINE MATERIAL START
237. ISOTROPIC CONCRETE
238. E 2.17185E+007
239. POISSON 0.17
240. DENSITY 25
241. ALPHA 1E-005
242. DAMP 0.05
243. TYPE CONCRETE
244. STRENGTH FCU 27579
245. END DEFINE MATERIAL
246. MEMBER PROPERTY AMERICAN
247. 48 49 1011 1012 PRIS YD 0.6 ZD 0.6
248. CONSTANTS
249. MATERIAL CONCRETE ALL
250. SUPPORTS
251. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
252. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
253. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
254. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
255. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
256. 1193 TO 1210 1213 TO 1216 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
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128	0.090000
129	0.090000
130	0.075000
131	0.075000
132	0.075000
133	0.090000
134	0.090000
135	0.090000
136	0.090000
137	0.090000
138	0.090000
139	0.075000
140	0.090000
141	0.075000
142	0.075000
143	0.075000
144	0.075000
145	0.062500
146	0.075000
147	0.075000
148	0.075000
149	0.075000
150	0.062500
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252	0.052500
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816	0.075000
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820	0.075000
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992	0.075000
993	0.075000
994	0.075000
995	0.075000
996	0.090000
997	0.090000
998	0.090000
999	0.090000
1000	0.090000
1006	0.075000
1007	0.075000
1008	0.075000
1009	0.075000
1010	0.075000
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1013	0.090000
1014	0.090000
1015	0.090000
1016	0.075000
1018	0.075000
1019	0.062500
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1021	0.075000
1022	0.075000
1023	0.062500
1025	0.052500
1026	0.052500
1027	0.052500
1028	0.052500
1029	0.052500
1030	0.052500
1031	0.043750
1032	0.052500
1033	0.052500
1034	0.052500
1035	0.052500
1036	0.052500
1037	0.052500
1038	0.043750
1079	0.045000
1080	0.045000
1083	0.045000
1085	0.045000
1087	0.045000
1089	0.045000
1091	0.037500
1151	0.060000

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1153	0.060000
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1155	0.050000
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1174	0.060000
1175	0.050000
1176	0.060000
1177	0.060000
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1179	0.060000
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1181	0.060000
1182	0.060000
1193	0.060000
1194	0.060000
1195	0.060000
1196	0.060000
1197	0.060000
1198	0.060000
1199	0.060000
1200	0.050000
1201	0.060000
1202	0.050000
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1204	0.060000
1205	0.060000
1206	0.050000
1207	0.060000
1208	0.050000
1209	0.060000
1210	0.060000
1211	0.060000
1212	0.050000
1213	0.035000
1214	0.035000
1216	0.030000
1226	0.040000
1227	0.040000
1228	0.060000
1229	0.060000
1230	0.060000
1231	0.060000
1232	0.050000
1233	0.050000
1234	0.060000
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1236	0.060000
1237	0.050000

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1239	0.050000
1240	0.035000
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1253	0.040000
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1259	0.025000
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1264	0.030000
1265	0.030000
1266	0.030000
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1270	0.030000
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1274	0.030000
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1277	0.030000
1278	0.025000
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1288	0.030000
1289	0.030000
1290	0.025000
1291	0.020000
1292	0.020000
1293	0.030000
1294	0.025000
1295	0.030000
1296	0.030000
1297	0.030000
1298	0.025000
1299	0.030000
1300	0.025000
1301	0.030000
1302	0.030000
1303	0.030000
1304	0.025000
1305	0.017500
1306	0.017500
1308	0.015000

```
      1310          0.020000
      1311          0.020000
      1312          0.010000
      1313          0.010000
      1314          0.030000
      1315          0.030000
      1316          0.030000
      1317          0.030000
      1318          0.025000
      1319          0.025000
      1320          0.030000
      1321          0.030000
      1322          0.030000
      1323          0.025000
      1324          0.030000
      1325          0.025000
      1326          0.017500
      1327          0.017500
      1329          0.015000
      1331          0.020000
      1332          0.020000
      1333          0.010000
      1334          0.010000
    TOTAL          22.499998
257. ***
258. ************************************************
259. ***LOAD COMBINATION FOR STRESS DESIGN
260. ************************************************
261. ***
262. ***
263. LOAD 1 LOADTYPE DEAD  TITLE DL
264. ***
265. SELFWEIGHT Y -1.1
266. ***FROM GALLERY-1
267. JOINT LOAD
268. 1 FX -0.198 FY -16.572 FZ -0.156
269. 2 FX -0.195 FY -16.993 FZ 0.235
270. 3 FX 0.195 FY -6.408 FZ -0.233
271. 4 FX 0.198 FY -6.44 FZ 0.237
274. ELEMENT LOAD
275. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
276. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
277. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
278. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
279. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
280. 1193 TO 1210 1213 TO 1216 PR GY -20
281. LOAD 2 LOADTYPE LIVE  TITLE LL
282. JOINT LOAD
283. 1 FX -0.426 FY 5.664 FZ 11.234
284. 2 FX 0.418 FY -84.061 FZ 11.231
285. 3 FX -0.423 FY 4.009 FZ 0.509
286. 4 FX 0.431 FY -4.009 FZ 0.509
288. *
289. LOAD 3 LOADTYPE LIVE  TITLE LL1
290. JOINT LOAD
291. 1 FX 0.356 FY -86.726 FZ -8.39
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292. 2 FX -0.365 FY 8.328 FZ -8.394
293. 3 FX 0.371 FY -3.524 FZ -0.436
294. 4 FX -0.363 FY 3.524 FZ -0.436
296. *
297. LOAD 11 LOADTYPE WIND  TITLE WL+Z
298. JOINT LOAD
299. 1 FX -0.361 FY 48.484 FZ 9.359
300. 2 FX 0.361 FY -48.484 FZ 9.359
301. 3 FX -0.367 FY 5.554 FZ 1.312
302. 4 FX 0.367 FY -5.554 FZ 1.312
304. *
305. LOAD 12 LOADTYPE WIND  TITLE WL-Z
306. JOINT LOAD
307. 1 FX 0.361 FY -48.484 FZ -9.359
308. 2 FX -0.361 FY 48.484 FZ -9.359
309. 3 FX 0.367 FY -5.554 FZ -1.312
310. 4 FX -0.367 FY 5.554 FZ -1.312
312. *
313. LOAD 13 LOADTYPE WIND  TITLE WL+X
314. JOINT LOAD
315. 1 FX 0.875 FY 2.226 FZ 0.002
316. 2 FX 0.875 FY 2.21 FZ 0.002
317. 3 FX 0.875 FY -2.449 FZ 0
318. 4 FX 0.875 FY -2.451 FZ 0
320. *
321. LOAD 14 LOADTYPE WIND  TITLE WL-X
322. JOINT LOAD
323. 1 FX -0.875 FY -2.226 FZ -0.002
324. 2 FX -0.875 FY -2.21 FZ -0.002
325. 3 FX -0.875 FY 2.449 FZ 0
326. 4 FX -0.875 FY 2.451 FZ 0
328. *
329. LOAD 20 LOADTYPE NONE  TITLE TL1
330. JOINT LOAD
331. 1 FX -0.014 FY 0 FZ -0.129
332. 2 FX -0.014 FY 0 FZ 0.129
333. 3 FX 0.014 FY 0 FZ -0.129
334. 4 FX 0.014 FY 0 FZ 0.129
336. *
337. LOAD 21 LOADTYPE NONE  TITLE TL2
338. JOINT LOAD
339. 1 FX 0.014 FY 0 FZ 0.129
340. 2 FX 0.014 FY 0 FZ -0.129
341. 3 FX -0.014 FY 0 FZ 0.129
342. 4 FX -0.014 FY 0 FZ -0.129
344. *
345. LOAD 30 EARTHQUAKE LOAD (+Z)
346. JOINT LOAD
347. 1 FX -0.024 FY 2.01 FZ 0.46
348. 2 FX 0.024 FY -2.01 FZ 0.461
349. 3 FX -0.024 FY 0.392 FZ 0.073
350. 4 FX 0.024 FY -0.392 FZ 0.073
352. *
353. LOAD 31 EARTHQUAKE LOAD (-Z)
354. JOINT LOAD
355. 1 FX 0.024 FY -2.01 FZ -0.46
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356. 2 FX -0.024 FY 2.01 FZ -0.461
357. 3 FX 0.024 FY -0.392 FZ -0.073
358. 4 FX -0.024 FY 0.392 FZ -0.073
360. *
361. LOAD 32 EARTHQUAKE LOAD (+X)
362. JOINT LOAD
363. 1 FX 0.29 FY 1.563 FZ -0.002
364. 2 FX 0.244 FY 1.32 FZ -0.003
365. 3 FX 0.29 FY -1.563 FZ 0.002
366. 4 FX 0.244 FY -1.32 FZ 0.003
368. *
369. LOAD 33 EARTHQUAKE LOAD (-X)
370. JOINT LOAD
371. 1 FX -0.29 FY -1.563 FZ 0.002
372. 2 FX -0.244 FY -1.32 FZ 0.003
373. 3 FX -0.29 FY 1.563 FZ -0.002
374. 4 FX -0.244 FY 1.32 FZ -0.003
375. *
376. ****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
377. *
378. LOAD COMB 201 D + L + TL1
379. 1 1.0 2 1.0 20 1.0
380. *
381. LOAD COMB 202 D + L1 + TL1
382. 1 1.0 3 1.0 20 1.0
383. *
384. LOAD COMB 203 D + L + TL2
385. 1 1.0 2 1.0 21 1.0
386. *
387. LOAD COMB 204 D + L1 + TL2
388. 1 1.0 3 1.0 21 1.0
389. *
390. LOAD COMB 205 D + 0.75XL + 0.75XTL1
391. 1 1.0 2 0.75 20 0.75
392. *
393. LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
394. 1 1.0 3 0.75 20 0.75
395. *
396. LOAD COMB 207 D + 0.75XL + 0.75XTL2
397. 1 1.0 2 0.75 21 0.75
398. *
399. LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
400. 1 1.0 3 0.75 21 0.75
401. *
402. LOAD COMB 209 D + WL(+Z)
403. 1 1.0 11 1.0
404. *
405. LOAD COMB 210 D + WL(+X)
406. 1 1.0 13 1.0
407. *
408. LOAD COMB 211 D + WL(-Z)
409. 1 1.0 12 1.0
410. *
411. LOAD COMB 212 D + WL(-X)
412. 1 1.0 14 1.0
413. *
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414. LOAD COMB 213 D + 0.7XEL(+Z)
415. 1 1.0 30 0.7
416. *
417. LOAD COMB 214 D + 0.7XEL(+X)
418. 1 1.0 32 0.7
419. *
420. LOAD COMB 215 D + 0.7XEL(-Z)
421. 1 1.0 31 0.7
422. *
423. LOAD COMB 216 D + 0.7XEL(-X)
424. 1 1.0 33 0.7
425. *
426. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
427. 1 1.0 2 0.75 11 0.75
428. *
429. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
430. 1 1.0 3 0.75 11 0.75
431. *
432. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
433. 1 1.0 2 0.75 12 0.75
434. *
435. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
436. 1 1.0 3 0.75 12 0.75
437. *
438. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
439. 1 1.0 2 0.75 13 0.75
440. *
441. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
442. 1 1.0 3 0.75 13 0.75
443. *
444. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
445. 1 1.0 2 0.75 14 0.75
446. *
447. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
448. 1 1.0 3 0.75 14 0.75
449. *
450. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
451. 1 1.0 2 0.75 30 0.525
452. *
453. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
454. 1 1.0 3 0.75 30 0.525
455. *
456. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
457. 1 1.0 2 0.75 32 0.525
458. *
459. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
460. 1 1.0 3 0.75 32 0.525
461. *
462. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
463. 1 1.0 2 0.75 31 0.525
464. *
465. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
466. 1 1.0 3 0.75 31 0.525
467. *
468. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
469. 1 1.0 2 0.75 33 0.525
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470. *
471. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
472. 1 1.0 3 0.75 33 0.525
473. *
474. LOAD COMB 233 0.6XD + WL(+Z)
475. 1 0.6 11 1.0
476. *
477. LOAD COMB 234 0.6XD + WL(+X)
478. 1 0.6 13 1.0
479. *
480. LOAD COMB 235 0.6XD + WL(-Z)
481. 1 0.6 12 1.0
482. *
483. LOAD COMB 236 0.6XD + WL(-X)
484. 1 0.6 14 1.0
485. *
486. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
487. 1 0.6 30 0.7
488. *
489. LOAD COMB 238 0.6XD + 0.7XEL(+X)
490. 1 0.6 32 0.7
491. *
492. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
493. 1 0.6 31 0.7
494. *
495. LOAD COMB 240 0.6XD + 0.7XEL(-X)
496. 1 0.6 33 0.7
497. *
498. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 365 NUMBER OF MEMBERS 4
NUMBER OF PLATES 324 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 361

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 265/ 32/ 99 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 1107
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 110 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.1/ 74008.8 MB

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499. CHANGE
500. SUPPORTS
501. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
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502. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
503. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
504. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
505. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
506. 1193 TO 1210 1213 TO 1215 -
507. 1216 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
23	0.090000
121	0.090000
122	0.090000
123	0.090000
124	0.090000
125	0.090000
126	0.090000
127	0.090000
128	0.090000
129	0.090000
130	0.075000
131	0.075000
132	0.075000
133	0.090000
134	0.090000
135	0.090000
136	0.090000
137	0.090000
138	0.090000
139	0.075000
140	0.090000
141	0.075000
142	0.075000
143	0.075000
144	0.075000
145	0.062500
146	0.075000
147	0.075000
148	0.075000
149	0.075000
150	0.062500
156	0.090000
157	0.090000
158	0.090000
159	0.090000
160	0.075000
161	0.075000
164	0.075000
165	0.090000
166	0.090000
167	0.090000
168	0.090000
169	0.075000
184	0.090000
185	0.090000
186	0.090000

187	0.090000
188	0.090000
189	0.090000
190	0.090000
191	0.090000
192	0.075000
193	0.075000
194	0.075000
195	0.090000
196	0.090000
197	0.090000
198	0.075000
199	0.075000
200	0.075000
201	0.075000
202	0.062500
203	0.090000
204	0.090000
205	0.090000
206	0.090000
207	0.075000
213	0.075000
214	0.075000
215	0.075000
216	0.075000
217	0.062500
218	0.075000
219	0.090000
220	0.090000
221	0.090000
222	0.090000
223	0.075000
225	0.075000
226	0.090000
227	0.090000
228	0.090000
229	0.090000
230	0.075000
246	0.052500
247	0.052500
248	0.052500
249	0.052500
250	0.043750
251	0.052500
252	0.052500
273	0.052500
274	0.052500
275	0.052500
276	0.052500
277	0.043750
278	0.052500
279	0.052500
527	0.045000
528	0.045000
531	0.045000
533	0.045000

535	0.037500
537	0.045000
539	0.045000
808	0.075000
810	0.090000
811	0.090000
812	0.090000
813	0.090000
814	0.075000
815	0.090000
816	0.075000
817	0.090000
818	0.090000
819	0.090000
820	0.075000
822	0.052500
823	0.052500
845	0.045000
927	0.090000
928	0.090000
929	0.090000
930	0.090000
931	0.090000
932	0.090000
933	0.090000
934	0.090000
935	0.090000
936	0.090000
937	0.090000
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940	0.090000
941	0.090000
942	0.090000
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944	0.090000
945	0.090000
946	0.090000
947	0.075000
948	0.075000
949	0.075000
950	0.075000
951	0.075000
952	0.075000
953	0.075000
954	0.075000
955	0.075000
956	0.075000
962	0.075000
964	0.090000
965	0.090000
966	0.090000
967	0.090000
968	0.075000
969	0.062500
971	0.075000

972	0.075000
973	0.075000
974	0.075000
975	0.062500
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983	0.090000
984	0.090000
985	0.090000
986	0.090000
987	0.090000
988	0.090000
989	0.090000
990	0.090000
991	0.075000
992	0.075000
993	0.075000
994	0.075000
995	0.075000
996	0.090000
997	0.090000
998	0.090000
999	0.090000
1000	0.090000
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1008	0.075000
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1010	0.075000
1011	0.090000
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1014	0.090000
1015	0.090000
1016	0.075000
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1019	0.062500
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1021	0.075000
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1023	0.062500
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1026	0.052500
1027	0.052500
1028	0.052500
1029	0.052500
1030	0.052500
1031	0.043750
1032	0.052500
1033	0.052500
1034	0.052500

1035	0.052500
1036	0.052500
1037	0.052500
1038	0.043750
1079	0.045000
1080	0.045000
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1085	0.045000
1087	0.045000
1089	0.045000
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1210	0.060000
1211	0.060000
1212	0.050000
1213	0.035000
1214	0.035000
1216	0.030000

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1227	0.040000
1228	0.060000
1229	0.060000
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1234	0.060000
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1240	0.035000
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1268	0.030000
1269	0.030000
1270	0.030000
1271	0.025000
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1277	0.030000
1278	0.025000
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1289	0.030000
1290	0.025000
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1292	0.020000
1293	0.030000
1294	0.025000
1295	0.030000

1296	0.030000
1297	0.030000
1298	0.025000
1299	0.030000
1300	0.025000
1301	0.030000
1302	0.030000
1303	0.030000
1304	0.025000
1305	0.017500
1306	0.017500
1308	0.015000
1310	0.020000
1311	0.020000
1312	0.010000
1313	0.010000
1314	0.030000
1315	0.030000
1316	0.030000
1317	0.030000
1318	0.025000
1319	0.025000
1320	0.030000
1321	0.030000
1322	0.030000
1323	0.025000
1324	0.030000
1325	0.025000
1326	0.017500
1327	0.017500
1329	0.015000
1331	0.020000
1332	0.020000
1333	0.010000
1334	0.010000
TOTAL	22.499998

```
508. *
509. *
510. *
511. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
512. LOAD COMB 100 1.4XD
513. 1 1.4
514. *
515. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
516. 1 1.2 20 1.2 2 1.6
517. *
518. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
519. 1 1.2 20 1.2 3 1.6
520. *
521. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
522. 1 1.2 21 1.2 2 1.6
523. *
524. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
525. 1 1.2 21 1.2 3 1.6
526. *
527. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
```

```
528. 1 1.2 11 0.8
529. *
530. LOAD COMB 106 1.2XD + 0.8XWL(+X)
531. 1 1.2 13 0.8
532. *
533. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
534. 1 1.2 12 0.8
535. *
536. LOAD COMB 108 1.2XD + 0.8XWL(-X)
537. 1 1.2 14 0.8
538. *
539. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
540. 1 0.9 11 1.6
541. *
542. LOAD COMB 110 0.9XD + 1.6XWL(+X)
543. 1 0.9 13 1.6
544. *
545. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
546. 1 0.9 12 1.6
547. *
548. LOAD COMB 112 0.9XD + 1.6XWL(-X)
549. 1 0.9 14 1.6
550. *
551. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
552. 1 1.2 2 0.5 1.6
553. *
554. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
555. 1 1.2 2 0.5 1.6
556. *
557. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
558. 1 1.2 2 0.5 1.6
559. *
560. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
561. 1 1.2 2 0.5 1.6
562. *
563. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
564. 1 1.2 3 0.5 1.6
565. *
566. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
567. 1 1.2 3 0.5 1.6
568. *
569. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
570. 1 1.2 3 0.5 1.6
571. *
572. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
573. 1 1.2 3 0.5 1.6
574. *
575. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
576. 1 1.2 2 0.5 1
577. *
578. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
579. 1 1.2 2 0.5 1
580. *
581. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
582. 1 1.2 2 0.5 1
583. *
```

```
584. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
585. 1 1.2 2 0.5 1
586. *
587. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
588. 1 1.2 3 0.5 1
589. *
590. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
591. 1 1.2 3 0.5 1
592. *
593. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
594. 1 1.2 3 0.5 1
595. *
596. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
597. 1 1.2 3 0.5 1
598. *
599. LOAD COMB 129 0.9XD + EL(+Z)
600. 1 0.9 30 1.0
601. *
602. LOAD COMB 130 0.9XD + EL(+X)
603. 1 0.9 32 1.0
604. *
605. LOAD COMB 131 0.9XD + EL(-Z)
606. 1 0.9 31 1.0
607. *
608. LOAD COMB 132 0.9XD + EL(-X)
609. 1 0.9 33 1.0
610. *
611. PERFORM ANALYSIS
612. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 17,2020 TIME= 11:23:15 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUL 17, 2020                 *
*           Time=    12: 5: 7                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\RCO1 _RCO2\CIVIL\Working\F4\F4.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-JULY-20
     4. *FOUNDATION F12 (TW-27)
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. SET NL 100
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 0 0.35 -2.5; 2 0 0.35 -0.5; 3 3 0.35 -2.5; 4 3 0.35 -0.5; 23 3 -1.5 -0.5
    11. 121 0 -1.5 -0.5; 122 0.3 -1.5 -0.5; 123 0.3 -1.5 -0.2; 124 0 -1.5 -0.2
    12. 125 0 -1.5 -0.8; 126 0.3 -1.5 -0.8; 127 -0.3 -1.5 -0.5; 128 -0.3 -1.5 -0.2
    13. 129 -0.3 -1.5 -0.8; 130 0.3 -1.5 0.1; 131 0 -1.5 0.1; 132 -0.3 -1.5 0.1
    14. 133 0 -1.5 -1.1; 134 0.3 -1.5 -1.1; 135 -0.3 -1.5 -1.1; 136 0.6 -1.5 -0.5
    15. 137 0.6 -1.5 -0.2; 138 0.6 -1.5 -0.8; 139 0.6 -1.5 0.1; 140 0.6 -1.5 -1.1
    16. 141 -0.6 -1.5 -1.1; 142 -0.6 -1.5 -0.8; 143 -0.6 -1.5 -0.5; 144 -0.6 -1.5 -0.2
    17. 145 -0.6 -1.5 0.1; 146 0.9 -1.5 -1.1; 147 0.9 -1.5 -0.8; 148 0.9 -1.5 -0.5
    18. 149 0.9 -1.5 -0.2; 150 0.9 -1.5 0.1; 156 0 -1.5 -1.4; 157 0.3 -1.5 -1.4
    19. 158 -0.3 -1.5 -1.4; 159 0.6 -1.5 -1.4; 160 -0.6 -1.5 -1.4; 161 0.9 -1.5 -1.4
    20. 164 -0.6 -1.5 -1.45; 165 -0.3 -1.5 -1.45; 166 0 -1.5 -1.45; 167 0.3 -1.5 -1.45
    21. 168 0.6 -1.5 -1.45; 169 0.9 -1.5 -1.45; 184 3.3 -1.5 -0.5; 185 3.3 -1.5 -0.2
    22. 186 3 -1.5 -0.2; 187 3 -1.5 -0.8; 188 3.3 -1.5 -0.8; 189 2.7 -1.5 -0.5
    23. 190 2.7 -1.5 -0.2; 191 2.7 -1.5 -0.8; 192 3 -1.5 0.1; 193 2.7 -1.5 0.1
    24. 194 3.3 -1.5 0.1; 195 2.7 -1.5 -1.1; 196 3 -1.5 -1.1; 197 3.3 -1.5 -1.1
    25. 198 3.6 -1.5 -1.1; 199 3.6 -1.5 -0.8; 200 3.6 -1.5 -0.5; 201 3.6 -1.5 -0.2
    26. 202 3.6 -1.5 0.1; 203 2.4 -1.5 -1.1; 204 2.4 -1.5 -0.8; 205 2.4 -1.5 -0.5
    27. 206 2.4 -1.5 -0.2; 207 2.4 -1.5 0.1; 213 2.1 -1.5 -1.1; 214 2.1 -1.5 -0.8
    28. 215 2.1 -1.5 -0.5; 216 2.1 -1.5 -0.2; 217 2.1 -1.5 0.1; 218 2.1 -1.5 -1.4
    29. 219 2.4 -1.5 -1.4; 220 2.7 -1.5 -1.4; 221 3 -1.5 -1.4; 222 3.3 -1.5 -1.4
    30. 223 3.6 -1.5 -1.4; 225 2.1 -1.5 -1.45; 226 2.4 -1.5 -1.45; 227 2.7 -1.5 -1.45
    31. 228 3 -1.5 -1.45; 229 3.3 -1.5 -1.45; 230 3.6 -1.5 -1.45; 246 1.2 -1.5 -1.1
    32. 247 1.2 -1.5 -0.8; 248 1.2 -1.5 -0.5; 249 1.2 -1.5 -0.2; 250 1.2 -1.5 0.1
    33. 251 1.2 -1.5 -1.4; 252 1.2 -1.5 -1.45; 273 1.8 -1.5 -1.1; 274 1.8 -1.5 -0.8
    34. 275 1.8 -1.5 -0.5; 276 1.8 -1.5 -0.2; 277 1.8 -1.5 0.1; 278 1.8 -1.5 -1.4
    35. 279 1.8 -1.5 -1.45; 527 1.5 -1.5 -0.8; 528 1.5 -1.5 -1.1; 531 1.5 -1.5 -0.5
    36. 533 1.5 -1.5 -0.2; 535 1.5 -1.5 0.1; 537 1.5 -1.5 -1.4; 539 1.5 -1.5 -1.45
    37. 808 -0.6 -1.5 -1.5; 810 -0.3 -1.5 -1.5; 811 0 -1.5 -1.5; 812 0.3 -1.5 -1.5
    38. 813 0.6 -1.5 -1.5; 814 0.9 -1.5 -1.5; 815 2.4 -1.5 -1.5; 816 2.1 -1.5 -1.5
```

39. 817 2.7 -1.5 -1.5; 818 3 -1.5 -1.5; 819 3.3 -1.5 -1.5; 820 3.6 -1.5 -1.5
40. 822 1.2 -1.5 -1.5; 823 1.8 -1.5 -1.5; 845 1.5 -1.5 -1.5; 927 0 -1.5 -1.6
41. 928 0.3 -1.5 -1.6; 929 0.3 -1.5 -1.9; 930 0 -1.5 -1.9; 931 0.3 -1.5 -2.2
42. 932 0 -1.5 -2.2; 933 -0.3 -1.5 -1.6; 934 -0.3 -1.5 -1.9; 935 -0.3 -1.5 -2.2
43. 936 -0.3 -1.5 -1.55; 937 0 -1.5 -1.55; 938 0.3 -1.5 -1.55; 939 0 -1.5 -2.5
44. 940 -0.3 -1.5 -2.5; 941 0.3 -1.5 -2.5; 942 0.6 -1.5 -2.2; 943 0.6 -1.5 -2.5
45. 944 0.6 -1.5 -1.9; 945 0.6 -1.5 -1.6; 946 0.6 -1.5 -1.55; 947 -0.6 -1.5 -2.2
46. 948 -0.6 -1.5 -2.5; 949 -0.6 -1.5 -1.9; 950 -0.6 -1.5 -1.6
47. 951 -0.6 -1.5 -1.55; 952 0.9 -1.5 -2.2; 953 0.9 -1.5 -2.5; 954 0.9 -1.5 -1.9
48. 955 0.9 -1.5 -1.6; 956 0.9 -1.5 -1.55; 962 -0.6 -1.5 -2.8; 964 -0.3 -1.5 -2.8
49. 965 0 -1.5 -2.8; 966 0.3 -1.5 -2.8; 967 0.6 -1.5 -2.8; 968 0.9 -1.5 -2.8
50. 969 -0.6 -1.5 -3.1; 971 -0.3 -1.5 -3.1; 972 0 -1.5 -3.1; 973 0.3 -1.5 -3.1
51. 974 0.6 -1.5 -3.1; 975 0.9 -1.5 -3.1; 976 3 -1.5 -1.6; 977 3.3 -1.5 -1.6
52. 978 3.3 -1.5 -1.9; 979 3 -1.5 -1.9; 980 3.3 -1.5 -2.2; 981 3 -1.5 -2.2
53. 982 2.7 -1.5 -1.6; 983 2.7 -1.5 -1.9; 984 2.7 -1.5 -2.2; 985 2.7 -1.5 -1.55
54. 986 3 -1.5 -1.55; 987 3.3 -1.5 -1.55; 988 3 -1.5 -2.5; 989 2.7 -1.5 -2.5
55. 990 3.3 -1.5 -2.5; 991 3.6 -1.5 -2.2; 992 3.6 -1.5 -2.5; 993 3.6 -1.5 -1.9
56. 994 3.6 -1.5 -1.6; 995 3.6 -1.5 -1.55; 996 2.4 -1.5 -2.2; 997 2.4 -1.5 -2.5
57. 998 2.4 -1.5 -1.9; 999 2.4 -1.5 -1.6; 1000 2.4 -1.5 -1.55; 1006 2.1 -1.5 -2.2
58. 1007 2.1 -1.5 -2.5; 1008 2.1 -1.5 -1.9; 1009 2.1 -1.5 -1.6
59. 1010 2.1 -1.5 -1.55; 1011 2.4 -1.5 -2.8; 1012 2.1 -1.5 -2.8
60. 1013 2.7 -1.5 -2.8; 1014 3 -1.5 -2.8; 1015 3.3 -1.5 -2.8; 1016 3.6 -1.5 -2.8
61. 1018 2.4 -1.5 -3.1; 1019 2.1 -1.5 -3.1; 1020 2.7 -1.5 -3.1; 1021 3 -1.5 -3.1
62. 1022 3.3 -1.5 -3.1; 1023 3.6 -1.5 -3.1; 1025 1.2 -1.5 -2.2; 1026 1.2 -1.5 -2.5
63. 1027 1.2 -1.5 -1.9; 1028 1.2 -1.5 -1.6; 1029 1.2 -1.5 -1.55
64. 1030 1.2 -1.5 -2.8; 1031 1.2 -1.5 -3.1; 1032 1.8 -1.5 -2.2; 1033 1.8 -1.5 -2.5
65. 1034 1.8 -1.5 -1.9; 1035 1.8 -1.5 -1.6; 1036 1.8 -1.5 -1.55
66. 1037 1.8 -1.5 -2.8; 1038 1.8 -1.5 -3.1; 1079 1.5 -1.5 -2.2; 1080 1.5 -1.5 -2.5
67. 1083 1.5 -1.5 -1.9; 1085 1.5 -1.5 -1.6; 1087 1.5 -1.5 -1.55
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69. 1153 3.8 -1.5 -0.5; 1154 3.8 -1.5 -0.2; 1155 3.8 -1.5 0.1; 1156 3.8 -1.5 -1.4
70. 1157 3.8 -1.5 -1.45; 1168 3.8 -1.5 -1.5; 1169 3.8 -1.5 -2.2
71. 1170 3.8 -1.5 -2.5; 1171 3.8 -1.5 -1.9; 1172 3.8 -1.5 -1.6
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74. 1179 -0.8 -1.5 -0.2; 1180 -0.8 -1.5 0.1; 1181 -0.8 -1.5 -1.4
75. 1182 -0.8 -1.5 -1.45; 1193 -0.8 -1.5 -1.5; 1194 -0.8 -1.5 -2.2
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77. 1198 -0.8 -1.5 -1.55; 1199 -0.8 -1.5 -2.8; 1200 -0.8 -1.5 -3.1
78. 1201 -0.3 -1.5 -3.3; 1202 -0.6 -1.5 -3.3; 1203 0 -1.5 -3.3; 1204 0.3 -1.5 -3.3
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83. 1231 0.6 -1.5 0.3; 1232 -0.6 -1.5 0.3; 1233 0.9 -1.5 0.3; 1234 2.7 -1.5 0.3
84. 1235 3 -1.5 0.3; 1236 3.3 -1.5 0.3; 1237 3.6 -1.5 0.3; 1238 2.4 -1.5 0.3
85. 1239 2.1 -1.5 0.3; 1240 1.2 -1.5 0.3; 1241 1.8 -1.5 0.3; 1243 1.5 -1.5 0.3
86. 1253 3.8 -1.5 0.3; 1254 -0.8 -1.5 0.3; 1255 4 -1.5 -0.8; 1256 4 -1.5 -1.1
87. 1257 4 -1.5 -0.5; 1258 4 -1.5 -0.2; 1259 4 -1.5 0.1; 1260 4 -1.5 -1.4
88. 1261 4 -1.5 -1.45; 1264 4 -1.5 -1.5; 1265 4 -1.5 -2.2; 1266 4 -1.5 -2.5
89. 1267 4 -1.5 -1.9; 1268 4 -1.5 -1.6; 1269 4 -1.5 -1.55; 1270 4 -1.5 -2.8
90. 1271 4 -1.5 -3.1; 1272 4 -1.5 -3.3; 1273 4 -1.5 0.3; 1274 -1 -1.5 -0.8
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 95. 1293 -0.3 -1.5 -3.5; 1294 -0.6 -1.5 -3.5; 1295 0 -1.5 -3.5; 1296 0.3 -1.5 -3.5
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 97. 1301 2.7 -1.5 -3.5; 1302 3 -1.5 -3.5; 1303 3.3 -1.5 -3.5; 1304 3.6 -1.5 -3.5
 98. 1305 1.2 -1.5 -3.5; 1306 1.8 -1.5 -3.5; 1308 1.5 -1.5 -3.5; 1310 3.8 -1.5 -3.5
 99. 1311 -0.8 -1.5 -3.5; 1312 4 -1.5 -3.5; 1313 -1 -1.5 -3.5; 1314 -0.3 -1.5 0.5
100. 1315 0 -1.5 0.5; 1316 0.3 -1.5 0.5; 1317 0.6 -1.5 0.5; 1318 -0.6 -1.5 0.5
101. 1319 0.9 -1.5 0.5; 1320 2.7 -1.5 0.5; 1321 3 -1.5 0.5; 1322 3.3 -1.5 0.5
102. 1323 3.6 -1.5 0.5; 1324 2.4 -1.5 0.5; 1325 2.1 -1.5 0.5; 1326 1.2 -1.5 0.5
103. 1327 1.8 -1.5 0.5; 1329 1.5 -1.5 0.5; 1331 3.8 -1.5 0.5; 1332 -0.8 -1.5 0.5
104. 1333 4 -1.5 0.5; 1334 -1 -1.5 0.5
105. MEMBER INCIDENCES
106. 48 23 4; 49 121 2; 1011 988 3; 1012 939 1
107. ELEMENT INCIDENCES SHELL
108. 50 124 123 122 121; 51 121 122 126 125; 52 128 124 121 127; 53 127 121 125 129
109. 54 132 131 124 128; 55 131 130 123 124; 56 129 125 133 135; 57 125 126 134 133
110. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
111. 62 142 129 135 141; 63 143 127 129 142; 64 144 128 127 143; 65 145 132 128 144
112. 66 138 147 146 140; 67 136 148 147 138; 68 137 149 148 136; 69 139 150 149 137
113. 75 141 135 158 160; 76 135 133 156 158; 77 133 134 157 156; 78 134 140 159 157
114. 79 140 146 161 159; 81 160 158 165 164; 82 158 156 166 165; 83 156 157 167 166
115. 84 157 159 168 167; 85 159 161 169 168; 98 186 185 184 23; 99 23 184 188 187
116. 100 190 186 23 189; 101 189 23 187 191; 102 193 192 186 190
117. 103 192 194 185 186; 104 191 187 196 195; 105 187 188 197 196
118. 106 188 199 198 197; 107 184 200 199 188; 108 185 201 200 184
119. 109 194 202 201 185; 110 204 191 195 203; 111 205 189 191 204
120. 112 206 190 189 205; 113 207 193 190 206; 118 214 204 203 213
121. 119 215 205 204 214; 120 216 206 205 215; 121 217 207 206 216
122. 122 213 203 219 218; 123 203 195 220 219; 124 195 196 221 220
123. 125 196 197 222 221; 126 197 198 223 222; 128 218 219 226 225
124. 129 219 220 227 226; 130 220 221 228 227; 131 221 222 229 228
125. 132 222 223 230 229; 146 147 247 246 146; 147 148 248 247 147
126. 148 149 249 248 148; 149 150 250 249 149; 150 146 246 251 161
127. 151 161 251 252 169; 170 274 214 213 273; 171 275 215 214 274
128. 172 276 216 215 275; 173 277 217 216 276; 174 273 213 218 278
129. 175 278 218 225 279; 178 527 274 273 528; 179 531 275 274 527
130. 180 533 276 275 531; 181 535 277 276 533; 182 528 273 278 537
131. 183 537 278 279 539; 243 164 165 810 808; 244 165 166 811 810
132. 245 166 167 812 811; 246 167 168 813 812; 247 168 169 814 813
133. 248 225 226 815 816; 249 226 227 817 815; 250 227 228 818 817
134. 251 228 229 819 818; 252 229 230 820 819; 254 169 252 822 814
135. 257 279 225 816 823; 258 539 279 823 845; 415 247 527 528 246
136. 418 248 531 527 247; 421 249 533 531 248; 424 250 535 533 249
137. 427 246 528 537 251; 430 251 537 539 252; 439 252 539 845 822
138. 815 927 928 929 930; 816 930 929 931 932; 817 933 927 930 934
139. 818 934 930 932 935; 819 936 937 927 933; 820 937 938 928 927
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141. 824 929 944 942 931; 825 928 945 944 929; 826 938 946 945 928
142. 827 947 935 940 948; 828 949 934 935 947; 829 950 933 934 949
143. 830 951 936 933 950; 831 942 952 953 943; 832 944 954 952 942
144. 833 945 955 954 944; 834 946 956 955 945; 840 948 940 964 962
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146. 844 943 953 968 967; 846 962 964 971 969; 847 964 965 972 971
147. 848 965 966 973 972; 849 966 967 974 973; 850 967 968 975 974
148. 851 976 977 978 979; 852 979 978 980 981; 853 982 976 979 983
149. 854 983 979 981 984; 855 985 986 976 982; 856 986 987 977 976
150. 857 984 981 988 989; 858 981 980 990 988; 859 980 991 992 990
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151. 860 978 993 991 980; 861 977 994 993 978; 862 987 995 994 977
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153. 866 1000 985 982 999; 871 1006 996 997 1007; 872 1008 998 996 1006
154. 873 1009 999 998 1008; 874 1010 1000 999 1009; 875 1007 997 1011 1012
155. 876 997 989 1013 1011; 877 989 988 1014 1013; 878 988 990 1015 1014
156. 879 990 992 1016 1015; 881 1012 1011 1018 1019; 882 1011 1013 1020 1018
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158. 887 952 1025 1026 953; 888 954 1027 1025 952; 889 955 1028 1027 954
159. 890 956 1029 1028 955; 891 953 1026 1030 968; 892 968 1030 1031 975
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161. 896 1036 1010 1009 1035; 897 1033 1007 1012 1037; 898 1037 1012 1019 1038
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163. 902 1087 1036 1035 1085; 903 1080 1033 1037 1089; 904 1089 1037 1038 1091
164. 934 1025 1079 1080 1026; 936 1027 1083 1079 1025; 938 1028 1085 1083 1027
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166. 1014 808 810 936 951; 1015 810 811 937 936; 1016 811 812 938 937
167. 1017 812 813 946 938; 1018 813 814 956 946; 1019 814 822 1029 956
168. 1021 822 845 1087 1029; 1031 816 815 1000 1010; 1032 815 817 985 1000
169. 1033 817 818 986 985; 1034 818 819 987 986; 1035 819 820 995 987
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171. 1042 200 1153 1151 199; 1043 201 1154 1153 200; 1044 202 1155 1154 201
172. 1045 198 1152 1156 223; 1046 223 1156 1157 230; 1047 230 1157 1168 820
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174. 1061 995 1173 1172 994; 1062 992 1170 1174 1016; 1063 1016 1174 1175 1023
175. 1064 820 1168 1173 995; 1065 1176 142 141 1177; 1066 1178 143 142 1176
176. 1067 1179 144 143 1178; 1068 1180 145 144 1179; 1069 1177 141 160 1181
177. 1070 1181 160 164 1182; 1071 1182 164 808 1193; 1082 1194 947 948 1195
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183. 1098 1022 1023 1212 1211; 1099 975 1031 1213 1206; 1100 1038 1019 1208 1214
184. 1101 1091 1038 1214 1216; 1102 1031 1091 1216 1213; 1113 1023 1175 1226 1212
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186. 1117 1230 1231 139 130; 1118 1232 1228 132 145; 1119 1231 1233 150 139
187. 1120 1234 1235 192 193; 1121 1235 1236 194 192; 1122 1236 1237 202 194
188. 1123 1238 1234 193 207; 1124 1239 1238 207 217; 1125 1233 1240 250 150
189. 1126 1241 1239 217 277; 1127 1243 1241 277 535; 1128 1240 1243 535 250
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191. 1142 1153 1257 1255 1151; 1143 1154 1258 1257 1153; 1144 1155 1259 1258 1154
192. 1145 1152 1256 1260 1156; 1146 1156 1260 1261 1157; 1147 1157 1261 1264 1168
193. 1150 1169 1265 1266 1170; 1151 1171 1267 1265 1169; 1152 1172 1268 1267 1171
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201. 1176 1292 1254 1180 1278; 1177 1202 1201 1293 1294; 1178 1201 1203 1295 1293
202. 1179 1203 1204 1296 1295; 1180 1204 1205 1297 1296; 1181 1205 1206 1298 1297
203. 1182 1208 1207 1299 1300; 1183 1207 1209 1301 1299; 1184 1209 1210 1302 1301
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206. 1193 1212 1226 1310 1304; 1194 1227 1202 1294 1311; 1195 1226 1272 1312 1310

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208. 1199 1316 1317 1231 1230; 1200 1318 1314 1228 1232; 1201 1317 1319 1233 1231
209. 1202 1320 1321 1235 1234; 1203 1321 1322 1236 1235; 1204 1322 1323 1237 1236
210. 1205 1324 1320 1234 1238; 1206 1325 1324 1238 1239; 1207 1319 1326 1240 1233
211. 1208 1327 1325 1239 1241; 1209 1329 1327 1241 1243; 1210 1326 1329 1243 1240
212. 1213 1323 1331 1253 1237; 1214 1332 1318 1232 1254; 1215 1331 1333 1273 1253
213. 1216 1334 1332 1254 1292
214. START GROUP DEFINITION
215. FLOOR
216. ELEMENT
217. _FDN_12_MOMENT 54 TO 69 75 TO 79 81 TO 85 102 TO 113 118 TO 126 128 TO 132 -
218. 146 TO 151 170 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 -
219. 430 439 815 TO 820 823 TO 834 840 843 844 846 TO 856 859 TO 866 871 TO 876 -
220. 879 881 TO 885 887 TO 904 934 936 938 940 942 944 1014 TO 1019 1021 1031 -
221. 1032 TO 1035 1037 1038 1041 TO 1047 1058 TO 1071 1082 TO 1102 1113 TO 1128 -
222. 1139 TO 1147 1150 TO 1165 1168 TO 1190 1193 TO 1210 1213 TO 1216
223. _FDN_12_SHEAR 81 TO 85 128 TO 132 146 TO 151 170 TO 175 178 TO 183 -
224. 243 TO 252 254 257 258 415 418 421 424 427 430 439 819 820 826 830 834 855 -
225. 856 862 866 874 887 TO 904 934 936 938 940 942 944 1014 TO 1019 1021 1031 -
226. 1032 TO 1035 1037 1038 1046 1047 1061 1064 1070 1071 1085 1088 1099 TO 1102 -
227. 1125 TO 1128 1141 TO 1147 1150 TO 1165 1168 TO 1190 1193 TO 1210 -
228. 1213 TO 1216
229. END GROUP DEFINITION
230. ELEMENT PROPERTY
231. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
232. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
233. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
234. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
235. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
236. 1193 TO 1210 1213 TO 1216 THICKNESS 0.5
237. DEFINE MATERIAL START
238. ISOTROPIC CONCRETE
239. E 2.17185E+007
240. POISSON 0.17
241. DENSITY 25
242. ALPHA 1E-005
243. DAMP 0.05
244. TYPE CONCRETE
245. STRENGTH FCU 27579
246. END DEFINE MATERIAL
247. MEMBER PROPERTY AMERICAN
248. 48 49 1011 1012 PRIS YD 0.6 ZD 0.6
249. CONSTANTS
250. MATERIAL CONCRETE ALL
251. SUPPORTS
252. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
253. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
254. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
255. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
256. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
257. 1193 TO 1210 1213 TO 1216 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
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127	0.090000
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130	0.075000
131	0.075000
132	0.075000
133	0.090000
134	0.090000
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138	0.090000
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149	0.090000
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160	0.043750
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164	0.012500
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166	0.015000
167	0.015000
168	0.015000
169	0.015000
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186	0.090000
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188	0.090000
189	0.090000
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191	0.090000
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194	0.075000
195	0.090000
196	0.090000
197	0.090000

198	0.075000
199	0.075000
200	0.075000
201	0.075000
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206	0.090000
207	0.075000
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215	0.090000
216	0.090000
217	0.075000
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226	0.015000
227	0.015000
228	0.015000
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533	0.090000
535	0.075000
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539	0.015000
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812	0.015000
813	0.015000
814	0.015000
815	0.015000
816	0.015000

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823	0.015000
845	0.015000
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944	0.090000
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998	0.090000
999	0.052500
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1010	0.015000
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1015	0.090000
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1322	0.030000
1323	0.025000
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1333	0.010000
1334	0.010000
TOTAL	20.000017

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***
************************************************
***LOAD COMBINATION FOR STRESS DESIGN
************************************************
***
***
LOAD 1 LOADTYPE DEAD  TITLE DL
***
SELFWEIGHT Y -1.1
***FROM GALLERY-1
JOINT LOAD
1 FX -0.3 FY -6.884 FZ -0.131
2 FX -0.233 FY -6.597 FZ 0.137
3 FX 0.233 FY -30.32 FZ -0.148
4 FX 0.3 FY -30.716 FZ 0.136
ELEMENT LOAD
50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
1193 TO 1210 1213 TO 1216 PR GY -20
LOAD 2 LOADTYPE LIVE  TITLE LL
JOINT LOAD
1 FX 38.552 FY 179.625 FZ -0.057
2 FX 19.244 FY 90.109 FZ -0.102
3 FX 38.547 FY -133.533 FZ 12.878
4 FX 19.257 FY -231.297 FZ 12.95
*
LOAD 3 LOADTYPE LIVE  TITLE LL1
JOINT LOAD
```

```
289. 1 FX 18.995 FY 88.972 FZ 0.11
290. 2 FX 38.801 FY 180.761 FZ 0.065
291. 3 FX 19.007 FY -232.432 FZ -13.052
292. 4 FX 38.797 FY -132.397 FZ -12.98
293. *
294. LOAD 11 LOADTYPE WIND  TITLE WL+Z
295. JOINT LOAD
296. 1 FX 26.178 FY 128.772 FZ 1.382
297. 2 FX -26.178 FY -128.772 FZ 1.382
298. 3 FX 26.137 FY 283.869 FZ 43.165
299. 4 FX -26.137 FY -283.869 FZ 43.165
300. *
301. LOAD 12 LOADTYPE WIND  TITLE WL-Z
302. JOINT LOAD
303. 1 FX -26.178 FY -128.772 FZ -1.382
304. 2 FX 26.178 FY 128.772 FZ -1.382
305. 3 FX -26.137 FY -283.869 FZ -43.165
306. 4 FX 26.137 FY 283.869 FZ -43.165
307. *
308. LOAD 13 LOADTYPE WIND  TITLE WL+X
309. JOINT LOAD
310. 1 FX 13.367 FY 60.335 FZ 0.011
311. 2 FX 13.393 FY 60.452 FZ -0.009
312. 3 FX 13.365 FY -57.299 FZ -0.011
313. 4 FX 13.391 FY -57.248 FZ 0.007
314. *
315. LOAD 14 LOADTYPE WIND  TITLE WL-X
316. JOINT LOAD
317. 1 FX -13.367 FY -60.335 FZ -0.011
318. 2 FX -13.393 FY -60.452 FZ 0.009
319. 3 FX -13.365 FY 57.299 FZ 0.011
320. 4 FX -13.391 FY 57.248 FZ -0.007
321. *
322. LOAD 20 LOADTYPE NONE  TITLE TL1
323. JOINT LOAD
324. 1 FX -0.167 FY -1.45 FZ -0.194
325. 2 FX 0.134 FY 1.45 FZ -0.007
326. 3 FX -0.134 FY 1.45 FZ 0.007
327. 4 FX 0.167 FY -1.45 FZ 0.194
328. *
329. LOAD 21 LOADTYPE NONE  TITLE TL2
330. JOINT LOAD
331. 1 FX 0.167 FY 1.45 FZ 0.194
332. 2 FX -0.134 FY -1.45 FZ 0.007
333. 3 FX 0.134 FY -1.45 FZ -0.007
334. 4 FX -0.167 FY 1.45 FZ -0.194
335. *
336. LOAD 30 EARTHQUAKE LOAD (+Z)
337. JOINT LOAD
338. 1 FX 0.052 FY 0.658 FZ 0.079
339. 2 FX -0.052 FY -0.658 FZ 0.079
340. 3 FX 0.051 FY 3.827 FZ 0.582
341. 4 FX -0.051 FY -3.827 FZ 0.581
342. *
343. LOAD 31 EARTHQUAKE LOAD (-Z)
344. JOINT LOAD
```

```
1 FX -0.052 FY -0.658 FZ -0.079
2 FX 0.052 FY 0.658 FZ -0.079
3 FX -0.051 FY -3.827 FZ -0.582
4 FX 0.051 FY 3.827 FZ -0.581
*
LOAD 32 EARTHQUAKE LOAD (+X)
JOINT LOAD
1 FX 0.318 FY 1.442 FZ 0.001
2 FX 0.343 FY 1.548 FZ 0.001
3 FX 0.318 FY -1.442 FZ -0.001
4 FX 0.343 FY -1.548 FZ -0.001
*
LOAD 33 EARTHQUAKE LOAD (-X)
JOINT LOAD
1 FX -0.318 FY -1.442 FZ -0.001
2 FX -0.343 FY -1.548 FZ -0.001
3 FX -0.318 FY 1.442 FZ 0.001
4 FX -0.343 FY 1.548 FZ 0.001
*
****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
*
LOAD COMB 201 D + L + TL1
1 1.0 2 1.0 20 1.0
*
LOAD COMB 202 D + L1 + TL1
1 1.0 3 1.0 20 1.0
*
LOAD COMB 203 D + L + TL2
1 1.0 2 1.0 21 1.0
*
LOAD COMB 204 D + L1 + TL2
1 1.0 3 1.0 21 1.0
*
LOAD COMB 205 D + 0.75XL + 0.75XTL1
1 1.0 2 0.75 20 0.75
*
LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
1 1.0 3 0.75 20 0.75
*
LOAD COMB 207 D + 0.75XL + 0.75XTL2
1 1.0 2 0.75 21 0.75
*
LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
1 1.0 3 0.75 21 0.75
*
LOAD COMB 209 D + WL(+Z)
1 1.0 11 1.0
*
LOAD COMB 210 D + WL(+X)
1 1.0 13 1.0
*
LOAD COMB 211 D + WL(-Z)
1 1.0 12 1.0
*
LOAD COMB 212 D + WL(-X)
1 1.0 14 1.0
```

```
401. *
402. LOAD COMB 213 D + 0.7XEL(+Z)
403. 1 1.0 30 0.7
404. *
405. LOAD COMB 214 D + 0.7XEL(+X)
406. 1 1.0 32 0.7
407. *
408. LOAD COMB 215 D + 0.7XEL(-Z)
409. 1 1.0 31 0.7
410. *
411. LOAD COMB 216 D + 0.7XEL(-X)
412. 1 1.0 33 0.7
413. *
414. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
415. 1 1.0 2 0.75 11 0.75
416. *
417. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
418. 1 1.0 3 0.75 11 0.75
419. *
420. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
421. 1 1.0 2 0.75 12 0.75
422. *
423. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
424. 1 1.0 3 0.75 12 0.75
425. *
426. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
427. 1 1.0 2 0.75 13 0.75
428. *
429. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
430. 1 1.0 3 0.75 13 0.75
431. *
432. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
433. 1 1.0 2 0.75 14 0.75
434. *
435. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
436. 1 1.0 3 0.75 14 0.75
437. *
438. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
439. 1 1.0 2 0.75 30 0.525
440. *
441. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
442. 1 1.0 3 0.75 30 0.525
443. *
444. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
445. 1 1.0 2 0.75 32 0.525
446. *
447. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
448. 1 1.0 3 0.75 32 0.525
449. *
450. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
451. 1 1.0 2 0.75 31 0.525
452. *
453. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
454. 1 1.0 3 0.75 31 0.525
455. *
456. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
```

```
457. 1 1.0 2 0.75 33 0.525
458. *
459. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
460. 1 1.0 3 0.75 33 0.525
461. *
462. LOAD COMB 233 0.6XD + WL(+Z)
463. 1 0.6 11 1.0
464. *
465. LOAD COMB 234 0.6XD + WL(+X)
466. 1 0.6 13 1.0
467. *
468. LOAD COMB 235 0.6XD + WL(-Z)
469. 1 0.6 12 1.0
470. *
471. LOAD COMB 236 0.6XD + WL(-X)
472. 1 0.6 14 1.0
473. *
474. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
475. 1 0.6 30 0.7
476. *
477. LOAD COMB 238 0.6XD + 0.7XEL(+X)
478. 1 0.6 32 0.7
479. *
480. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
481. 1 0.6 31 0.7
482. *
483. LOAD COMB 240 0.6XD + 0.7XEL(-X)
484. 1 0.6 33 0.7
485. *
486. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 365 NUMBER OF MEMBERS 4
NUMBER OF PLATES 324 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 361

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 265/ 32/ 99 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 1107
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 110 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.1/ 73966.3 MB

```
487. CHANGE
488. SUPPORTS
```

```
489. 50 TO 69 75 TO 79 81 TO 85 98 TO 113 118 TO 126 128 TO 132 146 TO 151 170 -
490. 171 TO 175 178 TO 183 243 TO 252 254 257 258 415 418 421 424 427 430 439 815 -
491. 816 TO 834 840 TO 844 846 TO 866 871 TO 879 881 TO 885 887 TO 904 934 936 -
492. 938 940 942 944 1014 TO 1019 1021 1031 TO 1035 1037 1038 1041 TO 1047 1058 -
493. 1059 TO 1071 1082 TO 1102 1113 TO 1128 1139 TO 1147 1150 TO 1165 1168 TO 1190 -
494. 1193 TO 1210 1213 TO 1215 -
495. 1216 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION
```

JOINT	INFLUENCE AREA
23	0.090000
121	0.090000
122	0.090000
123	0.090000
124	0.090000
125	0.090000
126	0.090000
127	0.090000
128	0.090000
129	0.090000
130	0.075000
131	0.075000
132	0.075000
133	0.090000
134	0.090000
135	0.090000
136	0.090000
137	0.090000
138	0.090000
139	0.075000
140	0.090000
141	0.075000
142	0.075000
143	0.075000
144	0.075000
145	0.062500
146	0.090000
147	0.090000
148	0.090000
149	0.090000
150	0.075000
156	0.052500
157	0.052500
158	0.052500
159	0.052500
160	0.043750
161	0.052500
164	0.012500
165	0.015000
166	0.015000
167	0.015000
168	0.015000
169	0.015000
184	0.090000
185	0.090000

186	0.090000
187	0.090000
188	0.090000
189	0.090000
190	0.090000
191	0.090000
192	0.075000
193	0.075000
194	0.075000
195	0.090000
196	0.090000
197	0.090000
198	0.075000
199	0.075000
200	0.075000
201	0.075000
202	0.062500
203	0.090000
204	0.090000
205	0.090000
206	0.090000
207	0.075000
213	0.090000
214	0.090000
215	0.090000
216	0.090000
217	0.075000
218	0.052500
219	0.052500
220	0.052500
221	0.052500
222	0.052500
223	0.043750
225	0.015000
226	0.015000
227	0.015000
228	0.015000
229	0.015000
230	0.012500
246	0.090000
247	0.090000
248	0.090000
249	0.090000
250	0.075000
251	0.052500
252	0.015000
273	0.090000
274	0.090000
275	0.090000
276	0.090000
277	0.075000
278	0.052500
279	0.015000
527	0.090000
528	0.090000
531	0.090000

533	0.090000
535	0.075000
537	0.052500
539	0.015000
808	0.012500
810	0.015000
811	0.015000
812	0.015000
813	0.015000
814	0.015000
815	0.015000
816	0.015000
817	0.015000
818	0.015000
819	0.015000
820	0.012500
822	0.015000
823	0.015000
845	0.015000
927	0.052500
928	0.052500
929	0.090000
930	0.090000
931	0.090000
932	0.090000
933	0.052500
934	0.090000
935	0.090000
936	0.015000
937	0.015000
938	0.015000
939	0.090000
940	0.090000
941	0.090000
942	0.090000
943	0.090000
944	0.090000
945	0.052500
946	0.015000
947	0.075000
948	0.075000
949	0.075000
950	0.043750
951	0.012500
952	0.090000
953	0.090000
954	0.090000
955	0.052500
956	0.015000
962	0.075000
964	0.090000
965	0.090000
966	0.090000
967	0.090000
968	0.090000
969	0.062500

971	0.075000
972	0.075000
973	0.075000
974	0.075000
975	0.075000
976	0.052500
977	0.052500
978	0.090000
979	0.090000
980	0.090000
981	0.090000
982	0.052500
983	0.090000
984	0.090000
985	0.015000
986	0.015000
987	0.015000
988	0.090000
989	0.090000
990	0.090000
991	0.075000
992	0.075000
993	0.075000
994	0.043750
995	0.012500
996	0.090000
997	0.090000
998	0.090000
999	0.052500
1000	0.015000
1006	0.090000
1007	0.090000
1008	0.090000
1009	0.052500
1010	0.015000
1011	0.090000
1012	0.090000
1013	0.090000
1014	0.090000
1015	0.090000
1016	0.075000
1018	0.075000
1019	0.075000
1020	0.075000
1021	0.075000
1022	0.075000
1023	0.062500
1025	0.090000
1026	0.090000
1027	0.090000
1028	0.052500
1029	0.015000
1030	0.090000
1031	0.075000
1032	0.090000
1033	0.090000

1034	0.090000
1035	0.052500
1036	0.015000
1037	0.090000
1038	0.075000
1079	0.090000
1080	0.090000
1083	0.090000
1085	0.052500
1087	0.015000
1089	0.090000
1091	0.075000
1151	0.060000
1152	0.060000
1153	0.060000
1154	0.060000
1155	0.050000
1156	0.035000
1157	0.010000
1168	0.010000
1169	0.060000
1170	0.060000
1171	0.060000
1172	0.035000
1173	0.010000
1174	0.060000
1175	0.050000
1176	0.060000
1177	0.060000
1178	0.060000
1179	0.060000
1180	0.050000
1181	0.035000
1182	0.010000
1193	0.010000
1194	0.060000
1195	0.060000
1196	0.060000
1197	0.035000
1198	0.010000
1199	0.060000
1200	0.050000
1201	0.060000
1202	0.050000
1203	0.060000
1204	0.060000
1205	0.060000
1206	0.060000
1207	0.060000
1208	0.060000
1209	0.060000
1210	0.060000
1211	0.060000
1212	0.050000
1213	0.060000
1214	0.060000

1216	0.060000
1226	0.040000
1227	0.040000
1228	0.060000
1229	0.060000
1230	0.060000
1231	0.060000
1232	0.050000
1233	0.060000
1234	0.060000
1235	0.060000
1236	0.060000
1237	0.050000
1238	0.060000
1239	0.060000
1240	0.060000
1241	0.060000
1243	0.060000
1253	0.040000
1254	0.040000
1255	0.030000
1256	0.030000
1257	0.030000
1258	0.030000
1259	0.025000
1260	0.017500
1261	0.005000
1264	0.005000
1265	0.030000
1266	0.030000
1267	0.030000
1268	0.017500
1269	0.005000
1270	0.030000
1271	0.025000
1272	0.020000
1273	0.020000
1274	0.030000
1275	0.030000
1276	0.030000
1277	0.030000
1278	0.025000
1279	0.017500
1280	0.005000
1283	0.005000
1284	0.030000
1285	0.030000
1286	0.030000
1287	0.017500
1288	0.005000
1289	0.030000
1290	0.025000
1291	0.020000
1292	0.020000
1293	0.030000
1294	0.025000

1295	0.030000
1296	0.030000
1297	0.030000
1298	0.030000
1299	0.030000
1300	0.030000
1301	0.030000
1302	0.030000
1303	0.030000
1304	0.025000
1305	0.030000
1306	0.030000
1308	0.030000
1310	0.020000
1311	0.020000
1312	0.010000
1313	0.010000
1314	0.030000
1315	0.030000
1316	0.030000
1317	0.030000
1318	0.025000
1319	0.030000
1320	0.030000
1321	0.030000
1322	0.030000
1323	0.025000
1324	0.030000
1325	0.030000
1326	0.030000
1327	0.030000
1329	0.030000
1331	0.020000
1332	0.020000
1333	0.010000
1334	0.010000
TOTAL	20.000017

```
496. *
497. *
498. *
499. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
500. LOAD COMB 100 1.4XD
501. 1 1.4
502. *
503. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
504. 1 1.2 20 1.2 2 1.6
505. *
506. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
507. 1 1.2 20 1.2 3 1.6
508. *
509. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
510. 1 1.2 21 1.2 2 1.6
511. *
512. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
513. 1 1.2 21 1.2 3 1.6
514. *
```

```
515. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
516. 1 1.2 11 0.8
517. *
518. LOAD COMB 106 1.2XD + 0.8XWL(+X)
519. 1 1.2 13 0.8
520. *
521. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
522. 1 1.2 12 0.8
523. *
524. LOAD COMB 108 1.2XD + 0.8XWL(-X)
525. 1 1.2 14 0.8
526. *
527. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
528. 1 0.9 11 1.6
529. *
530. LOAD COMB 110 0.9XD + 1.6XWL(+X)
531. 1 0.9 13 1.6
532. *
533. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
534. 1 0.9 12 1.6
535. *
536. LOAD COMB 112 0.9XD + 1.6XWL(-X)
537. 1 0.9 14 1.6
538. *
539. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
540. 1 1.2 2 0.5 1.6
541. *
542. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
543. 1 1.2 2 0.5 1.6
544. *
545. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
546. 1 1.2 2 0.5 1.6
547. *
548. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
549. 1 1.2 2 0.5 1.6
550. *
551. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
552. 1 1.2 3 0.5 1.6
553. *
554. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
555. 1 1.2 3 0.5 1.6
556. *
557. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
558. 1 1.2 3 0.5 1.6
559. *
560. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
561. 1 1.2 3 0.5 1.6
562. *
563. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
564. 1 1.2 2 0.5 1
565. *
566. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
567. 1 1.2 2 0.5 1
568. *
569. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
570. 1 1.2 2 0.5 1
```

```
571. *
572. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
573. 1 1.2 2 0.5 1
574. *
575. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
576. 1 1.2 3 0.5 1
577. *
578. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
579. 1 1.2 3 0.5 1
580. *
581. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
582. 1 1.2 3 0.5 1
583. *
584. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
585. 1 1.2 3 0.5 1
586. *
587. LOAD COMB 129 0.9XD + EL(+Z)
588. 1 0.9 30 1.0
589. *
590. LOAD COMB 130 0.9XD + EL(+X)
591. 1 0.9 32 1.0
592. *
593. LOAD COMB 131 0.9XD + EL(-Z)
594. 1 0.9 31 1.0
595. *
596. LOAD COMB 132 0.9XD + EL(-X)
597. 1 0.9 33 1.0
598. *
599. PERFORM ANALYSIS
600. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 17,2020 TIME= 12: 5:15 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUL 17, 2020                 *
*           Time=    15:49:30                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\RCO1 _RCO2\CIVIL\Working\F5\F5.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18-MARCH-20
     4. *FOUNDATION F5
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. SET NL 100
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 -0.175 0.35 -1; 2 -0.175 0.35 -3; 23 -0.175 -2 -3; 121 -0.175 -2 -1.
    11. 122 -0.175 -2 -1.3; 123 0.075 -2 -1.3; 124 0.075 -2 -1; 125 -0.425 -2 -1.
    12. 126 -0.425 -2 -1.3; 127 -0.175 -2 -0.7; 128 0.075 -2 -0.7; 129 -0.425 -2 -0.7
    13. 130 0.25 -2 -1.3; 131 0.25 -2 -1; 132 0.25 -2 -0.7; 133 -0.6 -2 -1.
    14. 134 -0.6 -2 -1.3; 135 -0.6 -2 -0.7; 136 -0.175 -2 -1.55; 137 0.075 -2 -1.55
    15. 138 -0.425 -2 -1.55; 139 0.25 -2 -1.55; 140 -0.6 -2 -1.55; 141 -0.6 -2 -0.45
    16. 142 -0.425 -2 -0.45; 143 -0.175 -2 -0.45; 144 0.075 -2 -0.45
    17. 145 0.25 -2 -0.45; 146 -0.6 -2 -1.8; 147 -0.425 -2 -1.8; 148 -0.175 -2 -1.8
    18. 149 0.075 -2 -1.8; 150 0.25 -2 -1.8; 151 -0.6 -2 -0.2; 152 -0.425 -2 -0.2
    19. 153 -0.175 -2 -0.2; 154 0.075 -2 -0.2; 155 0.25 -2 -0.2; 156 -0.925 -2 -1.
    20. 157 -0.925 -2 -1.3; 158 -0.925 -2 -0.7; 159 -0.925 -2 -1.55
    21. 160 -0.925 -2 -0.45; 161 -0.925 -2 -1.8; 162 -0.925 -2 -0.2
    22. 170 0.575 -2 -0.45; 171 0.575 -2 -0.2; 172 0.575 -2 -0.7; 173 0.575 -2 -1.
    23. 174 0.575 -2 -1.3; 175 0.575 -2 -1.55; 176 0.575 -2 -1.8; 184 -0.175 -2 -3.3
    24. 185 0.075 -2 -3.3; 186 0.075 -2 -3; 187 -0.425 -2 -3; 188 -0.425 -2 -3.3
    25. 189 -0.175 -2 -2.7; 190 0.075 -2 -2.7; 191 -0.425 -2 -2.7; 192 0.25 -2 -3
    26. 193 0.25 -2 -2.7; 194 0.25 -2 -3.3; 195 -0.6 -2 -2.7; 196 -0.6 -2 -3
    27. 197 -0.6 -2 -3.3; 198 -0.6 -2 -3.55; 199 -0.425 -2 -3.55; 200 -0.175 -2 -3.55
    28. 201 0.075 -2 -3.55; 202 0.25 -2 -3.55; 203 -0.6 -2 -2.45; 204 -0.425 -2 -2.45
    29. 205 -0.175 -2 -2.45; 206 0.075 -2 -2.45; 207 0.25 -2 -2.45; 208 -0.6 -2 -3.8
    30. 209 -0.425 -2 -3.8; 210 -0.175 -2 -3.8; 211 0.075 -2 -3.8; 212 0.25 -2 -3.8
    31. 213 -0.6 -2 -2.2; 214 -0.425 -2 -2.2; 215 -0.175 -2 -2.2; 216 0.075 -2 -2.2
    32. 217 0.25 -2 -2.2; 218 -0.925 -2 -2.2; 219 -0.925 -2 -2.45; 220 -0.925 -2 -2.7
    33. 221 -0.925 -2 -3; 222 -0.925 -2 -3.3; 223 -0.925 -2 -3.55; 224 -0.925 -2 -3.8
    34. 232 0.575 -2 -2.45; 233 0.575 -2 -2.2; 234 0.575 -2 -2.7; 235 0.575 -2 -3
    35. 236 0.575 -2 -3.3; 237 0.575 -2 -3.55; 238 0.575 -2 -3.8; 246 -0.6 -2 -2
    36. 247 -0.425 -2 -2; 248 -0.175 -2 -2; 249 0.075 -2 -2; 250 0.25 -2 -2
    37. 251 -0.925 -2 -2; 253 0.575 -2 -2; 309 -0.6 -2 -0.15; 310 -0.425 -2 -0.15
    38. 311 -0.175 -2 -0.15; 312 0.075 -2 -0.15; 313 0.25 -2 -0.15
```

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39. 314 -0.925 -2 -0.15; 316 0.575 -2 -0.15; 327 -0.6 -2 -3.85
40. 328 -0.425 -2 -3.85; 329 -0.175 -2 -3.85; 330 0.075 -2 -3.85
41. 331 0.25 -2 -3.85; 332 -0.925 -2 -3.85; 334 0.575 -2 -3.85; 461 -0.425 -2 -3.9
42. 462 -0.6 -2 -3.9; 463 -0.175 -2 -3.9; 464 0.075 -2 -3.9; 465 0.25 -2 -3.9
43. 466 -0.925 -2 -3.9; 468 0.575 -2 -3.9; 474 -0.425 -2 -0.1; 475 -0.6 -2 -0.1
44. 476 -0.175 -2 -0.1; 477 0.075 -2 -0.1; 478 0.25 -2 -0.1; 479 -0.925 -2 -0.1
45. 481 0.575 -2 -0.1; 487 -0.425 -2 -4; 488 -0.6 -2 -4; 489 -0.175 -2 -4
46. 490 0.075 -2 -4; 491 0.25 -2 -4; 492 -0.925 -2 -4; 494 0.575 -2 -4
47. 500 -0.425 -2 -1.11759E-008; 501 -0.6 -2 -1.11759E-008
48. 502 -0.175 -2 -1.11759E-008; 503 0.075 -2 -1.11759E-008
49. 504 0.25 -2 -1.11759E-008; 505 -0.925 -2 -1.11759E-008
50. 507 0.575 -2 -1.11759E-008; 508 -1.175 -2 -0.45; 509 -1.175 -2 -0.2
51. 510 -1.175 -2 -0.7; 511 -1.175 -2 -1; 512 -1.175 -2 -1.3; 513 -1.175 -2 -1.55
52. 514 -1.175 -2 -1.8; 515 -1.175 -2 -2.45; 516 -1.175 -2 -2.2
53. 517 -1.175 -2 -2.7; 518 -1.175 -2 -3; 519 -1.175 -2 -3.3; 520 -1.175 -2 -3.55
54. 521 -1.175 -2 -3.8; 522 -1.175 -2 -2; 523 -1.175 -2 -0.15; 524 -1.175 -2 -3.85
55. 525 -1.175 -2 -3.9; 526 -1.175 -2 -0.1; 527 -1.175 -2 -4
56. 528 -1.175 -2 -1.11759E-008; 529 -1.425 -2 -0.45; 530 -1.425 -2 -0.2
57. 531 -1.425 -2 -0.7; 532 -1.425 -2 -1; 533 -1.425 -2 -1.3; 534 -1.425 -2 -1.55
58. 535 -1.425 -2 -1.8; 536 -1.425 -2 -2.45; 537 -1.425 -2 -2.2
59. 538 -1.425 -2 -2.7; 539 -1.425 -2 -3; 540 -1.425 -2 -3.3; 541 -1.425 -2 -3.55
60. 542 -1.425 -2 -3.8; 543 -1.425 -2 -2; 544 -1.425 -2 -0.15; 545 -1.425 -2 -3.85
61. 546 -1.425 -2 -3.9; 547 -1.425 -2 -0.1; 548 -1.425 -2 -4
62. 549 -1.425 -2 -1.11759E-008; 550 1.075 -2 -0.2; 551 1.075 -2 -0.45
63. 552 0.825 -2 -0.45; 553 0.825 -2 -0.2; 554 1.075 -2 -0.7; 555 0.825 -2 -0.7
64. 556 1.075 -2 -1; 557 0.825 -2 -1; 558 1.075 -2 -1.3; 559 0.825 -2 -1.3
65. 560 1.075 -2 -1.55; 561 0.825 -2 -1.55; 562 1.075 -2 -1.8; 563 0.825 -2 -1.8
66. 564 1.075 -2 -2.2; 565 1.075 -2 -2.45; 566 0.825 -2 -2.45; 567 0.825 -2 -2.2
67. 568 1.075 -2 -2.7; 569 0.825 -2 -2.7; 570 1.075 -2 -3; 571 0.825 -2 -3
68. 572 1.075 -2 -3.3; 573 0.825 -2 -3.3; 574 1.075 -2 -3.55; 575 0.825 -2 -3.55
69. 576 1.075 -2 -3.8; 577 0.825 -2 -3.8; 578 1.075 -2 -2; 579 0.825 -2 -2
70. 580 1.075 -2 -0.15; 581 0.825 -2 -0.15; 582 1.075 -2 -3.85; 583 0.825 -2 -3.85
71. 584 1.075 -2 -3.9; 585 0.825 -2 -3.9; 586 1.075 -2 -0.1; 587 0.825 -2 -0.1
72. 588 1.075 -2 -4; 589 0.825 -2 -4; 590 1.075 -2 -1.11759E-008
73. 591 0.825 -2 -1.11759E-008
74. MEMBER INCIDENCES
75. 48 23 2; 49 121 1
76. ELEMENT INCIDENCES SHELL
77. 50 124 123 122 121; 51 121 122 126 125; 52 128 124 121 127; 53 127 121 125 129
78. 54 132 131 124 128; 55 131 130 123 124; 56 129 125 133 135; 57 125 126 134 133
79. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
80. 62 142 129 135 141; 63 143 127 129 142; 64 144 128 127 143; 65 145 132 128 144
81. 66 138 147 146 140; 67 136 148 147 138; 68 137 149 148 136; 69 139 150 149 137
82. 70 152 142 141 151; 71 153 143 142 152; 72 154 144 143 153; 73 155 145 144 154
83. 74 151 141 160 162; 75 141 135 158 160; 76 135 133 156 158; 77 133 134 157 156
84. 78 134 140 159 157; 79 140 146 161 159; 86 171 170 145 155; 87 170 172 132 145
85. 88 172 173 131 132; 89 173 174 130 131; 90 174 175 139 130; 91 175 176 150 139
86. 98 186 185 184 23; 99 23 184 188 187; 100 190 186 23 189; 101 189 23 187 191
87. 102 193 192 186 190; 103 192 194 185 186; 104 191 187 196 195
88. 105 187 188 197 196; 106 188 199 198 197; 107 184 200 199 188
89. 108 185 201 200 184; 109 194 202 201 185; 110 204 191 195 203
90. 111 205 189 191 204; 112 206 190 189 205; 113 207 193 190 206
91. 114 199 209 208 198; 115 200 210 209 199; 116 201 211 210 200
92. 117 202 212 211 201; 118 214 204 203 213; 119 215 205 204 214
93. 120 216 206 205 215; 121 217 207 206 216; 122 213 203 219 218
94. 123 203 195 220 219; 124 195 196 221 220; 125 196 197 222 221
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 95. 126 197 198 223 222; 127 198 208 224 223; 134 233 232 207 217
 96. 135 232 234 193 207; 136 234 235 192 193; 137 235 236 194 192
 97. 138 236 237 202 194; 139 237 238 212 202; 146 147 247 246 146
 98. 147 148 248 247 147; 148 149 249 248 148; 149 150 250 249 149
 99. 150 146 246 251 161; 152 176 253 250 150; 170 247 214 213 246
100. 171 248 215 214 247; 172 249 216 215 248; 173 250 217 216 249
101. 174 246 213 218 251; 176 253 233 217 250; 210 310 152 151 309
102. 211 311 153 152 310; 212 312 154 153 311; 213 313 155 154 312
103. 214 309 151 162 314; 216 316 171 155 313; 226 209 328 327 208
104. 227 210 329 328 209; 228 211 330 329 210; 229 212 331 330 211
105. 230 208 327 332 224; 232 238 334 331 212; 353 328 461 462 327
106. 354 329 463 461 328; 355 330 464 463 329; 356 331 465 464 330
107. 357 327 462 466 332; 359 334 468 465 331; 365 474 310 309 475
108. 366 476 311 310 474; 367 477 312 311 476; 368 478 313 312 477
109. 369 475 309 314 479; 371 481 316 313 478; 377 461 487 488 462
110. 378 463 489 487 461; 379 464 490 489 463; 380 465 491 490 464
111. 381 462 488 492 466; 383 468 494 491 465; 389 500 474 475 501
112. 390 502 476 474 500; 391 503 477 476 502; 392 504 478 477 503
113. 393 501 475 479 505; 395 507 481 478 504; 396 162 160 508 509
114. 397 160 158 510 508; 398 158 156 511 510; 399 156 157 512 511
115. 400 157 159 513 512; 401 159 161 514 513; 402 218 219 515 516
116. 403 219 220 517 515; 404 220 221 518 517; 405 221 222 519 518
117. 406 222 223 520 519; 407 223 224 521 520; 408 161 251 522 514
118. 409 251 218 516 522; 410 314 162 509 523; 411 224 332 524 521
119. 412 332 466 525 524; 413 479 314 523 526; 414 466 492 527 525
120. 415 505 479 526 528; 416 509 508 529 530; 417 508 510 531 529
121. 418 510 511 532 531; 419 511 512 533 532; 420 512 513 534 533
122. 421 513 514 535 534; 422 516 515 536 537; 423 515 517 538 536
123. 424 517 518 539 538; 425 518 519 540 539; 426 519 520 541 540
124. 427 520 521 542 541; 428 514 522 543 535; 429 522 516 537 543
125. 430 523 509 530 544; 431 521 524 545 542; 432 524 525 546 545
126. 433 526 523 544 547; 434 525 527 548 546; 435 528 526 547 549
127. 436 550 551 552 553; 437 551 554 555 552; 438 554 556 557 555
128. 439 556 558 559 557; 440 558 560 561 559; 441 560 562 563 561
129. 442 564 565 566 567; 443 565 568 569 566; 444 568 570 571 569
130. 445 570 572 573 571; 446 572 574 575 573; 447 574 576 577 575
131. 448 562 578 579 563; 449 578 564 567 579; 450 580 550 553 581
132. 451 576 582 583 577; 452 582 584 585 583; 453 586 580 581 587
133. 454 584 588 589 585; 455 590 586 587 591; 456 553 552 170 171
134. 457 552 555 172 170; 458 555 557 173 172; 459 557 559 174 173
135. 460 559 561 175 174; 461 561 563 176 175; 462 567 566 232 233
136. 463 566 569 234 232; 464 569 571 235 234; 465 571 573 236 235
137. 466 573 575 237 236; 467 575 577 238 237; 468 563 579 253 176
138. 469 579 567 233 253; 470 581 553 171 316; 471 577 583 334 238
139. 472 583 585 468 334; 473 587 581 316 481; 474 585 589 494 468
140. 475 591 587 481 507
141. START GROUP DEFINITION
142. FLOOR
143. ELEMENT
144. _FDN_5_MOMENT 58 TO 79 86 TO 91 106 TO 127 134 TO 139 146 TO 150 152 -
145. 170 TO 174 176 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 -
146. 377 TO 381 383 389 TO 393 395 TO 475
147. _FDN_5_SHEAR 146 TO 150 152 170 TO 174 176 210 TO 214 216 226 TO 230 232 353 -
148. 354 TO 357 359 365 TO 369 371 377 TO 381 383 389 TO 393 395 408 TO 415 428 -
149. 429 TO 435 448 TO 455 468 TO 475
150. END GROUP DEFINITION
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151. ELEMENT PROPERTY
152. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
153. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
154. 389 TO 393 395 TO 475 THICKNESS 0.5
155. DEFINE MATERIAL START
156. ISOTROPIC CONCRETE
157. E 2.17185E+007
158. POISSON 0.17
159. DENSITY 25
160. ALPHA 1E-005
161. DAMP 0.05
162. TYPE CONCRETE
163. STRENGTH FCU 27579
164. END DEFINE MATERIAL
165. MEMBER PROPERTY AMERICAN
166. 48 49 PRIS YD 0.85 ZD 0.6
167. CONSTANTS
168. MATERIAL CONCRETE ALL
169. SUPPORTS
170. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
171. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
172. 389 TO 393 395 TO 475 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
```

JOINT	INFLUENCE AREA
23	0.075000
121	0.075000
122	0.068750
123	0.058437
124	0.063750
125	0.063750
126	0.058437
127	0.068750
128	0.058437
129	0.058437
130	0.068750
131	0.075000
132	0.068750
133	0.075000
134	0.068750
135	0.068750
136	0.062500
137	0.053125
138	0.053125
139	0.062500
140	0.062500
141	0.062500
142	0.053125
143	0.062500
144	0.053125
145	0.062500
146	0.056250
147	0.047813
148	0.056250
149	0.047813

150	0.056250
151	0.037500
152	0.031875
153	0.037500
154	0.031875
155	0.037500
156	0.086250
157	0.079062
158	0.079062
159	0.071875
160	0.071875
161	0.064688
162	0.043125
170	0.071875
171	0.043125
172	0.079063
173	0.086250
174	0.079062
175	0.071875
176	0.064688
184	0.068750
185	0.058437
186	0.063750
187	0.063750
188	0.058437
189	0.068750
190	0.058437
191	0.058437
192	0.075000
193	0.068750
194	0.068750
195	0.068750
196	0.075000
197	0.068750
198	0.062500
199	0.053125
200	0.062500
201	0.053125
202	0.062500
203	0.062500
204	0.053125
205	0.062500
206	0.053125
207	0.062500
208	0.037500
209	0.031875
210	0.037500
211	0.031875
212	0.037500
213	0.056250
214	0.047813
215	0.056250
216	0.047813
217	0.056250
218	0.064688
219	0.071875

220	0.079062
221	0.086250
222	0.079062
223	0.071875
224	0.043125
232	0.071875
233	0.064688
234	0.079062
235	0.086250
236	0.079062
237	0.071875
238	0.043125
246	0.050000
247	0.042500
248	0.050000
249	0.042500
250	0.050000
251	0.057500
253	0.057500
309	0.012500
310	0.010625
311	0.012500
312	0.010625
313	0.012500
314	0.014375
316	0.014375
327	0.012500
328	0.010625
329	0.012500
330	0.010625
331	0.012500
332	0.014375
334	0.014375
461	0.015938
462	0.018750
463	0.018750
464	0.015938
465	0.018750
466	0.021563
468	0.021563
474	0.015937
475	0.018750
476	0.018750
477	0.015937
478	0.018750
479	0.021562
481	0.021562
487	0.010625
488	0.012500
489	0.012500
490	0.010625
491	0.012500
492	0.014375
494	0.014375
500	0.010625
501	0.012500

502	0.012500
503	0.010625
504	0.012500
505	0.014375
507	0.014375
508	0.062500
509	0.037500
510	0.068750
511	0.075000
512	0.068750
513	0.062500
514	0.056250
515	0.062500
516	0.056250
517	0.068750
518	0.075000
519	0.068750
520	0.062500
521	0.037500
522	0.050000
523	0.012500
524	0.012500
525	0.018750
526	0.018750
527	0.012500
528	0.012500
529	0.031250
530	0.018750
531	0.034375
532	0.037500
533	0.034375
534	0.031250
535	0.028125
536	0.031250
537	0.028125
538	0.034375
539	0.037500
540	0.034375
541	0.031250
542	0.018750
543	0.025000
544	0.006250
545	0.006250
546	0.009375
547	0.009375
548	0.006250
549	0.006250
550	0.018750
551	0.031250
552	0.062500
553	0.037500
554	0.034375
555	0.068750
556	0.037500
557	0.075000
558	0.034375

559	0.068750
560	0.031250
561	0.062500
562	0.028125
563	0.056250
564	0.028125
565	0.031250
566	0.062500
567	0.056250
568	0.034375
569	0.068750
570	0.037500
571	0.075000
572	0.034375
573	0.068750
574	0.031250
575	0.062500
576	0.018750
577	0.037500
578	0.025000
579	0.050000
580	0.006250
581	0.012500
582	0.006250
583	0.012500
584	0.009375
585	0.018750
586	0.009375
587	0.018750
588	0.006250
589	0.012500
590	0.006250
591	0.012500
TOTAL	10.000004

```
173. ***
174. ************************************************
175. ***LOAD COMBINATION FOR STRESS DESIGN
176. ************************************************
177. ***
178. ***
179. LOAD 1 LOADTYPE DEAD  TITLE DL
180. ***
181. SELFWEIGHT Y -1.1
182. ***FROM GALLERY-1
183. JOINT LOAD
184. 1 FX 0 FY -33.997 FZ -3.482
185. 2 FX -0.059 FY -34.045 FZ 3.513
186. ELEMENT LOAD
187. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
188. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
189. 389 TO 393 395 TO 475 PR GY -20
190. *
191. LOAD 2 LOADTYPE LIVE  TITLE LL
192. JOINT LOAD
193. 1 FX -0.001 FY 30.641 FZ 5.879
194. 2 FX -0.902 FY -132.312 FZ 16.214
```

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195. *
196. LOAD 3 LOADTYPE LIVE  TITLE LL1
197. JOINT LOAD
198. 1 FX -0.001 FY -132.707 FZ -16.123
199. 2 FX -0.855 FY 31.042 FZ -5.719
200. *
201. LOAD 11 LOADTYPE WIND  TITLE WL+Z
202. JOINT LOAD
203. 1 FX 0 FY 359.861 FZ 35.833
204. 2 FX -0.115 FY -359.93 FZ 35.662
205. *
206. LOAD 12 LOADTYPE WIND  TITLE WL-Z
207. JOINT LOAD
208. 1 FX 0 FY -359.861 FZ -35.833
209. 2 FX 0.115 FY 359.93 FZ -35.662
210. *
211. LOAD 13 LOADTYPE WIND  TITLE WL+X
212. JOINT LOAD
213. 1 FX -0.001 FY -3.04 FZ -0.316
214. 2 FX -0.47 FY -3.195 FZ 0.318
215. *
216. LOAD 14 LOADTYPE WIND  TITLE WL-X
217. JOINT LOAD
218. 1 FX 0.001 FY 3.04 FZ 0.316
219. 2 FX 0.47 FY 3.195 FZ -0.318
220. *
221. LOAD 20 LOADTYPE NONE  TITLE TL1
222. JOINT LOAD
223. 1 FX 0 FY 0 FZ -10.4
224. 2 FX 0 FY 0 FZ 10.4
225. *
226. LOAD 21 LOADTYPE NONE  TITLE TL2
227. JOINT LOAD
228. 1 FX 0 FY 0 FZ 10.4
229. 2 FX 0 FY 0 FZ -10.4
230. *
231. LOAD 30 EARTHQUAKE LOAD (+Z)
232. JOINT LOAD
233. 1 FX 0 FY 2.521 FZ 0.37
234. 2 FX 0 FY -2.521 FZ 0.369
235. *
236. LOAD 31 EARTHQUAKE LOAD (-Z)
237. JOINT LOAD
238. 1 FX 0 FY -2.521 FZ -0.37
239. 2 FX 0 FY 2.521 FZ -0.369
240. *
241. LOAD 32 EARTHQUAKE LOAD (+X)
242. JOINT LOAD
243. 1 FX 0.367 FY 0 FZ 0
244. 2 FX 0.372 FY 0 FZ 0
245. *
246. LOAD 33 EARTHQUAKE LOAD (-X)
247. JOINT LOAD
248. 1 FX -0.367 FY 0 FZ 0
249. 2 FX -0.372 FY 0 FZ 0
250. *
```

```
****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
*
LOAD COMB 201 D + L + TL1
1 1.0 2 1.0 20 1.0
*
LOAD COMB 202 D + L1 + TL1
1 1.0 3 1.0 20 1.0
*
LOAD COMB 203 D + L + TL2
1 1.0 2 1.0 21 1.0
*
LOAD COMB 204 D + L1 + TL2
1 1.0 3 1.0 21 1.0
*
LOAD COMB 205 D + 0.75XL + 0.75XTL1
1 1.0 2 0.75 20 0.75
*
LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
1 1.0 3 0.75 20 0.75
*
LOAD COMB 207 D + 0.75XL + 0.75XTL2
1 1.0 2 0.75 21 0.75
*
LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
1 1.0 3 0.75 21 0.75
*
LOAD COMB 209 D + WL(+Z)
1 1.0 11 1.0
*
LOAD COMB 210 D + WL(+X)
1 1.0 13 1.0
*
LOAD COMB 211 D + WL(-Z)
1 1.0 12 1.0
*
LOAD COMB 212 D + WL(-X)
1 1.0 14 1.0
*
LOAD COMB 213 D + 0.7XEL(+Z)
1 1.0 30 0.7
*
LOAD COMB 214 D + 0.7XEL(+X)
1 1.0 32 0.7
*
LOAD COMB 215 D + 0.7XEL(-Z)
1 1.0 31 0.7
*
LOAD COMB 216 D + 0.7XEL(-X)
1 1.0 33 0.7
*
LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
1 1.0 2 0.75 11 0.75
*
LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
1 1.0 3 0.75 11 0.75
*
```

```
307. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
308. 1 1.0 2 0.75 12 0.75
309. *
310. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
311. 1 1.0 3 0.75 12 0.75
312. *
313. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
314. 1 1.0 2 0.75 13 0.75
315. *
316. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
317. 1 1.0 3 0.75 13 0.75
318. *
319. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
320. 1 1.0 2 0.75 14 0.75
321. *
322. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
323. 1 1.0 3 0.75 14 0.75
324. *
325. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
326. 1 1.0 2 0.75 30 0.525
327. *
328. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
329. 1 1.0 3 0.75 30 0.525
330. *
331. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
332. 1 1.0 2 0.75 32 0.525
333. *
334. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
335. 1 1.0 3 0.75 32 0.525
336. *
337. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
338. 1 1.0 2 0.75 31 0.525
339. *
340. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
341. 1 1.0 3 0.75 31 0.525
342. *
343. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
344. 1 1.0 2 0.75 33 0.525
345. *
346. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
347. 1 1.0 3 0.75 33 0.525
348. *
349. LOAD COMB 233 0.6XD + WL(+Z)
350. 1 0.6 11 1.0
351. *
352. LOAD COMB 234 0.6XD + WL(+X)
353. 1 0.6 13 1.0
354. *
355. LOAD COMB 235 0.6XD + WL(-Z)
356. 1 0.6 12 1.0
357. *
358. LOAD COMB 236 0.6XD + WL(-X)
359. 1 0.6 14 1.0
360. *
361. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
362. 1 0.6 30 0.7
```

```
363. *
364. LOAD COMB 238 0.6XD + 0.7XEL(+X)
365. 1 0.6 32 0.7
366. *
367. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
368. 1 0.6 31 0.7
369. *
370. LOAD COMB 240 0.6XD + 0.7XEL(-X)
371. 1 0.6 33 0.7
372. *
373. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**233**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**200**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**231**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 176/ 20/ 66 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 705
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 47 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.7/ 73877.7 MB

```
374. CHANGE
375. SUPPORTS
376. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
377. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
378. 389 TO 393 395 TO 475 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION
```

JOINT	INFLUENCE AREA
23	0.075000
121	0.075000
122	0.068750
123	0.058437
124	0.063750
125	0.063750
126	0.058437
127	0.068750
128	0.058437
129	0.058437
130	0.068750
131	0.075000

132	0.068750
133	0.075000
134	0.068750
135	0.068750
136	0.062500
137	0.053125
138	0.053125
139	0.062500
140	0.062500
141	0.062500
142	0.053125
143	0.062500
144	0.053125
145	0.062500
146	0.056250
147	0.047813
148	0.056250
149	0.047813
150	0.056250
151	0.037500
152	0.031875
153	0.037500
154	0.031875
155	0.037500
156	0.086250
157	0.079062
158	0.079062
159	0.071875
160	0.071875
161	0.064688
162	0.043125
170	0.071875
171	0.043125
172	0.079063
173	0.086250
174	0.079062
175	0.071875
176	0.064688
184	0.068750
185	0.058437
186	0.063750
187	0.063750
188	0.058437
189	0.068750
190	0.058437
191	0.058437
192	0.075000
193	0.068750
194	0.068750
195	0.068750
196	0.075000
197	0.068750
198	0.062500
199	0.053125
200	0.062500
201	0.053125

202	0.062500
203	0.062500
204	0.053125
205	0.062500
206	0.053125
207	0.062500
208	0.037500
209	0.031875
210	0.037500
211	0.031875
212	0.037500
213	0.056250
214	0.047813
215	0.056250
216	0.047813
217	0.056250
218	0.064688
219	0.071875
220	0.079062
221	0.086250
222	0.079062
223	0.071875
224	0.043125
232	0.071875
233	0.064688
234	0.079062
235	0.086250
236	0.079062
237	0.071875
238	0.043125
246	0.050000
247	0.042500
248	0.050000
249	0.042500
250	0.050000
251	0.057500
253	0.057500
309	0.012500
310	0.010625
311	0.012500
312	0.010625
313	0.012500
314	0.014375
316	0.014375
327	0.012500
328	0.010625
329	0.012500
330	0.010625
331	0.012500
332	0.014375
334	0.014375
461	0.015938
462	0.018750
463	0.018750
464	0.015938
465	0.018750

466	0.021563
468	0.021563
474	0.015937
475	0.018750
476	0.018750
477	0.015937
478	0.018750
479	0.021562
481	0.021562
487	0.010625
488	0.012500
489	0.012500
490	0.010625
491	0.012500
492	0.014375
494	0.014375
500	0.010625
501	0.012500
502	0.012500
503	0.010625
504	0.012500
505	0.014375
507	0.014375
508	0.062500
509	0.037500
510	0.068750
511	0.075000
512	0.068750
513	0.062500
514	0.056250
515	0.062500
516	0.056250
517	0.068750
518	0.075000
519	0.068750
520	0.062500
521	0.037500
522	0.050000
523	0.012500
524	0.012500
525	0.018750
526	0.018750
527	0.012500
528	0.012500
529	0.031250
530	0.018750
531	0.034375
532	0.037500
533	0.034375
534	0.031250
535	0.028125
536	0.031250
537	0.028125
538	0.034375
539	0.037500
540	0.034375

541	0.031250
542	0.018750
543	0.025000
544	0.006250
545	0.006250
546	0.009375
547	0.009375
548	0.006250
549	0.006250
550	0.018750
551	0.031250
552	0.062500
553	0.037500
554	0.034375
555	0.068750
556	0.037500
557	0.075000
558	0.034375
559	0.068750
560	0.031250
561	0.062500
562	0.028125
563	0.056250
564	0.028125
565	0.031250
566	0.062500
567	0.056250
568	0.034375
569	0.068750
570	0.037500
571	0.075000
572	0.034375
573	0.068750
574	0.031250
575	0.062500
576	0.018750
577	0.037500
578	0.025000
579	0.050000
580	0.006250
581	0.012500
582	0.006250
583	0.012500
584	0.009375
585	0.018750
586	0.009375
587	0.018750
588	0.006250
589	0.012500
590	0.006250
591	0.012500
TOTAL	10.000004

```
379. *
380. *
381. *
382. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
```

```
383. LOAD COMB 100 1.4XD
384. 1 1.4
385. *
386. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
387. 1 1.2 20 1.2 2 1.6
388. *
389. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
390. 1 1.2 20 1.2 3 1.6
391. *
392. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
393. 1 1.2 21 1.2 2 1.6
394. *
395. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
396. 1 1.2 21 1.2 3 1.6
397. *
398. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
399. 1 1.2 11 0.8
400. *
401. LOAD COMB 106 1.2XD + 0.8XWL(+X)
402. 1 1.2 13 0.8
403. *
404. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
405. 1 1.2 12 0.8
406. *
407. LOAD COMB 108 1.2XD + 0.8XWL(-X)
408. 1 1.2 14 0.8
409. *
410. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
411. 1 0.9 11 1.6
412. *
413. LOAD COMB 110 0.9XD + 1.6XWL(+X)
414. 1 0.9 13 1.6
415. *
416. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
417. 1 0.9 12 1.6
418. *
419. LOAD COMB 112 0.9XD + 1.6XWL(-X)
420. 1 0.9 14 1.6
421. *
422. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
423. 1 1.2 2 0.5 1.6
424. *
425. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
426. 1 1.2 2 0.5 1.6
427. *
428. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
429. 1 1.2 2 0.5 1.6
430. *
431. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
432. 1 1.2 2 0.5 1.6
433. *
434. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
435. 1 1.2 3 0.5 1.6
436. *
437. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
438. 1 1.2 3 0.5 1.6
```

```
439. *
440. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
441. 1 1.2 3 0.5 1.6
442. *
443. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
444. 1 1.2 3 0.5 1.6
445. *
446. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
447. 1 1.2 2 0.5 1
448. *
449. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
450. 1 1.2 2 0.5 1
451. *
452. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
453. 1 1.2 2 0.5 1
454. *
455. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
456. 1 1.2 2 0.5 1
457. *
458. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
459. 1 1.2 3 0.5 1
460. *
461. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
462. 1 1.2 3 0.5 1
463. *
464. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
465. 1 1.2 3 0.5 1
466. *
467. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
468. 1 1.2 3 0.5 1
469. *
470. LOAD COMB 129 0.9XD + EL(+Z)
471. 1 0.9 30 1.0
472. *
473. LOAD COMB 130 0.9XD + EL(+X)
474. 1 0.9 32 1.0
475. *
476. LOAD COMB 131 0.9XD + EL(-Z)
477. 1 0.9 31 1.0
478. *
479. LOAD COMB 132 0.9XD + EL(-X)
480. 1 0.9 33 1.0
481. PERFORM ANALYSIS
482. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 17,2020 TIME= 15:49:35 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    JUL 17, 2020                 *
*           Time=    15: 8:43                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\RCO1 _RCO2\CIVIL\Working\F6\F6.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18-MARCH-20
     4. *FOUNDATION F5
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. SET NL 100
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 -0.175 0.35 -1; 2 -0.175 0.35 -3; 23 -0.175 -2 -3; 121 -0.175 -2 -1.
    11. 122 -0.175 -2 -1.3; 123 0.075 -2 -1.3; 124 0.075 -2 -1; 125 -0.425 -2 -1.
    12. 126 -0.425 -2 -1.3; 127 -0.175 -2 -0.7; 128 0.075 -2 -0.7; 129 -0.425 -2 -0.7
    13. 130 0.25 -2 -1.3; 131 0.25 -2 -1; 132 0.25 -2 -0.7; 133 -0.6 -2 -1.
    14. 134 -0.6 -2 -1.3; 135 -0.6 -2 -0.7; 136 -0.175 -2 -1.55; 137 0.075 -2 -1.55
    15. 138 -0.425 -2 -1.55; 139 0.25 -2 -1.55; 140 -0.6 -2 -1.55; 141 -0.6 -2 -0.45
    16. 142 -0.425 -2 -0.45; 143 -0.175 -2 -0.45; 144 0.075 -2 -0.45
    17. 145 0.25 -2 -0.45; 146 -0.6 -2 -1.8; 147 -0.425 -2 -1.8; 148 -0.175 -2 -1.8
    18. 149 0.075 -2 -1.8; 150 0.25 -2 -1.8; 151 -0.6 -2 -0.2; 152 -0.425 -2 -0.2
    19. 153 -0.175 -2 -0.2; 154 0.075 -2 -0.2; 155 0.25 -2 -0.2; 156 -0.925 -2 -1.
    20. 157 -0.925 -2 -1.3; 158 -0.925 -2 -0.7; 159 -0.925 -2 -1.55
    21. 160 -0.925 -2 -0.45; 161 -0.925 -2 -1.8; 162 -0.925 -2 -0.2
    22. 170 0.575 -2 -0.45; 171 0.575 -2 -0.2; 172 0.575 -2 -0.7; 173 0.575 -2 -1.
    23. 174 0.575 -2 -1.3; 175 0.575 -2 -1.55; 176 0.575 -2 -1.8; 184 -0.175 -2 -3.3
    24. 185 0.075 -2 -3.3; 186 0.075 -2 -3; 187 -0.425 -2 -3; 188 -0.425 -2 -3.3
    25. 189 -0.175 -2 -2.7; 190 0.075 -2 -2.7; 191 -0.425 -2 -2.7; 192 0.25 -2 -3
    26. 193 0.25 -2 -2.7; 194 0.25 -2 -3.3; 195 -0.6 -2 -2.7; 196 -0.6 -2 -3
    27. 197 -0.6 -2 -3.3; 198 -0.6 -2 -3.55; 199 -0.425 -2 -3.55; 200 -0.175 -2 -3.55
    28. 201 0.075 -2 -3.55; 202 0.25 -2 -3.55; 203 -0.6 -2 -2.45; 204 -0.425 -2 -2.45
    29. 205 -0.175 -2 -2.45; 206 0.075 -2 -2.45; 207 0.25 -2 -2.45; 208 -0.6 -2 -3.8
    30. 209 -0.425 -2 -3.8; 210 -0.175 -2 -3.8; 211 0.075 -2 -3.8; 212 0.25 -2 -3.8
    31. 213 -0.6 -2 -2.2; 214 -0.425 -2 -2.2; 215 -0.175 -2 -2.2; 216 0.075 -2 -2.2
    32. 217 0.25 -2 -2.2; 218 -0.925 -2 -2.2; 219 -0.925 -2 -2.45; 220 -0.925 -2 -2.7
    33. 221 -0.925 -2 -3; 222 -0.925 -2 -3.3; 223 -0.925 -2 -3.55; 224 -0.925 -2 -3.8
    34. 232 0.575 -2 -2.45; 233 0.575 -2 -2.2; 234 0.575 -2 -2.7; 235 0.575 -2 -3
    35. 236 0.575 -2 -3.3; 237 0.575 -2 -3.55; 238 0.575 -2 -3.8; 246 -0.6 -2 -2
    36. 247 -0.425 -2 -2; 248 -0.175 -2 -2; 249 0.075 -2 -2; 250 0.25 -2 -2
    37. 251 -0.925 -2 -2; 253 0.575 -2 -2; 309 -0.6 -2 -0.15; 310 -0.425 -2 -0.15
    38. 311 -0.175 -2 -0.15; 312 0.075 -2 -0.15; 313 0.25 -2 -0.15
```

```
39. 314 -0.925 -2 -0.15; 316 0.575 -2 -0.15; 327 -0.6 -2 -3.85
40. 328 -0.425 -2 -3.85; 329 -0.175 -2 -3.85; 330 0.075 -2 -3.85
41. 331 0.25 -2 -3.85; 332 -0.925 -2 -3.85; 334 0.575 -2 -3.85; 461 -0.425 -2 -3.9
42. 462 -0.6 -2 -3.9; 463 -0.175 -2 -3.9; 464 0.075 -2 -3.9; 465 0.25 -2 -3.9
43. 466 -0.925 -2 -3.9; 468 0.575 -2 -3.9; 474 -0.425 -2 -0.1; 475 -0.6 -2 -0.1
44. 476 -0.175 -2 -0.1; 477 0.075 -2 -0.1; 478 0.25 -2 -0.1; 479 -0.925 -2 -0.1
45. 481 0.575 -2 -0.1; 487 -0.425 -2 -4; 488 -0.6 -2 -4; 489 -0.175 -2 -4
46. 490 0.075 -2 -4; 491 0.25 -2 -4; 492 -0.925 -2 -4; 494 0.575 -2 -4
47. 500 -0.425 -2 -1.11759E-008; 501 -0.6 -2 -1.11759E-008
48. 502 -0.175 -2 -1.11759E-008; 503 0.075 -2 -1.11759E-008
49. 504 0.25 -2 -1.11759E-008; 505 -0.925 -2 -1.11759E-008
50. 507 0.575 -2 -1.11759E-008
51. MEMBER INCIDENCES
52. 48 23 2; 49 121 1
53. ELEMENT INCIDENCES SHELL
54. 50 124 123 122 121; 51 121 122 126 125; 52 128 124 121 127; 53 127 121 125 129
55. 54 132 131 124 128; 55 131 130 123 124; 56 129 125 133 135; 57 125 126 134 133
56. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
57. 62 142 129 135 141; 63 143 127 129 142; 64 144 128 127 143; 65 145 132 128 144
58. 66 138 147 146 140; 67 136 148 147 138; 68 137 149 148 136; 69 139 150 149 137
59. 70 152 142 141 151; 71 153 143 142 152; 72 154 144 143 153; 73 155 145 144 154
60. 74 151 141 160 162; 75 141 135 158 160; 76 135 133 156 158; 77 133 134 157 156
61. 78 134 140 159 157; 79 140 146 161 159; 86 171 170 145 155; 87 170 172 132 145
62. 88 172 173 131 132; 89 173 174 130 131; 90 174 175 139 130; 91 175 176 150 139
63. 98 186 185 184 23; 99 23 184 188 187; 100 190 186 23 189; 101 189 23 187 191
64. 102 193 192 186 190; 103 192 194 185 186; 104 191 187 196 195
65. 105 187 188 197 196; 106 188 199 198 197; 107 184 200 199 188
66. 108 185 201 200 184; 109 194 202 201 185; 110 204 191 195 203
67. 111 205 189 191 204; 112 206 190 189 205; 113 207 193 190 206
68. 114 199 209 208 198; 115 200 210 209 199; 116 201 211 210 200
69. 117 202 212 211 201; 118 214 204 203 213; 119 215 205 204 214
70. 120 216 206 205 215; 121 217 207 206 216; 122 213 203 219 218
71. 123 203 195 220 219; 124 195 196 221 220; 125 196 197 222 221
72. 126 197 198 223 222; 127 198 208 224 223; 134 233 232 207 217
73. 135 232 234 193 207; 136 234 235 192 193; 137 235 236 194 192
74. 138 236 237 202 194; 139 237 238 212 202; 146 147 247 246 146
75. 147 148 248 247 147; 148 149 249 248 148; 149 150 250 249 149
76. 150 146 246 251 161; 152 176 253 250 150; 170 247 214 213 246
77. 171 248 215 214 247; 172 249 216 215 248; 173 250 217 216 249
78. 174 246 213 218 251; 176 253 233 217 250; 210 310 152 151 309
79. 211 311 153 152 310; 212 312 154 153 311; 213 313 155 154 312
80. 214 309 151 162 314; 216 316 171 155 313; 226 209 328 327 208
81. 227 210 329 328 209; 228 211 330 329 210; 229 212 331 330 211
82. 230 208 327 332 224; 232 238 334 331 212; 353 328 461 462 327
83. 354 329 463 461 328; 355 330 464 463 329; 356 331 465 464 330
84. 357 327 462 466 332; 359 334 468 465 331; 365 474 310 309 475
85. 366 476 311 310 474; 367 477 312 311 476; 368 478 313 312 477
86. 369 475 309 314 479; 371 481 316 313 478; 377 461 487 488 462
87. 378 463 489 487 461; 379 464 490 489 463; 380 465 491 490 464
88. 381 462 488 492 466; 383 468 494 491 465; 389 500 474 475 501
89. 390 502 476 474 500; 391 503 477 476 502; 392 504 478 477 503
90. 393 501 475 479 505; 395 507 481 478 504
91. START GROUP DEFINITION
92. FLOOR
93. ELEMENT
94. _FDN_5_MOMENT 58 TO 79 86 TO 91 106 TO 127 134 TO 139 146 TO 150 152 -
```

```
 95. 170 TO 174 176 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 -
 96. 377 TO 381 383 389 TO 393 395
 97. _FDN_5_SHEAR 146 TO 150 152 170 TO 174 176 210 TO 214 216 226 TO 230 232 353 -
 98. 354 TO 357 359 365 TO 369 371 377 TO 381 383 389 TO 393 395
 99. END GROUP DEFINITION
100. ELEMENT PROPERTY
101. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
102. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
103. 389 TO 393 395 THICKNESS 0.5
104. DEFINE MATERIAL START
105. ISOTROPIC CONCRETE
106. E 2.17185E+007
107. POISSON 0.17
108. DENSITY 25
109. ALPHA 1E-005
110. DAMP 0.05
111. TYPE CONCRETE
112. STRENGTH FCU 27579
113. END DEFINE MATERIAL
114. MEMBER PROPERTY AMERICAN
115. 48 49 PRIS YD 0.85 ZD 0.6
116. CONSTANTS
117. MATERIAL CONCRETE ALL
118. SUPPORTS
119. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
120. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
121. 389 TO 393 395 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
```

JOINT	INFLUENCE AREA
23	0.075000
121	0.075000
122	0.068750
123	0.058437
124	0.063750
125	0.063750
126	0.058437
127	0.068750
128	0.058437
129	0.058437
130	0.068750
131	0.075000
132	0.068750
133	0.075000
134	0.068750
135	0.068750
136	0.062500
137	0.053125
138	0.053125
139	0.062500
140	0.062500
141	0.062500
142	0.053125
143	0.062500
144	0.053125

145	0.062500
146	0.056250
147	0.047813
148	0.056250
149	0.047813
150	0.056250
151	0.037500
152	0.031875
153	0.037500
154	0.031875
155	0.037500
156	0.048750
157	0.044687
158	0.044688
159	0.040625
160	0.040625
161	0.036563
162	0.024375
170	0.040625
171	0.024375
172	0.044687
173	0.048750
174	0.044687
175	0.040625
176	0.036563
184	0.068750
185	0.058437
186	0.063750
187	0.063750
188	0.058437
189	0.068750
190	0.058437
191	0.058437
192	0.075000
193	0.068750
194	0.068750
195	0.068750
196	0.075000
197	0.068750
198	0.062500
199	0.053125
200	0.062500
201	0.053125
202	0.062500
203	0.062500
204	0.053125
205	0.062500
206	0.053125
207	0.062500
208	0.037500
209	0.031875
210	0.037500
211	0.031875
212	0.037500
213	0.056250
214	0.047813

215	0.056250
216	0.047813
217	0.056250
218	0.036563
219	0.040625
220	0.044687
221	0.048750
222	0.044687
223	0.040625
224	0.024375
232	0.040625
233	0.036563
234	0.044687
235	0.048750
236	0.044687
237	0.040625
238	0.024375
246	0.050000
247	0.042500
248	0.050000
249	0.042500
250	0.050000
251	0.032500
253	0.032500
309	0.012500
310	0.010625
311	0.012500
312	0.010625
313	0.012500
314	0.008125
316	0.008125
327	0.012500
328	0.010625
329	0.012500
330	0.010625
331	0.012500
332	0.008125
334	0.008125
461	0.015938
462	0.018750
463	0.018750
464	0.015938
465	0.018750
466	0.012188
468	0.012188
474	0.015937
475	0.018750
476	0.018750
477	0.015937
478	0.018750
479	0.012187
481	0.012187
487	0.010625
488	0.012500
489	0.012500
490	0.010625

```
       491          0.012500
       492          0.008125
       494          0.008125
       500          0.010625
       501          0.012500
       502          0.012500
       503          0.010625
       504          0.012500
       505          0.008125
       507          0.008125
    TOTAL           5.999995
122. ***
123. **************************************************
124. ***LOAD COMBINATION FOR STRESS DESIGN
125. **************************************************
126. ***
127. ***
128. LOAD 1 LOADTYPE DEAD  TITLE DL
129. ***
130. SELFWEIGHT Y -1.1
131. ***FROM GALLERY-1
132. JOINT LOAD
133. 1 FX 0 FY -24.386 FZ -2.569
134. 2 FX 0 FY -25.552 FZ 2.585
137. ELEMENT LOAD
138. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
139. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
140. 389 TO 393 395 PR GY -20
141. *
142. LOAD 2 LOADTYPE LIVE  TITLE LL
143. JOINT LOAD
144. 1 FX 0 FY 21.777 FZ 3.884
145. 2 FX 0 FY -85.599 FZ 10.384
147. *
148. LOAD 3 LOADTYPE LIVE  TITLE LL1
149. JOINT LOAD
150. 1 FX 0 FY -83.836 FZ -10.313
151. 2 FX 0 FY 20.014 FZ -3.769
153. *
154. LOAD 11 LOADTYPE WIND  TITLE WL+Z
155. JOINT LOAD
156. 1 FX 0 FY 221.985 FZ 22.666
157. 2 FX 0 FY -221.985 FZ 22.565
159. *
160. LOAD 12 LOADTYPE WIND  TITLE WL-Z
161. JOINT LOAD
162. 1 FX 0 FY -221.985 FZ -22.666
163. 2 FX 0 FY 221.985 FZ -22.565
165. *
166. LOAD 13 LOADTYPE WIND  TITLE WL+X
167. JOINT LOAD
168. 1 FX 0 FY -3.008 FZ -0.317
169. 2 FX 0 FY -3.232 FZ 0.317
171. *
172. LOAD 14 LOADTYPE WIND  TITLE WL-X
173. JOINT LOAD
```

```
174. 1 FX 0 FY 3.008 FZ 0.317
175. 2 FX 0 FY 3.232 FZ -0.317
177. *
178. LOAD 20 LOADTYPE NONE  TITLE TL1
179. JOINT LOAD
180. 1 FX 0 FY 0 FZ -10.4
181. 2 FX 0 FY 0 FZ 10.4
183. *
184. LOAD 21 LOADTYPE NONE  TITLE TL2
185. JOINT LOAD
186. 1 FX 0 FY 0 FZ 10.4
187. 2 FX 0 FY 0 FZ -10.4
189. *
190. LOAD 30 EARTHQUAKE LOAD (+Z)
191. JOINT LOAD
192. 1 FX 0 FY 2.521 FZ 0.37
193. 2 FX 0 FY -2.521 FZ 0.369
195. *
196. LOAD 31 EARTHQUAKE LOAD (-Z)
197. JOINT LOAD
198. 1 FX 0 FY -2.521 FZ -0.37
199. 2 FX 0 FY 2.521 FZ -0.369
201. *
202. LOAD 32 EARTHQUAKE LOAD (+X)
203. JOINT LOAD
204. 1 FX 0.367 FY 0 FZ 0
205. 2 FX 0.372 FY 0 FZ 0
207. *
208. LOAD 33 EARTHQUAKE LOAD (-X)
209. JOINT LOAD
210. 1 FX -0.367 FY 0 FZ 0
211. 2 FX -0.372 FY 0 FZ 0
212. *
213. ****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
214. *
215. LOAD COMB 201 D + L + TL1
216. 1 1.0 2 1.0 20 1.0
217. *
218. LOAD COMB 202 D + L1 + TL1
219. 1 1.0 3 1.0 20 1.0
220. *
221. LOAD COMB 203 D + L + TL2
222. 1 1.0 2 1.0 21 1.0
223. *
224. LOAD COMB 204 D + L1 + TL2
225. 1 1.0 3 1.0 21 1.0
226. *
227. LOAD COMB 205 D + 0.75XL + 0.75XTL1
228. 1 1.0 2 0.75 20 0.75
229. *
230. LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
231. 1 1.0 3 0.75 20 0.75
232. *
233. LOAD COMB 207 D + 0.75XL + 0.75XTL2
234. 1 1.0 2 0.75 21 0.75
235. *
```

```
236. LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
237. 1 1.0 3 0.75 21 0.75
238. *
239. LOAD COMB 209 D + WL(+Z)
240. 1 1.0 11 1.0
241. *
242. LOAD COMB 210 D + WL(+X)
243. 1 1.0 13 1.0
244. *
245. LOAD COMB 211 D + WL(-Z)
246. 1 1.0 12 1.0
247. *
248. LOAD COMB 212 D + WL(-X)
249. 1 1.0 14 1.0
250. *
251. LOAD COMB 213 D + 0.7XEL(+Z)
252. 1 1.0 30 0.7
253. *
254. LOAD COMB 214 D + 0.7XEL(+X)
255. 1 1.0 32 0.7
256. *
257. LOAD COMB 215 D + 0.7XEL(-Z)
258. 1 1.0 31 0.7
259. *
260. LOAD COMB 216 D + 0.7XEL(-X)
261. 1 1.0 33 0.7
262. *
263. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
264. 1 1.0 2 0.75 11 0.75
265. *
266. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
267. 1 1.0 3 0.75 11 0.75
268. *
269. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
270. 1 1.0 2 0.75 12 0.75
271. *
272. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
273. 1 1.0 3 0.75 12 0.75
274. *
275. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
276. 1 1.0 2 0.75 13 0.75
277. *
278. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
279. 1 1.0 3 0.75 13 0.75
280. *
281. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
282. 1 1.0 2 0.75 14 0.75
283. *
284. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
285. 1 1.0 3 0.75 14 0.75
286. *
287. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
288. 1 1.0 2 0.75 30 0.525
289. *
290. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
291. 1 1.0 3 0.75 30 0.525
```

```
292. *
293. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
294. 1 1.0 2 0.75 32 0.525
295. *
296. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
297. 1 1.0 3 0.75 32 0.525
298. *
299. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
300. 1 1.0 2 0.75 31 0.525
301. *
302. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
303. 1 1.0 3 0.75 31 0.525
304. *
305. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
306. 1 1.0 2 0.75 33 0.525
307. *
308. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
309. 1 1.0 3 0.75 33 0.525
310. *
311. LOAD COMB 233 0.6XD + WL(+Z)
312. 1 0.6 11 1.0
313. *
314. LOAD COMB 234 0.6XD + WL(+X)
315. 1 0.6 13 1.0
316. *
317. LOAD COMB 235 0.6XD + WL(-Z)
318. 1 0.6 12 1.0
319. *
320. LOAD COMB 236 0.6XD + WL(-X)
321. 1 0.6 14 1.0
322. *
323. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
324. 1 0.6 30 0.7
325. *
326. LOAD COMB 238 0.6XD + 0.7XEL(+X)
327. 1 0.6 32 0.7
328. *
329. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
330. 1 0.6 31 0.7
331. *
332. LOAD COMB 240 0.6XD + 0.7XEL(-X)
333. 1 0.6 33 0.7
334. *
335. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 149 NUMBER OF MEMBERS 2
NUMBER OF PLATES 120 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 147

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 79/ 13/ 42 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 453
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 20 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.9/ 73920.3 MB

```
336. CHANGE
337. SUPPORTS
338. 50 TO 79 86 TO 91 98 TO 127 134 TO 139 146 TO 150 152 170 TO 174 176 -
339. 210 TO 214 216 226 TO 230 232 353 TO 357 359 365 TO 369 371 377 TO 381 383 -
340. 389 TO 393 395 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION
```

JOINT	INFLUENCE AREA
23	0.075000
121	0.075000
122	0.068750
123	0.058437
124	0.063750
125	0.063750
126	0.058437
127	0.068750
128	0.058437
129	0.058437
130	0.068750
131	0.075000
132	0.068750
133	0.075000
134	0.068750
135	0.068750
136	0.062500
137	0.053125
138	0.053125
139	0.062500
140	0.062500
141	0.062500
142	0.053125
143	0.062500
144	0.053125
145	0.062500
146	0.056250
147	0.047813
148	0.056250
149	0.047813
150	0.056250
151	0.037500
152	0.031875
153	0.037500
154	0.031875

155	0.037500
156	0.048750
157	0.044687
158	0.044688
159	0.040625
160	0.040625
161	0.036563
162	0.024375
170	0.040625
171	0.024375
172	0.044687
173	0.048750
174	0.044687
175	0.040625
176	0.036563
184	0.068750
185	0.058437
186	0.063750
187	0.063750
188	0.058437
189	0.068750
190	0.058437
191	0.058437
192	0.075000
193	0.068750
194	0.068750
195	0.068750
196	0.075000
197	0.068750
198	0.062500
199	0.053125
200	0.062500
201	0.053125
202	0.062500
203	0.062500
204	0.053125
205	0.062500
206	0.053125
207	0.062500
208	0.037500
209	0.031875
210	0.037500
211	0.031875
212	0.037500
213	0.056250
214	0.047813
215	0.056250
216	0.047813
217	0.056250
218	0.036563
219	0.040625
220	0.044687
221	0.048750
222	0.044687
223	0.040625
224	0.024375

232	0.040625
233	0.036563
234	0.044687
235	0.048750
236	0.044687
237	0.040625
238	0.024375
246	0.050000
247	0.042500
248	0.050000
249	0.042500
250	0.050000
251	0.032500
253	0.032500
309	0.012500
310	0.010625
311	0.012500
312	0.010625
313	0.012500
314	0.008125
316	0.008125
327	0.012500
328	0.010625
329	0.012500
330	0.010625
331	0.012500
332	0.008125
334	0.008125
461	0.015938
462	0.018750
463	0.018750
464	0.015938
465	0.018750
466	0.012188
468	0.012188
474	0.015937
475	0.018750
476	0.018750
477	0.015937
478	0.018750
479	0.012187
481	0.012187
487	0.010625
488	0.012500
489	0.012500
490	0.010625
491	0.012500
492	0.008125
494	0.008125
500	0.010625
501	0.012500
502	0.012500
503	0.010625
504	0.012500
505	0.008125
507	0.008125

```
    TOTAL           5.999995
341. *
342. *
343. *
344. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
345. LOAD COMB 100 1.4XD
346. 1 1.4
347. *
348. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
349. 1 1.2 20 1.2 2 1.6
350. *
351. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
352. 1 1.2 20 1.2 3 1.6
353. *
354. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
355. 1 1.2 21 1.2 2 1.6
356. *
357. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
358. 1 1.2 21 1.2 3 1.6
359. *
360. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
361. 1 1.2 11 0.8
362. *
363. LOAD COMB 106 1.2XD + 0.8XWL(+X)
364. 1 1.2 13 0.8
365. *
366. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
367. 1 1.2 12 0.8
368. *
369. LOAD COMB 108 1.2XD + 0.8XWL(-X)
370. 1 1.2 14 0.8
371. *
372. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
373. 1 0.9 11 1.6
374. *
375. LOAD COMB 110 0.9XD + 1.6XWL(+X)
376. 1 0.9 13 1.6
377. *
378. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
379. 1 0.9 12 1.6
380. *
381. LOAD COMB 112 0.9XD + 1.6XWL(-X)
382. 1 0.9 14 1.6
383. *
384. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
385. 1 1.2 2 0.5 1.6
386. *
387. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
388. 1 1.2 2 0.5 1.6
389. *
390. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
391. 1 1.2 2 0.5 1.6
392. *
393. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
394. 1 1.2 2 0.5 1.6
395. *
```

```
396. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
397. 1 1.2 3 0.5 1.6
398. *
399. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
400. 1 1.2 3 0.5 1.6
401. *
402. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
403. 1 1.2 3 0.5 1.6
404. *
405. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
406. 1 1.2 3 0.5 1.6
407. *
408. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
409. 1 1.2 2 0.5 1
410. *
411. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
412. 1 1.2 2 0.5 1
413. *
414. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
415. 1 1.2 2 0.5 1
416. *
417. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
418. 1 1.2 2 0.5 1
419. *
420. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
421. 1 1.2 3 0.5 1
422. *
423. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
424. 1 1.2 3 0.5 1
425. *
426. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
427. 1 1.2 3 0.5 1
428. *
429. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
430. 1 1.2 3 0.5 1
431. *
432. LOAD COMB 129 0.9XD + EL(+Z)
433. 1 0.9 30 1.0
434. *
435. LOAD COMB 130 0.9XD + EL(+X)
436. 1 0.9 32 1.0
437. *
438. LOAD COMB 131 0.9XD + EL(-Z)
439. 1 0.9 31 1.0
440. *
441. LOAD COMB 132 0.9XD + EL(-X)
442. 1 0.9 33 1.0
443. PERFORM ANALYSIS
444. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 17,2020 TIME= 15: 8:48 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
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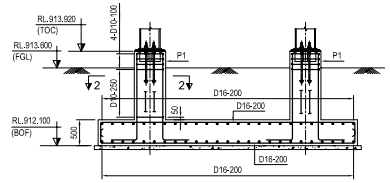
ATTACHMENT 2 CIVIL DRAWINGS

PROPRIETARY & CONFIDENTIAL

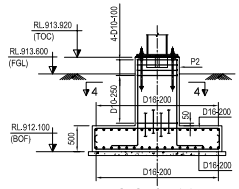
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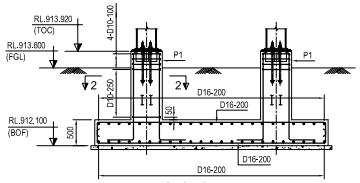
DRAWING NUMBER: MD - 216 - 49PR- EG - CV - GAD - 1601
REV: D01



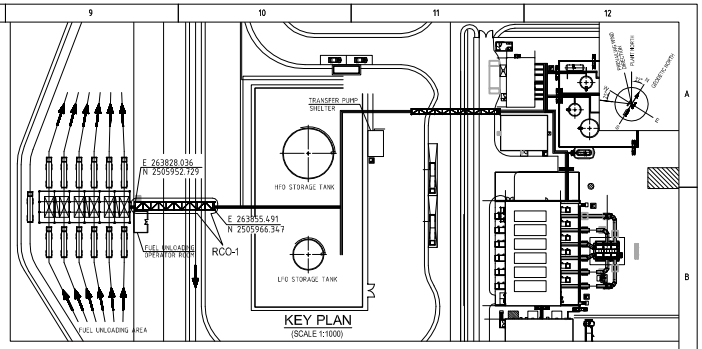
SECTION 1-1
(SCALE - 1:50)



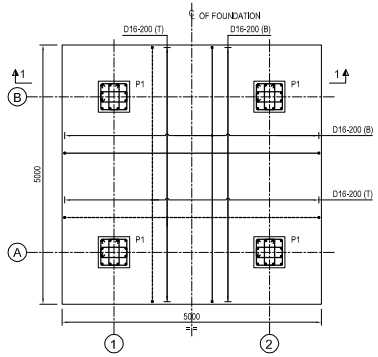
SECTION 3-3
(SCALE - 1:50)



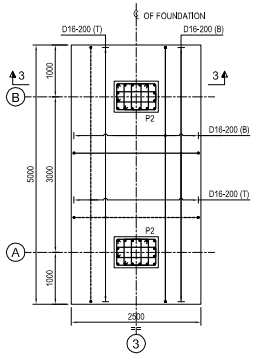
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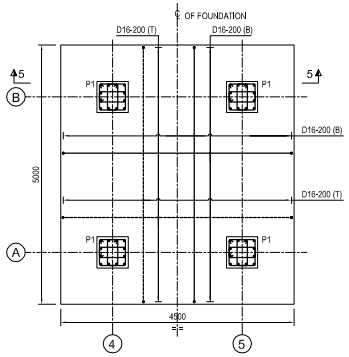
KEY PLAN
(SCALE 1:1000)



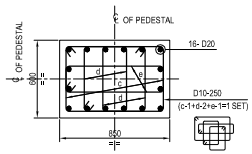
DETAIL OF FOUNDATION MKD. F1
(SCALE - 1:50)



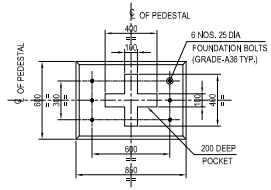
DETAIL OF FOUNDATION MKD. F2
(SCALE - 1:50)



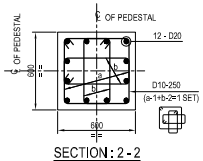
DETAIL OF FOUNDATION MKD. F3
(SCALE - 1:50)



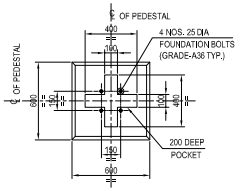
SECTION : 4 - 4
(TYP. RC DETAIL OF PEDESTAL MKD. P2)
(SCALE- 1:20)



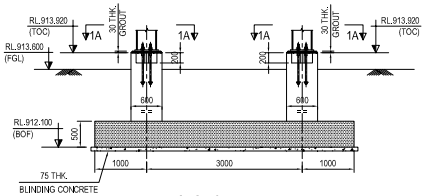
SECTION : 1B - 1B
(TYP. DETAIL OF PEDESTAL MKD. P2)
(SCALE- 1:20)



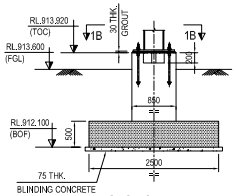
SECTION : 2 - 2
(TYP. RC DETAIL OF PEDESTAL MKD. P1)
(SCALE- 1:20)



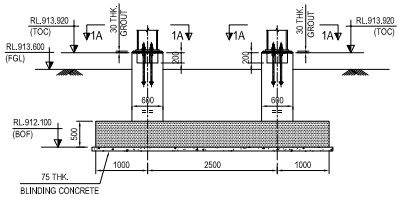
SECTION : 1A - 1A
(TYP. DETAIL OF PEDESTAL MKD. P1)
(SCALE- 1:20)



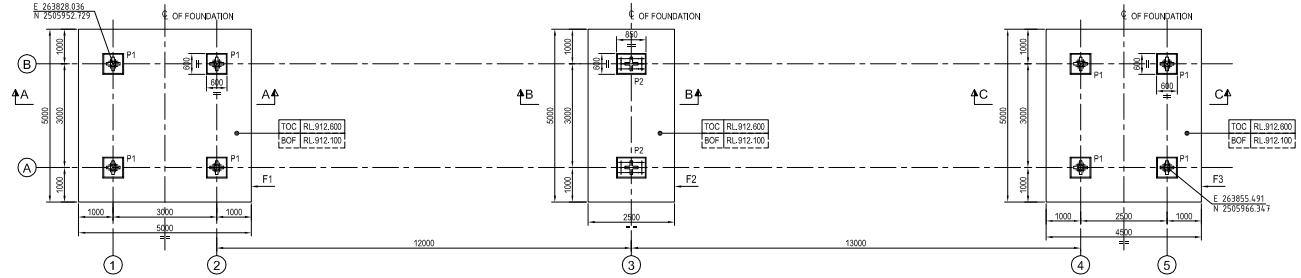
SECTION A-A
(SCALE - 1:50)



SECTION B-B
(SCALE - 1:50)



SECTION C-C
(SCALE - 1:50)



FOUNDATION LAYOUT PLAN FOR DUCT SUPPORT
(SCALE 1:75)

LEGEND :-

TOC	= TOP OF CONCRETE
BOC	= BOTTOM OF CONCRETE
FGL	= FINISHED GROUND LEVEL
BOF	= BOTTOM OF FOUNDATION
TYP	= TYPICAL
MKD	= MARKED
℄	= CENTRE LINE
THK	= THICK
GA	= GENERAL ARRANGEMENT.
RC	= REINFORCED CONCRETE
PCC	= PLAIN CEMENT CONCRETE
DIA	= DIAMETER
T	= TOP
B	= BOTTOM
RL	= REDUCED LEVEL
- - - -	= TOP REINFORCEMENT
———	= BOTTOM REINFORCEMENT

SCHEDULE OF BOLTS

BOLT DIA- (MM)	BOLT NUMBERS REQUIRED
25	44

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATION & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 450KN SQ. M.
5. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. FOR ALL OTHER NOTES REFER DRAWING NO. MD-101-SMEM-EG-ST-DST-0001 TO 0008.
8. FOR GENERAL NOTES OF CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	26/10/2023	AS BUILT	ABH	UR	SVK	AG	
MD-216-49PR-EG-EL-LAD-V553-1002	POWER GENERATION – PLAN	C01	13/08/2020	ISSUED FOR CONSTRUCTION	ABH	UR	SVK	AG	
MD-216-49PR-EG-ST-GAD-2401	PLAN, ELEVATION & DETAILS OF PIPE RACK IN RCO-1	A01	17/07/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

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Saudi Arabian Mining Company

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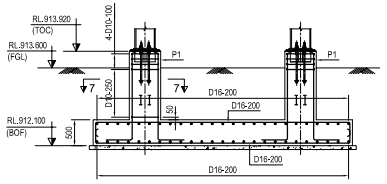
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OUTOTEC DRAWING NUMBER: 900707__GAU01__WG01__MTL01__45057 — DOCUMENT ID: 0U100682333 — REVISION: 2

DRAWN	ABH	07/07/2020
CHECKED	UR	09/07/2020
ENGINEER	SVK	07/07/2020
APPROVED	AG	17/07/2020
EMPLOYER		
SCALE	1:75, 50, 20	A1

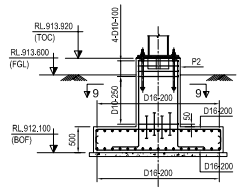
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT PIPE RACK
G.A. AND DETAIL OF FOUNDATION FOR RCO-1

DRAWING NUMBER: MD - 216 - 49PR- EG - CV - GAD - 1601 — REV: D01

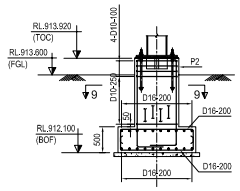
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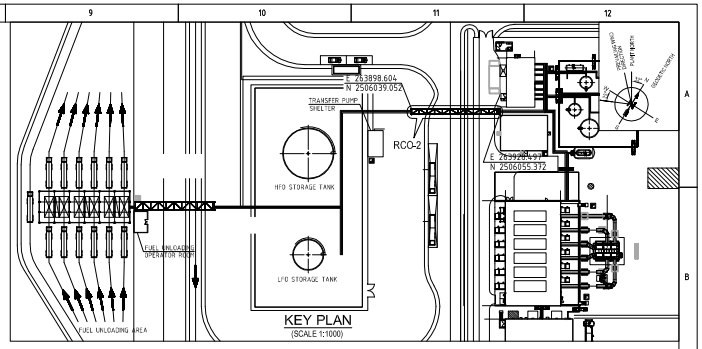
SECTION : 6-6
(SCALE - 1:50)



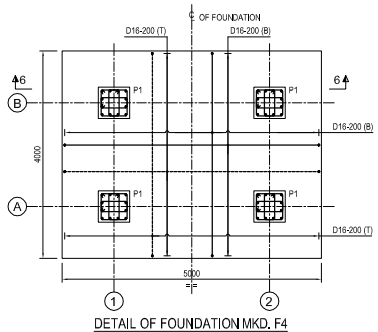
SECTION : 8-8
(SCALE - 1:50)



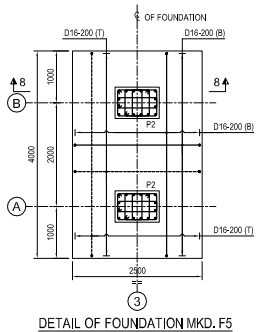
SECTION : 10-10
(SCALE - 1:50)



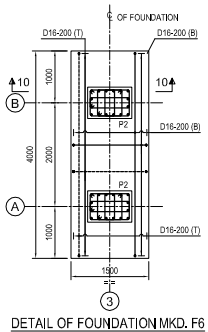
KEY PLAN
(SCALE 1:1000)



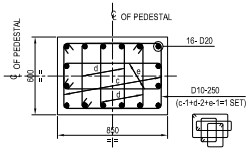
DETAIL OF FOUNDATION MKD. F4
(SCALE - 1:50)



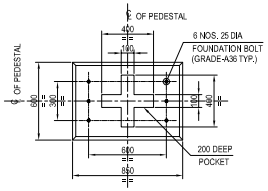
DETAIL OF FOUNDATION MKD. F5
(SCALE - 1:50)



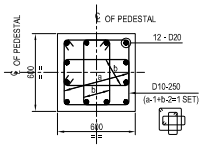
DETAIL OF FOUNDATION MKD. F6
(SCALE - 1:50)



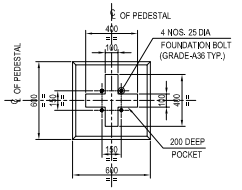
SECTION : 9-9
(TYP. RC DETAIL OF PEDESTAL MKD. P2)
(SCALE- 1:20)



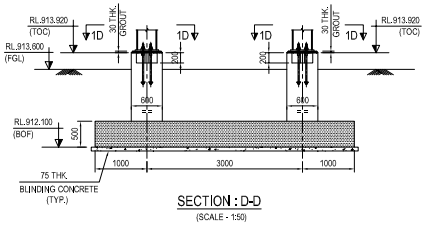
SECTION : 1E-1E
(TYP. DETAIL OF PEDESTAL MKD. P2)
(SCALE- 1:20)



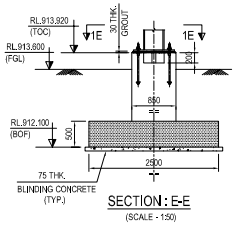
SECTION : 7-7
(TYP. RC DETAIL OF PEDESTAL MKD. P1)
(SCALE- 1:20)



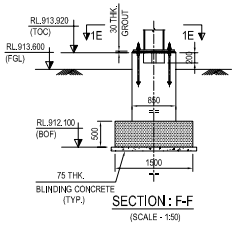
SECTION : 1D-1D
(TYP. DETAIL OF PEDESTAL MKD. P1)
(SCALE- 1:20)



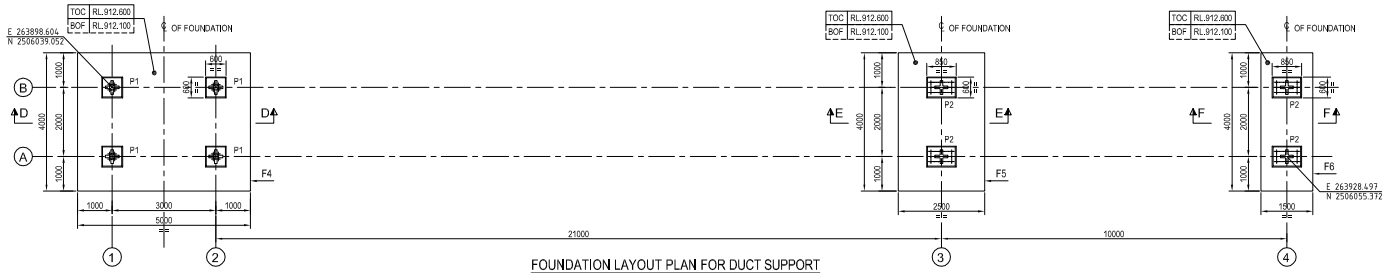
SECTION : D-D
(SCALE - 1:50)



SECTION : E-E
(SCALE - 1:50)



SECTION : F-F
(SCALE - 1:50)



FOUNDATION LAYOUT PLAN FOR DUCT SUPPORT
(SCALE 1:75)

LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOUNDATION
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
THK = THICK
GA = GENERAL ARRANGEMENT.
RC = REINFORCED CONCRETE
PCC = PLAIN CEMENT CONCRETE
DIA = DIAMETER
T = TOP
B = BOTTOM
RL = REDUCED LEVEL
– – – = TOP REINFORCEMENT
——— = BOTTOM REINFORCEMENT

SCHEDULE OF BOLTS

BOLT DIA. (MM)	BOLT NUMBERS REQUIRED
25	40

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
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3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 450KN/ SQ. M.
5. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
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7. FOR ALL OTHER NOTES REFER DRAWING NO. MD-101-SMEM-EG-ST-DST-0001 TO 0008.
8. FOR GENERAL NOTES OF CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	26/10/2023	AS BUILT	ABH	UP	SVK	AG	
MD-216-49PR-EG-EL-LAD-V553-1002	POWER GENERATION – PLAN	C01	13/08/2020	ISSUED FOR CONSTRUCTION	ABH	UR	SVK	AG	
MD-216-49PR-EG-ST-GAD-2402	PLAN, ELEVATION & DETAILS OF PIPE RACK IN RCO-2	A01	17/07/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_008_C_G_24292_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__WG01__MTL01__45077 — DOCUMENT ID: 0U100688135 — REVISION: 2

DRAWN	ABH	04/07/2020
CHECKED	UR	07/07/2020
ENGINEER	SVK	04/07/2020
APPROVED	AG	17/07/2020
EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT PIPE RACK
G.A. AND DETAIL OF FOUNDATION FOR RCO-2

DRAWING NUMBER: MD - 216 - 49PR- EG - CV - GAD - 2003 — REV: D01